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UNIVERSITY OF CALIFORNIA

Los Angeles

The Origin and Development of Adpositions and Adpositional Phrases in the Indo-European Languages

A dissertation submitted in partial satisfaction of the requirements for the degree Doctor of Philosophy in Indo-European Studies

by

Christopher Wilhelm

2001

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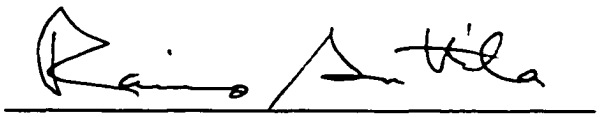
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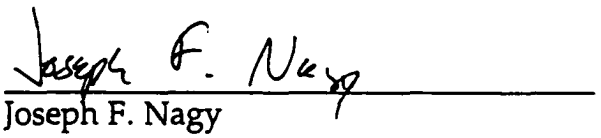
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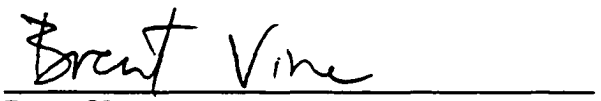
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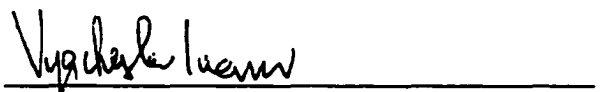
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DEDICATION

to Laura-

...with whom all good things are possible.

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ABSTRACT OF THE DISSERTATION

The Origin and Development
of Adpositions and Adpositional Phrases
in the Indo-European Languages

by

Christopher Wilhelm
Doctor of Philosophy in Indo-European Studies
University of California, Los Angeles, 2001
Professor Vyacheslav V. Ivanov, Chair

Prepositions and postpositions are among the most heavily-employed lexemes in any given Indo-European language. And their lack of inflection in a language family in which morphosyntactic relations are in great measure expressed through elaborate inflectional systems is immediately striking. How these lexemes arrived at such a state, and how they are affected by and participate in the grammatical and syntactic changes experienced by all languages, are questions calling out for investigation. While A. Meillet has

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demonstrated the nominal origins of many prominent Indo-European adpositions, complete with firm evidence that several attested adpositions represent various noun case-forms or even significant portions of nominal paradigms, but a great many details have yet to be explained on this point.

Furthermore, a particularly fascinating feature of the various branches of the Indo-European family has barely been addressed: While most of the earliest attested daughter languages, are almost exclusively prepositional in their syntactic placement of adpositional phrases, others, are mostly postpositional. A small number, such as Luwian, Umbrian and Old Lithuanian, show a substantial number of both varieties. and evidence in these languages indicates that a change in word order from postpositions to prepositions is in progress. The processes involved in this rather surprising state of affairs call for detailed study, especially given evidence that change in the word order of the adpositional phrase is one of the first steps in a general syntactic change among the Indo-European languages.

The present investigation, then, will seek a greater understanding of the steps that marked this process of word order change, as well as, insofar as is possible to ascertain, the conditions that set it in motion. Possible implications for linguistic and especially syntactic universal will also be eagerly sought after. For the present exercise, concentration will be focused upon those languages offering particular insight into the origins of these items, above all Hittite, or presenting substantial evidence of an adpositional word order shift in progress. These include Luwian and Lycian, Umbrian and other Italic languages, and Old Lithuanian and the rest of the Baltic subgroup.

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Chapter I
Introduction

1.0. The Problem

Prepositions and postpositions, along with their attendant phrases, have received rather little attention in discussions of the development of the Indo-European morphological and syntactic systems. The evolutions of the nominal and verbal paradigms have of course commanded the lion's share of scholarly interest, while adpositions, without the extensive permutations shown by other word classes, have not drawn lengthy analysis. However, these little words tend to be among the most heavily-employed lexemes in any given Indo-European language. And their lack of inflection in a language family in which morphosyntactic relations are in great measure expressed through elaborate inflectional systems is in itself immediately striking[1]. How these lexemes arrived at such a state, and how they are affected by and participate in the grammatical and syntactic changes experienced by all languages, are questions calling out for investigation.

In many respects, however, the state of knowledge of the Indo-European languages had not developed sufficiently to support such a study until

---

[1] It is true that in some Indo-European languages discovered in the past century, above all Old Hittite, adpositions show inflectional endings and other nominal characteristics (see Chapter II, especially sections 4, 5, and 6.2). And A. Meillet has reconstructed partial nominal paradigms for the antecedents of many Indo-European adpositions (see Section 2.3, this chapter). But these features were not apparent in the nineteenth century when comparative Indo-European studies first flourished, and when the general direction of investigation of the field became established.

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contemporary times, as will be elaborated. The nineteenth century, the classical age of Indo-European exploration, also knew nothing of Hittite, or J. Greenberg's universals, and had no notion of relational nouns as delineated by C. Fillmore (1968), so that there was far less understanding of the importance of these items. Moreover, since A. Meillet's demonstration of the nominal origins of many prominent Indo-European adpositions, complete with firm evidence that several attested adpositions represent various noun case-forms or even significant portions of nominal paradigms (1937:350-2), the question of these segments' sources, insofar as it has been discussed at all, has often been regarded as essentially solved. However, a great many details have yet to be explained on this point. Such questions remain most prominently in Hittite, where relational nouns cognate with adpositions found both within that language and beyond it occur with some frequency, but also in some of the other daughter languages.

Furthermore, a particularly fascinating feature of the various branches of the Indo-European family has barely been addressed: While most of the earliest attested daughter languages, such as Greek, Latin, Gothic, and Old Church Slavic, are almost exclusively prepositional in their syntactic placement of adpositional phrases, others, especially Hittite but also Tocharian and Sanskrit, at least by the Classical period, are mostly postpositional, and a small number, such as Umbrian and, to a somewhat lesser extent, Old Lithuanian, show a substantial number of both varieties.

The processes leading to this rather surprising state of affairs call for a detailed study. The great synchronic stability of word order in adpositional

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phrases within the syntax of most of the languages under consideration highlights the significance of these units. In the set of grammatical universals set forth by Greenberg, a strong correlation is found between adpositional word order and the unmarked order of the sentence as a whole. Specifically, languages showing a verb-final word order are nearly always postpositional, while those in which the verb normally occurs before the object tend also to be prepositional (1963:61-2).

Moreover, and in contrast to the other correlates of sentential word order in Greenberg's schema, such as the order of noun-genitive and noun-adjective pairs, adpositional phrases within the attested Indo-European languages are far less susceptible to variation; such items normally postposed in a given language are seldom also preposed in the same tongue, and likewise prepositions are rarely found thus postposed. In other words, one must normally admit a great number of exceptions in any discussion of unmarked word order, but in adpositional phrases, such exceptions are highly marked indeed. Within the different languages, variation in the order of adpositional phrases is usually much rarer than variation in the order of the sentence as a whole. In addition, V. Ambrazas notes for Lithuanian (1982:113-4, 118), as does G. Holland for Greek (1980:161), that the shift from one adpositional alignment to the other has clearly taken place earlier than others seemingly more central to a language's syntax, such as that between object-verb and verb-object. Holland goes on to hypothesize that this shift, in both cases from postpositions to prepositions, is one of the first steps in a general syntactic change among the Indo-European

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languages (187). One may suppose, then, that adpositional shift is of especially great consequence in this language family.

Given the broad disparity in adpositional alignment over the attested Indo-European language family, then, clearly some mechanism has been at work to effect changes in word order for such phrases from one direction to the other. In addition, the resistance of such syntactic units to variation suggests that, where such a change has in fact taken place, it is indicative of a greater, more comprehensive change in a language's syntactic system. With the correlation between postpositions and subject-object-verb (SOV) sentence order on the one hand and prepositions and verb-object (VO) order on the other, one may safely suppose that a shift from one type of adpositional-phrase order to the other definitively marks a shift between verb-final and object-final syntax. Therefore, regardless of the unmarked word order prevailing in the language of the Indo-Europeans, for the sentence as a whole and for the adpositional phrase in particular, the typological distribution of such syntactic units conclusively demonstrates that in some subset of these languages, a complete shift in the order of meaningful elements took place.

1.1. Purpose

A major focus of the present effort will be to attempt to discover the mechanisms or processes by which a change in syntactic orientation between postposing and preposing of adpositions took place in the Indo-European languages, as well as the speed with which such changes were realized. The

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avenue of inquiry for this attempt will be an examination of adpositions and adpositional phrases in some branches of the Indo-European family where these units have diverged in especially striking ways. By comparing the organization and formation of these units in the languages in question, it should be possible to gain some insight into the different paths of development followed by these various offshoots of Proto-Indo-European.

2.0 Preliminaria

2.1. Terminology

Before this examination may proceed further, a definition of the terms under discussion is an essential foundation to any scholarly endeavor. Somewhat surprisingly, many of the terms central to this effort, such as adposition and preverb, are often discussed only in passing or not at all in such glossaries as E. Hamp (1966) or W. Bright, ed. (1992). This omission may in part be because some of the terms seem clear enough not to require definition, but also possibly as a consequence of their importance being overlooked. Still, Bright at least offers sufficiently copious references to guide the inquiring researcher toward treatments of such matters in works on general and theoretical linguistics. With these and such works focusing on Indo-European syntax as W. Lehmann (1974) and P. Friedrich (1975), it is possible to produce working definitions for such terms to be used over the course of the present effort.

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2.1.1. Adpositions

Adposition is simply a blanket term covering both prepositions and postpositions, with the former occurring before and the latter coming after the noun or substantive with which they associate. ***Prepositions*** are defined by D. Crystal as "Items that govern and typically precede nouns, pronouns, and certain other forms (428)," while a ***postposition*** is "A particle [the definition of which will be discussed below] that follows the noun it governs (428)." P. Schachter (1985), however, in his investigation of parts-of-speech systems, offers a more thorough examination of these adposed items. He places adpositions in the category of "role markers", which may serve as case markers, discourse markers, or "other" role indicators (35). He elaborates: "Case markers are words[2] that indicate the syntactic and/or semantic role (e.g. subject and/or agent) of the noun phrase to which they belong. Discourse markers are words that indicate the discourse role (e.g. topic) of the associated noun phrase" (36). He offers by way of example a Japanese sentence presenting postpositions in both functions:

1) Type de wa John ga Mary ni tegami o kaita.
typewriter instr. topic subj. ind. obj. letter dir. obj. wrote
'John wrote Mary a letter on the typewriter.' (36)

On the nebulous heading of "other" role markers, Schachter observes,

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[2] Discrete words, it must be specified, and therefore distinct from case affixes.

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> There are certain adpositions that are clearly not discourse markers, but that are not ordinarily identified as case markers... The distinction between case-marking and other adpositions seems to be a somewhat arbitrary one, however, based in part on the traditional identification of only certain types of grammatical or semantic roles with the label 'case' (36).

Those roles tending not to be considered "case" relations, according to Schachter, are purely directional concepts such as 'under' or 'beside'. The connection between adpositions and the general subject of case relations will then need to be at least briefly addressed. One may nevertheless conclude that adpositions are words that assign a grammatical, semantic, or discourse role to their associated noun phrase in relation to the sentence or clause in which that phrase occurs. Moreover, since adpositions in the Indo-European languages are not known for conveying discourse functions, this investigation will concern itself rather with the grammatical and semantic roles marked by these words. For the purposes of the present investigation, then, adpositions are considered discrete words that convey grammatical or adverbial relationships between the substantives they govern and the sentence as a whole.

2.1.2. Case

As for the concept of *case*, Schachter's use of the term follows the theory outlined in C. Fillmore's (1968) treatise "The Case for Case." Fillmore postulates a set, presumably finite in number, of specific relationships binding a noun

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phrase to the sentence as a whole; these relationships function at the level of a given language's deep structure. They then find expression at the surface identifiable by such labels as "Manner, Extent, Location, Agent," and so forth (17). Fillmore is careful, however, to distinguish between these underlying relationships and the use of the word "case" normally found in discussions of inflecting languages such as those of the Indo-European family:

> [I]f there are recognizable intrasentence relationships of the types discussed in studies of case systems (whether they are inflected in case affixes or not),... if these same relationships can be shown to be comparable across languages, and... if there is some predictive or explanatory use to which assumptions concerning the universality of these relations can be put, then surely there is no meaningful objection to using the word case, in a clearly-understood deep-structure sense, to identify these relationships (20).

These case relations, then, may be conveyed by a number of different means in various languages or indeed within any one language. Most Indo-European languages, especially the earliest-attested ones, employ both case affixes and adpositions in expressing these various concepts. For example, the relationship Instrument in most of these languages is expressed by an instrumental case suffix, although where such is unavailable through sound change and morphological syncretism, as in Latin or Greek, another morpheme, such as that normally identifying the ablative or dative case, may be pressed into service. The case relation Place Where may be conveyed by a case morpheme,

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as in Hittite, Sanskrit, or Lithuanian, or by an adposition, as again in Latin or Greek. Finally, the concept Place Before Which or In Front of Which is in all these languages expressed by an adposition. In this respect, then, adpositions and case affixes are similar in functioning. For the purposes of the present effort, therefore, a distinction will be observed at all times between case relations and case affixes, which like adpositions are vehicles for the expression of such relations. The term *case* will be considered to describe these relations, either strictly grammatical or more broadly adverbial, which may be expressed either by adpositions or by case affixes.

2.1.3. Relational Nouns

In addition to adpositions and case affixes, case relations in some Indo-European languages, most notably Hittite and Luwian, are also found delineated by one other set of forms. These are ***relational nouns***, such as Hittite *píran* '[at the] front, before' and *appan* '[at the] back, behind, after'. Indeed, such English phrases as 'in front of,' 'in back of', 'at the head of', vividly convey the senses of such relational nouns. Such a noun serves to relate another noun, which it also governs, to the greater unit of the sentence, generally in some spatial or temporal sense.

2.1.4. Preverbs

A counterpart set to that of the adpositions throughout the Indo-European family is that of the ***preverbs***, which, according to Lehmann, "are

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distinctively characterized by being closely associated with verbs and modifying their meaning. In their normal position they stand directly before verbs" (117). Similarly, J. Friedrich, writing on Hittite, declares, "Wie in den anderen indogermanischen Sprachen wird auch im Hethitischen die Bedeutung des einfachen Verbalstammes durch Präverbien modifiziert" (1960:75). P. Friedrich, like Lehmann (212), observes that the same segments frequently serve both adpositional and preverbal capacities:

> Homeric Greek has 24 particles which Chantraine has accurately described as 'little invariable words, originally autonomous, which make space-time relations more precise;' his key words are 'autonomous' and 'space-time.' Such words are essentially or potentially autonomous but tend to precede a modified verb. Their semantic function is mainly to disambiguate (1) the case subcategories of substantive, or (2) the aspectual subcategories of verbs, or (3) both. (35)

As such, preverbs must be considered to modify and therefore to govern verbs just as adpositions govern noun phrases; the modification of the verbal sense referred to by Lehman is generally spatial or temporal. A. Goetze (1963) nevertheless offers criteria for distinguishing postpositions from preverbs, at least in Hittite, but P. Friedrich argues that these two sets of items should simply be classified together, along with local adverbs, as ***locative auxiliaries***. The question of whether a distinction between adpositions and preverbs is valid and necessary is thus raised, and F. Starke, as will be observed in the following section, also addresses it.

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2.1.5. Adverbs

While P. Friedrich adopts the umbrella term of locative auxiliary for adpositions and preverbs in the earliest-attested Indo-European languages, Starke classifies them even more broadly, placing them under the rubric of ***adverbs*** (1977:133). Crystal offers as a definition for the adverb, "A word whose main function is to specify the kind of action expressed by a verb; other functions include acting as an intensifier and as a sentence connector" (424). Schachter, however, proposes a more encompassing analysis:

> The usual functional definition of adverbs identifies them as modifiers of verbs, adjectives, and other adverbs. ... In order to expand this definition so as to include sentence adverbs like *unfortunately* (which are in fact modifiers of entire sentences), and to allow for certain other possibilities (such as adverbs that modify entire verb phrases), we can say that adverbs function as modifiers of constituents other than nouns. (20)

Certainly both preverbs and adpositional phrases serve at least some functions similar to those of adverbs as outlined by Schachter, whose definition is a cogent one. Still, in both cases, their possible roles are more circumscribed than those of general adverbs, although, the line between the categories is often blurry. Nevertheless, there are generally clear differences in function between when such an item takes a nominal complement and thus serves as an adposition; or modifies a verb and is therefore a preverb (especially in those languages, as in Italic and Baltic, where auxiliary and verb are regarded as

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forming a single word); or fills neither of these roles and may therefore be considered an adverb. It is true that in languages, such as those of Anatolia, where preverbs are not bound to their verbs, that the line between the categories may become quite blurred; see in particular Chapter I, Sections 1.1.2, 3.1, 6-6.1, and 10.0. But the overall functional differences between adpositions, preverbs and adverbs remains significant enough that it would be a mistake to ignore them., even if in some circumstances the distinction seems somewhat artificial.

2.1.6. Government

If adpositions and preverbs are to be considered to govern nouns (or noun phrases) and verbs respectively in similar manners, the concept of ***government*** requires some explanation. The concept is defined by Crystal as "A type of grammatical linkage in which one word requires a specific form of another," as exemplified by the requirement in Latin that *ad* be followed by an accusative (422). In more general terms, a definition cited by M. Saito as standard is that of J. Aoun and D. Sportiche 1982: "α governs β: Every maximal projection dominating α dominates β and conversely" (1992:75). Saito then lists as maximal projections such categories as the Sentence as a whole, along with Verb, Noun, Adjective, and Prepositional Phrases, with of course the understanding that, at least for the first three of these, a phrase may consist of a single word. The use by Aoun and Sportiche of the word "dominate" reflects the nodes on tree diagrams favored by Transformational Grammar (see, for

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example, Radford 1988). These same diagrams also inspire the term ***head*** to designate the governing item or projection.

The relationship between dominating and dominated constituents may for the present purpose be better characterized then as that of head and complement, that is, the specific form required under the linkage of government alluded to by Crystal. As N. Chomsky observes, "The central notion of government theory is the relation between the head of a construction and the categories dependent on it" (1981:5), while "a category governs its complements in a construction of which it is the head" (50). However, while a head word governs its complement, it is also modified by that complement; Hawkins for instance consistently identifies complements or dependent constituents and modifiers and heads with modified categories (1983:2, 97-8). As such, throughout the present undertaking, the relations of governing and governed constituent, of head and complement or dependent, and of modified and modifier will be regarded as all being essentially equivalent: Each pair of terms describes the same grammatical and semantic relationships, wherein the former item affects the latter one.

2.1.7. Particles

The term ***particle*** has been invoked in a couple of definitions here, and therefore it too must be addressed. These are, as the name implies, generally small words, and their roles are vaguely described: "An invariable word with a grammatical function", as Crystal has it (427). A. Zwicky observes that the

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concept of a particle has proven attractive to many scholars investigating inflectional languages such as those of the Indo-European family: "Especially in older works (e.g. W. Whitney 1889...) the word covers any indeclinable, or uninflectable, item; this is especially common for languages, like Sanskrit, in which most words have inflected forms" (1985:290). More generally, Zwicky declares, "The particle is a ubiquitous notion in syntax. The most common use of the term is to label items which, in contrast to those established word classes in a language, have (a) peculiar semantics and (b) idiosyncratic distributions. Thus 'particle' is a cover term for items that do not easily fit into syntactic and sematic generalizations about the language" (290). Indeed, Zwicky's position is that, "'[P]article', in its customary broad usage, is a pretheoretical notion that has no translation into a theoretical concept of linguistics, and must be eliminated in favor of such constructs" (284). He argues that many of these items may be classified as discourse markers (303), but by no mean all of them; these latter forms will however fall into other established grammatical categories.

However, the concept of particles as discrete grammatical units was in large measure rehabilitated by G. Dunkel in the final decades of the twentieth century, above all in Dunkel 1992. He argues that, while such morphological elements as roots, suffixes, and endings are universally recognized as essential components of Indo-European grammar, there remains a set of items that do not fall into any of these categories. These comprise "die Indeklinabilia, oder Partikeln" (1992:153). This definition then represents a return to the older position held by such grammarians as Whitney. Dunkel then breaks this

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category down into several subclasses:

> Semantisch sind die Partikeln im weitesten Sinne adverbial, ob nun jeweils *deiktisch*, einschließlich demonstrativ und pronominal...; *konjunktional*, einschließlich disjunktiv und adversativ...; *modal*, einschließlich irreal..., verallgemeinernd..., meliorativ..., peiorativ..., negativ..., und interrogativ; *emphatisch*..., einschließlich hortativ... und affirmativ...; oder *lokal*, ob räumlich oder zeitlich (**nú, *sóm, *pró, *ápo, *dṷís*) (1992:154).

Adpositions, then, would in this schema fall into the final subcategory, and indeed, the term *Lokalpartikeln* is adopted by such scholars as H. Hettrich (1991, 1993). Dunkel describes the characteristics of particles, including the simplification of consonant clusters (156) and their tendency to be derived in the diachronic sense from declined nominal forms (158-60), but the fundamental qualities that define the segments under examination in this effort is their indeclinability or invariability, along with their local orientation. The term "local" here may of course include connotations of both stationary location or motion in a given direction, and in either a spatial or temporal sense.

In short, the term "local particle" is slightly broader in sense than J. Friedrich's locative auxiliary, in that a lexeme may be the former without necessarily also being the latter, but the set of locative auxiliaries is invariably drawn from the set of local particles. Clearly, there is no widespread scholarly consensus as yet as to what constitutes a particle. For the present effort, where the term is used, it refers to an indeclinable word with a grammatical function, *but* which is not an adposition, preverb, or adverb.

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2.1.8. Clitics

One other concept calling for preliminary definition is that of ***clitic***. Clitics are occasionally lumped in with particles and, like that set rejected by Zwicky, may serve a variety of usages. Hamp labels them "Morphemes... intermediate between words and affixes, when these morphemes are grammatically loosely bound, but phonologically tightly bound to a free word to which they are adjacent" (22); or, in Crystal's words, "A form that resembles a word but that cannot stand on its own as a normal utterance because it is structurally dependent on a neighbouring word" (417). Beyond these postulations, however, Zwicky argues that no more precise definition is possible: "[I]n the case of terms (like 'word' and 'clitic'[3]) which function as theoretical primitives, ONLY lists of symptoms can be provided" (285) as diagnostics in determining to which basic category an item belongs. To this end, he proposes numerous tests, too extensive to reproduce here, regarding such qualities as accentuation, internal and external sandhi, variability in order, and the like, to determine whether a segment ought to be considered an affix, a clitic, or an independent word (286-90; see also 1992:269-70). A clitic, then, is not an affix, but it is not entirely an independent word, either. Hamp is accurate in observing the tight phonological binding of the clitic to another word, but the grammatical connection is usually more substantial than his definition implies.

Subsets of clitics include ***proclitics***, which precede the word with which

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[3] As well, presumably, as 'affix'.

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they are semi-bound, and *enclitics*, which follow. Clitics are of interest to the study of adpositions because of the clitic positioning of many such items, not only to individual nouns, but, as J. Klavans in particular notes, they may be like other such segments cliticized to whole phrases as well (1985:117). Such characteristics, as will be elaborated subsequently in this endeavor, evidently represent a significant step in adpositional development, whereby they may become morphologized affixes, usually indicating case functions.

2.2. Previous Scholarship

In many respects, awareness of the importance of adpositions and their attendant phrases to the grammar and syntax of a given language has come relatively late. J. Kuryłowicz relates Otto Jespersen's syntactic analysis of the principal parts of speech, classified as "primaries (substantives), secondaries (adjectives and verbs), and tertiaries (adverbs)" (1964:23); evidently such entities as adpositions did not register significantly enough to be addressed. Nevertheless, these items have been discussed here and there even in the early stages of linguistic study. Moreover, questions of the origin and development of such seemingly unobtrusive forms have at least in general terms been considered since the changeability of language was first observed. A summary of various approaches to and ideas on how adpositions come into being and how their usage evolves over time, then, is in order before the central problem and hypothesis for the present effort may be adequately set forth.

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2.2.1. First Efforts

B. Heine, U. Claudi, and F. Hünnemeyer credit the fourteenth-century Chinese scholar Zhou Bo-qi with being the first to make the case that "empty," linguistic symbols, that is, abstract or grammatical ones, had formerly been "full," or lexical items with a concrete semantic content (1991:5). In the western linguistic tradition, this notion was first advanced by Étienne Bonnot de Condillac, although according to Heine, Claudi, and Hünnemeyer, Horne Tooke around the turn of the nineteenth century was the first to propose a formal theory that "language in its original state is 'concrete' and 'abstract' phenomena are derived from concrete ones," while "inflectional and derivational forms are... fragments of earlier independent words agglutinated to the root word" (5-6).

2.2.2. The Nineteenth Century

The nineteenth century saw the first great flowering of linguistic study in general and of the investigation of the Indo-European family in particular. Towards the end of the century, however, in 1891, a remarkably comprehensive formulation of patterns of language evolution was offered by Georg von der Gabelentz, most notably in his *Die Sprachwissenschaft: Ihre Aufgaben, Methoden, und bisherigen Ergebnisse* (1891). While earlier scholars had observed a tendency for abstract items with largely grammatical content to be derived from more pedestrian words, Gabelentz expanded on this understanding and applied it to the development of languages over great periods of time. P. Hopper and E. Traugott in fact credit Gabelentz with the insight that linguistic evolution is "not a

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linear process, but rather a cyclical one," in contrast with the outlook of many of the great nineteenth-century linguists such as Wilhelm von Humboldt, that "synthetic (inflectional) languages like the classical Indo-European languages represented an evolutionary endpoint" (20). Gabelentz on the other hand offers a more universal schema of the general course of language development: "Was erst neu und selten war, wird dann alltäglich und damit verliert es an Kraft, verblasst, rückt schliesslich wohl gar in die Reihe jener abstracten Bestandteile der Rede, die es hatte verbessernd und verstärkend ergänzen sollen, und die es am Ende wohl gar verdrängte und ersetzte" (238)[4].

Gabelentz in addition postulated the "driving forces" (Heine, Claudi, and Hünnemeyer 8) that set this cycle or spiral of linguistic evolution in motion. Speculating on the motivations for grammaticalization, the process by which "full" or concrete lexemes evolve into grammatical markers, whether as independent words, clitics, or affixes, he proposed that it "is a result of two competing tendencies, one tendency toward ease of articulation [resulting in 'Lautverschliffen'], the other toward distinctness ['Deutlichkeitstriebe']" (Hopper and Traugott 20; Gabelentz 240). In other words, distinctive grammatical morphemes tend to be eroded away over time through regular sound changes, particularly in rapid colloquial speech. As a result, the distinguishing features are gradually obscured, necessitating the replacement of these morphemes by other constructions, which undergo the same process. Gabelentz illustrates this

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[4] Curiously, Hopper and Traugott cite this discussion as occurring on pp. 241-2 of Gabelentz' work.

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process with a rather picturesque simile describing the development of such morphemes: "Es ist wie Staatsdienste: es wird angestellt, befördert, auf Wartegeld gesetzt, schliesslich wohl ganz pensionirt; und draussen vor den Pforten harrt eine Schaar Bewerber" (238).

2.2.2.1. Research on the Indo-European languages

At roughly the same time that Gabelentz was outlining his theory of general linguistic change, pioneering work specifically on the syntax specifically of the Indo-European languages was also being produced, most significantly by Berthold Delbrück and Jacob Wackernagel. Delbrück's great innovation was the application of the comparative method to syntactic studies, on the basis of which he attempted a reconstruction of Proto-Indo-European word order. His reconstruction draws heavily on the language of the Vedas, and thus postulates, "[I]n der Grundsprache... Das Verbum stand in dem unabhängigen Aussagensatz am Ende" (1900:83), while adpositions could be either preposed or postposed to their attendant phrases (1900:105, 108, 112). Wackernagel, in addition to his famous account of clitic order in Indo-European phrases, also made an initial attempt at explaining change in Indo-European word order, relying mainly on such phenomena as attraction or "assimilation" from one phrasal type to another (Lehmann 4).

Delbrück and Wackernagel both draw a distinction between "proper" and "improper" adpositions. The former are "those which may be used either as adpositions to nouns or as preverbs," such as Greek ἀπό, 'from', διά 'through', ἐν

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'in'; or Sanskrit *ánu* 'after', *ápi* 'upon', or *pári* 'around' (Holland 1980:151). The latter, on the other hand, "are used only with nouns," such as Greek ἕνεκα 'on account of' and χάριν 'for the sake of', or Sanskrit *parás* 'beyond' and *sahá* 'with' (Holland 151-2). This distinction represents the first attempt at describing degrees of adpositional functioning. Such segments are thus no longer regarded as a monolithic group; rather, some belong more purely to the set of adpositional words than others. The stage was also set for consideration of a relative chronology as to when these subsets developed into prepositions and postpositions, as well as, with the evident resemblances especially of the "improper" group to inflected forms, informed speculation on their origins and derivation.

2.2.3. The Early Twentieth Century - Meillet

Antoine Meillet succeeded in synthesizing Gabelentz' vision of repeated cycles of language evolution with the distinct groups of adposed Indo-European words observed by Delbrück and Wackernagel. In his groundbreaking 1912 work, "L'Évolution des formes grammaticales," Meillet argues that there are exactly two processes by which a language's morphological inventory develops. One is analogy, the shaping of a form "sur le modèle d'une autre" (1958:130), which often results in the regularization of an irregular paradigm. Beyond analogy, "L'autre procédé consiste dans le passage d'un mot autonome au rôle d'élément grammatical" (131). In contrast with the early treatments of words concrete in sense gradually growing more abstract, Meillet specifically declares

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that this gradual semantic bleaching is the source, the only source in fact, of new forms expressing grammatical content: "Tandis que l'analogie peut renouveler le détail des formes, mais laisse le plus souvent intact le plan d'ensemble du système existant, la «grammaticalisation» de certain mots crée des formes neuves, introduit des catégories qui n'avaient pas d'expression linguistique, transforme l'ensemble du système" (133). As such, the continual transformation of "autonomous" words into grammatical forms repeatedly renews and replenishes the grammatical and morphosyntactic infrastructure of a given language over the course of its history. The term "grammaticalization" is moreover Meillet's invention, and Heine, Claudi, and Hünnemeyer describe him as "the founder of modern grammaticalization studies" (8).

Meillet also proposed derivations for many of the grammatical words found among the Indo-European languages, such as the preverbs falling under the heading of "proper" adpositions suggested by Delbrück and Wackernagel. Meillet argues that these preverbs could be traced to inflected case-forms of Proto-Indo-European substantives. For example, from a root **perə* 'before', Meillet found locative forms, either in *-i* or from the old endingless locative: Skt. *pári*, Av. *pairi*, Gk. περί, Lat. *per*, Goth. *fair*, Lith. *per̃*, and OCS *prě*. From the genitive-ablative, Meillet drew Skt. *puráḥ*, Av. *paro*, and Gk. πάρος. And from a case in **-ŏ*, which Meillet thought possibly may have been an instrumental, but which thanks to the work of E. Laroche (1970) and of G. Dunkel (1994) we may confidently identify as the directive, there is Skt. *prá*, Av. *fra*, Gk. πρό, Lat. *prō*,

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Lith *pra-*, and OCS *pro* (1937:350). In short, Meillet proposed that Proto-Indo-European nominal paradigms underlay many groups of adpositions known from the attested languages. The connection between former inflected noun forms and the adpositions that clarify grammatical relations was thus made explicit by Meillet.

2.2.4. The Later Twentieth Century

2.2.4.1. Greenberg's "Language Universals"

A great deal of the work that established linguistics as a field of study was driven by investigations of the Indo-European languages. However, by the mid-twentieth century, there arose significant interest in qualities shared by languages in general, in universal characteristics held in common among language families throughout the world. Greenberg's 1966 treatment of this topic sparked much active discussion of the universals of grammar and word order, and indeed, it continues to do so. Of particular relevance to the present effort, Greenberg's sample of thirty languages scattered across the globe indicated a correlation parallel to Delbrück's conclusion that Proto-Indo-European had a basic sentential word order of the subject at the beginning of the sentence, with the "Akkusativ" or other predicate appearing before the verb, which generally came sentence finally, and with postpositions commonly occurring (see for example 1900:82-3):

Universal 4: With overwhelmingly greater than chance frequency,

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languages with normal SOV order are postpositional (62).

At the other end of the sentence-order spectrum, many languages, including Old Irish and most other attested Celtic tongues, normally place the verb at the beginning of the sentence, correlating with a preposed adpositional alignment.

> Universal 3: Languages with dominant VSO order are always prepositional (62).

Prepositions also tend to be associated with an SVO order, albeit less rigidly so. To summarize Greenberg's findings on this point:

	Type I (VSO)	Type II (SVO)	Type III (SOV)
Postpositional	0	3	11
Prepositional	6	10	0

(From Greenberg 61)

Greenberg also identified several other syntactic formations more likely to be found in one word order type than another. For example, genitives and adjectives are more likely to precede the noun they modify in OV languages and to follow them where a VO pattern is more prevalent (61-2). Similarly, in constructions of comparison, an order of adjective-marker-standard ('bigger than that one') generally occurs in Type I or Type II word-order patterns, while a

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Type III language will usually feature a standard-marker-adjective formation (69-70).

Greenberg's proposed universals have generated a considerable degree of controversy. Most often, they have been criticized for their unreliability as a predictive device, and for an over-reliance upon them on the part of many linguists in analyzing the syntax of particular languages (see for example C. Watkins 1976:305; W. Schmalstieg 1988:326-7). Nevertheless, they offer many useful rules of thumb as long as one does not expect every language to conform with them in every detail (and indeed, Greenberg readily acknowledges that he has uncovered general tendencies rather than hard and fast rules). The value of these universals to comparative Indo-European syntax is that they serve to underscore the broad disparity of word order types among the related languages and consequently the need to explain such a state of affairs.

2.2.4.2. Research on Word Order Change

One outgrowth of Greenberg's investigation of word order universals was a renewed interest in questions of word order change. This resurgence in curiosity also sparked a recollection of E. Sapir's musings on "linguistic drift," defined as "the unconscious selection on the part of [a language's] speakers of those individual variations that are cumulative in some special direction" (1921:155). Sapir had identified three such tendencies among the Indo-European languages: a levelling of the morphological distinction between subject and object; a tendency to fixed position in a sentence; and a drift toward the

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"invariable word" (163-8). Languages also exist, on the other hand, where drift appears to be operating in the opposite direction, as Li and Thompson for example argue is the case in Mandarin Chinese (1974); linguistic drift can also carry a language towards a richer morphology and a greater flexibility in word order.

Meanwhile, T. Givón undertook "an attempt to explain the genesis of affixal morphology, both derivational and inflectional" (1971:394). His conclusion was that such affixes originate from free segments that gradually fossilize in positions relative to a head word; become clitics and therefore less free; and finally, bound morphemes affixed to their former heads. This finding was, in large measure, anticipated by Meillet's earlier observations on grammaticalization. However, Givón's slogan that "Today's morphology is yesterday's syntax" (413) serves to reinforce the proposition that in grammatical markers may be found remnants of formerly productive lexical items.

Shortly thereafter T. Vennemann produced a series of articles addressing the specific topic of word order change. His argument emphasizes the relationship between morphological case marking and word order noted by Greenberg[5]. Vennemann proposed that a rigid verb-medial word order arises when an absence of morphological subject-object markers necessitates some other means of distinguishing the two (1974a:220; see also 1974b:354; 1975:284). On the other hand, this resulting rigidity "favors the positional reduction of full

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[5] Universal 41: If in a language the verb follows both the nominal subject and the nominal object as the dominant word order, the language almost always has a case system (75).

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words to a new morphology" (1973:359) arising from the fossilization of such words outlined by Meillet and Givón. This event in turn eventually prompts a return to case-marking and an OV order, albeit with greater flexibility (1973:359-60; 1975:292). This schema, perhaps overgeneralized, nevertheless represents an attempt to identify regular causes of syntactic change and therefore to gain the ability to predict it. This is a worthy if highly elusive goal in studies of diachronic grammar, and the present investigation will proceed with an eye toward this end.

2.2.4.3. Indo-European Historical Grammar

Further accomplishments in the area of historical grammar during the nineteen seventies include two book-length treatments of Proto-Indo-European syntax, W. Lehmann 1974 and P. Friedrich 1975. Both of these works represent attempted reconstructions of the proto-language, without giving much attention to explaining the changes that will have taken place between the reconstructed proto-syntax and the syntactic formations of the attested languages (Lehmann's brief chapter on the subject - 238-251 - observes the differences but does not hazard many explanations for them). Lehmann, drawing primarily on Vedic Sanskrit, postulates an OV proto-language, with the expected attendant features including postpositions (31-2, 212, 234). Friedrich, writing in part in response to Lehmann, argues against an over-reliance on any one or two branches of the Indo-European family (5-6), but where the various branches contradict one another, as they often do, he tends to gravitate to Homeric Greek for a

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resolution (as noted by Watkins 310). As such, he envisions the proto-language as having had a basic SVO order (31-4), with a tendency, although not a rigid one, towards prepositions (37).

Watkins, however, in criticizing both Lehmann and Friedrich, declares, "[T]he confirmation by Hittite of virtually every syntactic assertion about Indo-European word order patters made by Berthold Delbrück [is] in itself as dramatic as the surfacing of the laryngeals in that language" (306). This would once again imply a verb-final, postpositional sentential formation. Nevertheless, it then remains to account for the reverse to varying degrees in Greek as well as Celtic, Germanic, Slavic, and several of the other descendant branches of this source language.

Nevertheless, a valuable contribution by Friedrich as far as the present endeavor is concerned arises from the rather ambiguous status of many Homeric adpositional items as to whether they should be considered prepositions, postpositions, preverbs, or adverbs (as, of course, in Hittite and other Indo-European languages as well). As a consequence he proposes the umbrella term "locative auxiliary" (35) after V. Ivanov 1973:60, thereby introducing the concept to English-language scholarship.

One other work of note from this time and on this subject matter is J. Kahr's (1976) treatment on the renewal of case morphology drawing on adpositions, generally postposed. Among the members of the Indo-European family she considers are Oscan and Umbrian, Lithuanian, and some Iranian languages, particularly Ossetic. The proposition included in her thesis that such

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renewal is an ongoing and constantly-repeating progress in many if not all languages most definitely informs the present undertaking as well.

2.2.4.4. Holland on Word Order in Homeric Greek

A study undertaking to account for changes in word order reflected in the earliest-attested Indo-European languages is that of Holland 1980. Similarly to P. Friedrich, Holland finds data from Homeric Greek especially instructive. His analysis of the adpositional situation in the Homeric epics is quite valuable, given the previously-mentioned irregularities in that language's syntax. While Classical Greek is for all intents and purposes a prepositional language, Homer admits of a small but significant number of exceptions. As Holland reports after A. Haggert 1902:

> [O]f 8198 examples [of "proper prepositions" in the Homeric texts] some 255 were postpositions governing single nouns (Np)... about 3.1 percent of the total. With phrases consisting of an adjective and a noun, some 390 (4.7 percent) have the adposition between the noun and its modifier (NpA), while 600 (7.3 percent) have the adposition between the adjective and the noun (ApN). The remainder are prepositional and have the order pN, pAN, or pNA, etc... If the figures for all the non-preposed examples are combined, the total is 1245 (somewhat less than 15.2 percent) (Holland 158).

The absence of any sequences in NAp or ANp in so sizable a sample suggests to Holland that such were not allowable, and therefore the placement of these adpositions is not entirely free (160-1). In contrast, there is no such

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restriction in the Vedas (163). Holland notes, however, that Vedic adpositions are regularly accented differently from Greek prepositions; Vedic disyllabic forms are accented on the initial syllable (*ápa, ápi, pári, úpa*), while their Greek counterparts are accented on the final segment (ἀπό, ἐπί, περί, ὑπό) (159). There is, however, an exception: Accent is word-initial when the adposition is postposed to its nominal complement:

γαιὴς ἄπο πατρίδος	(Il.13.696)	'from his paternal land'
θεῶν ἄπο	(Od.6.12)	'from the gods
ἣν πέρι	(Il.15.142)	'about her'
χερσὶν ὕπο Τρώων	(Il.11.827)	'at the hands of the Trojans' (158-9)

Holland takes this accentual correspondence with Sanskrit as evidence that the postposing construction was the older one (160), with the accentual shift taking place as a consequence of the shift in positioning (169). The postpositional phrases, then, are the residue of an intermediate step in the shift to prepositions: "I would like to suggest that Adjective-postposition-Noun and Noun-postposition-Adjective sequences are... transitional between typologically more consistent ANp or pNA sequences. In [such] constructions, lengthy objects are broken up around the governing word order to preserve minimal OV and Np structure" (162). While Holland is not prepared to identify a specific driving force in this change in word order, it seems likely that the need to resolve cumbersome postpositional objects was a significant factor in this process.

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A point on which Holland is quite clear, however, is that "[T]he shift from object-verb to verb-object order cannot have caused the shift from postposition to preposition in Greek. According to the figures I have collected for six books of the Iliad and Odyssey, the ratio of OV to VO where there are full noun objects is about 2:1... As against the ratio of OV to VO, the ratio of proper preposition to postposition is about 13:1" (161). This overwhelming preponderance leads Holland to conclude, "[T]he shift from postposition to preposition has proceeded farther than the shift from OV to VO" (161). More explicitly, he declares, "One may establish a relative chronology for these types of changes in word order. The shift from postposition to preposition is one of the earliest changes. It is virtually complete when the shift from object-verb to verb-object is only in its initial stages" (187). This hypothesis, as mentioned, is echoed for another language sub-family by Ambrazas 1982. It must therefore follow that this shift is a major step in a general syntactic shift, not only for Greek but throughout the Indo-European languages, and therefore merits particularly close study.

2.2.4.5. Grammaticalization

A great deal of research in just the past decade has been conducted on grammaticalization, although broader notions of the concept have been under discussion since Meillet's introduction of the term. Heine, Claudi, and Hünnemeyer's examination of the subject is noteworthy for its description of the semantic changes undergone by a lexeme as its connotations grow increasingly abstract. Hopper and Traugott, in their extensive treatment of the subject, offer

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a description of the trajectories followed by morphemes in becoming grammaticalized. Following C. Lehmann (1985:304), they propose as a path along which adpositions commonly develop before possibly becoming further grammaticalized as nominal affixes:

> relational noun >
> secondary adposition >
> primary adposition >
> agglutinative case affix >
> fusional case affix (Hopper and Traugott 107).

Examples of an entire such path, or cline (Hopper and Traugott 105), are difficult to come by as the complete process takes far more time than the attested history of most languages. However, one fairly complete case presents itself over the long and well-documented history of the Iranian languages. In Old Persian, the noun *radiy* 'goal, purpose' takes on a relational sense ('by way of, by means of') and may function as a secondary postposition, in that it defines "concrete rather than grammatical relationship" (Hopper and Traugott 107), meaning 'on account of, by'[6]. By the time of Middle Persian, not directly descended from Old Persian but from the closely-related Median, the form has been reduced to *rāy* (or *rād* in Parthian). It serves as a postposition with a dative or benefactive sense, although it may also indicate cause or purpose (C.

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6 This postposition is also borrowed into the Slavic languages, appearing in both Old Church Slavic and Russian as *radi*. The presence of this item in Slavic will be discussed in greater detail in Chapter V, Section 7.3.

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Brunner 1971:168-9). As Hopper and Traugott note, such adpositions "are easily cliticized" (107), and such clisis is often a prelude to affixation. In Classical Persian, *-ra* is a dative suffix and also occurs with definite direct objects. By Modern Persian, also a Median descendant, this *-ra* may even mark indefinite direct objects (Hopper and Traugott 158-9; Kahr 1976:132). As such, modern *-ra* may not be directly derived from Old Persian *radiy*, but nevertheless a fairly complete picture of the changes occurring in grammaticalization emerges. One may thus observe that, if a lexeme becomes grammaticalized over time, its sense will become less semantic and more grammatical, it will become increasingly unlikely to appear as a separate and distinct word, and due to increasing frequency of use, it will likely undergo even greater phonological attrition than other segments of a given language.

3.0. Assumptions and Method

The mechanics and driving forces behind this change in the word order of adpositional phrases among the various daughter languages of Proto-Indo-European are indeed of greater interest to the present study than the question of the ordering of such phrases in the proto-language itself. The preponderance of evidence from the most archaic known Indo-European languages - Hittite, Luwian, Vedic Sanskrit, Gathic Avestan, as well as Tocharian - certainly suggests a postpositional structure. It is true that Mycenaean Greek, nearly as ancient as the Anatolian languages mentioned and committed to written documentation

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earlier than the Indo-Iranian pair, is unambiguously prepositional (see for example Holland 206). Still, it is widely acknowledged that Mycenaean syntax is in many respects rather less archaic than is the syntax of Homer (Holland 159). And, as discussed above, the Homeric texts do present some evidence of an earlier postpositional stage.

But even if the evidence of these languages were somehow to prove misleading, and Proto-Indo-European did in fact have a predominantly prepositional structure in such phrases, the modern investigator is still faced with the same question: How did the change in word order in the adpositional phrases in some subset of the Indo-European languages unfold? If the proto-language was postpositional, then a shift has taken place in the Hellenic, Armenian, Italic, Germanic, Celtic, Albanian, and Slavic branches, and if it was prepositional, then the Anatolian, Indo-Iranian, and Tocharian branches will have undergone a change. The present investigation, then, will seek a greater understanding of the steps that marked this process of word order change, as well as, insofar as is possible to ascertain, the conditions that set it in motion. Possible implications for linguistic and especially syntactic universals will also be eagerly sought after.

Also of interest to this study of the development of adpositions and their phrases is their origin, their relationship to preverbs in the syntax of the various languages, and the evolution of their usage and forms beyond their change in word order. This undertaking will consider the extent to which Meillet's observations comparing the various forms of Indo-European locative auxiliaries

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to the permutations of nominal paradigms can be supported and expanded upon. Particularly striking in this regard are the highly nominal characteristics of many postpositions in Hittite and Luwian. The similarities and differences in adpositional and preverbal function of these auxiliaries, as well as the extent and significance of the overlap in these functions, will also be considered, along with the question of the chronological precedence and evolution of these roles. Moreover, in some of the branches of the Indo-European family attracting close examination, such as Italic and Baltic, adpositions are taking on the role and appearance of case affixes. In any event, just as word order is not static, in adpositional phrases and elsewhere, neither is phonological shape or grammatical function. As such, the fate of Indo-European adpositions in these respects will also be discussed.

For the present exercise, concentration will be focused upon those languages offering particular insight into the origins of these items, above all Hittite (Chapter II), or presenting substantial evidence of an adpositional word order shift in progress. These include Luwian and Lycian (Chapter III), Umbrian and other Italic languages (Chapter IV), and Old Lithuanian and the rest of the Baltic subgroup (Chapter V). As for the adpositional systems of the Tocharian, Indo-Iranian, Celtic, Germanic, Armenian, and Albanian branches, these will be briefly addressed in Chapter VI, while the situation among the Slavic languages will be discussed in connection with Baltic in Chapter V. Moreover, it is doubtful that the present study can add significantly to Holland's excellent analysis of the adpositional situation in Greek.

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The over-arching goal of the present effort, then, is to gain a firmer grasp on the divergent evolutions of Indo-European adpositional segments and the forces behind these various courses of development. An analysis of such phrases in some of the earliest attested texts from Hittite and the Southern Anatolian, Italic, and Baltic languages, coupled with a survey of the relevant scholarship, should make possible conclusions on the nature of these subjects. Moreover, possible implications for the understanding of universals of word order and language change will be considered where the opportunity arises. Thus will a crucial yet all too often overlooked area of comparative Indo-European grammar be pursued.

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Chapter II:
Postpositions and Locative Auxiliaries from Old to Middle Hittite[1]

1.0. Introduction

A complicating factor in any discussion of adpositional usages in Hittite is that, similarly to many Indo-European languages, essentially all of the more prominent postpositions in the language also serve as adverbs and preverbs. Indeed, the demarcation between postposition and preverb has proven a rather vexing problem, so much so that P. Friedrich has proposed to classify both of these throughout the period ending with the earliest-attested Indo-European languages under the broader heading of "locative auxiliaries" (1975:35). On the specific topic of Hittite, F. Starke declares "Da es... keine stichhaltigen Argumente gibt, die für das Vorhandensein von Postpositionen und Präverbien im Hethititschen sprechen, gehe ich davon aus, daß nur eine Kategorie 'Adverb' besteht" (1977:130-31). Nevertheless, the distinction between postposition and preverb, as blurry as it can be, is worth attempting to preserve. There are differences in syntactic function, at least between clear postpositions and clear preverbs; the extraordinarily uncertain boundary between the two sets, as well as, in Old Hittite, that between postposition and noun, does not warrant ignoring these differences altogether. In the first place, as will be elaborated, many of

[1] Portions of this chapter were presented at the 2000 Berkeley Germanic Linguistics Roundtable, University of California, Berkeley and at the Nineteenth East-Coast Indo-European Conference, University of Georgia, in April and June, 2000, respectively. A version of the Berkeley presentation will appear in a forthcoming issue of the *Interdisciplinary Journal for Germanic Linguistics and Semiotic Analysis.*

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these segments appear far more often as preverbs, while others are manifestly more disposed to occur as postpositions. For the exceptions, the instances where the category designation is less clear, the distinctions among degrees of belonging to a given class of lexeme offered by Fuzzy Set or Prototype Theory will prove helpful.

1.1. Previous Scholarship

1.1.1. L. Zuntz

The first attempt to identify the circumstances under which these Hittite locative auxiliaries serve as preverbs, postpositions, or adverbs was that of Zuntz 1936. Her hypothesis was based entirely on the position of the item within the clause. Specifically, the auxiliary is a postposition "wo es hinter einem Nomen mit „regiertem" Cas. obl. steht", and a preverb "wo es vor einer Verbalform steht" (9). In any other place, according to Zuntz, it is an adverb, including at the beginning of the sentence or where it carries suffixed enclitic possessive pronouns (9). Particularly problematic to any such classification, as Zuntz recognized, are those constructions where an auxiliary occurs between a noun in a governed oblique case and a verb, which Zuntz terms "Mittelstellung" (10):

1) LUGAL-*uš* SAL.LUGAL-*ša ḫal-ma-aš-šu-it-ta-az pa-ra-a ú-en-zi.*
'The king and queen come out from the throne.'
(*Ein althethitisches Gewitterritual*, E. Neu 1970 Vs. I. 13)

In such cases, Zuntz attempts to draw on other sentences where the same

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auxiliary is employed in similar environments. For example, since *arḫa* 'away', like *parā* 'forth, from', may also be found directly before the verb without such inflected nouns in the vicinity, in a context such as in sentence 1), they should be considered preverbs. On the other hand, since *píran* 'from' alternates in some texts with the Akkadian preposition *PANI*, it should in potentially ambiguous environments be regarded as a postposition (10).

1.1.2. A. Goetze

Subsequently, Goetze 1963 lists five syntactic environments in which the items he terms "local adverbs" function as preverbs or postpositions, as opposed to simple adverbs:

(a) ... directional case [e.g., dative/locative or, in Old Hittite, genitive] - local adverb
(b) ... -local adverb-verb
(c) ... directional case - local adverb - verb
(d) local adverb ... - verb
(e) local adverb - directional case - verb. (98)

In this schema, (a) describes postpositional usages for the locative auxiliaries, while (b), (d), and (e), are most readily identifiable as preverbs. Goetze thus, in contrast with Zuntz, includes local adverbs in sentence-initial position under the heading of preverb as well: "Tmesis is of frequent occurrence in Hittite. The Old Latin *ob vōs sacrō, sub vōs plācō* has numerous analogues in

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that ancient language. Any compound verb may be split up so that the preverb begins the clause and the verb closes it" (100).

One is then left with the conclusion that any local adverb not occurring in any of the five environments outlined above must be considered neither a preverb nor a postposition, but simply an adverb. Goetze summarizes by observing, "Looking for the common denominator under which all these usages may be subsumed we might say: the local adverb is attached, either enclitically or proclitically, to that element of the sentence which is to be emphasized" (101). Auxiliaries in contexts other than these five will be found to emphasize or modify another item in the sentence, such as another adverbial element, or the clause as a whole rather than the verb in particular; as such these words are therefore pure adverbs.

It nevertheless remains to address Goetze's troublesome environment (c). Here, like Zuntz, Goetze refers to similar contexts, namely, "examples which insert a subject or an object in the decisive spot and thereby separate the local adverb either from the directional case or from the compound verb (99)":

2) *Ki-ma-kan tuppi A-NA* ^{d}UTU-*SI issaz para Hattilis aniyat*
"'H. took down this tablet from my majesty's mouth' (99)"

3) *itten-wa-kan IS-TU* É.GAL-*LIM kallar uddar para suwatten*[2]
"'go ye (and) chase out of the palace the spook' (99)"

For these usages of *parā*, Goetze observes that such instances as the first, "here

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[2] The transcription in both 2) and 3) is Goetze's.

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and presumably in all similar occurrences, are postpositions" with no evidence for compound verbs as would result from a verb and preverb, while the second in fact exemplifies just such a compound (99).

1.1.3. E. Laroche

Even as early as Zuntz' work, the impression was discernible that certain of the auxiliaries, specifically those ending in *-a*, were more likely to serve as preverbs, and that those in *-an* showed a greater tendency toward postpositional roles. The significance of these different endings became clearer with subsequent efforts. Laroche 1970, for example, first made the connection between the pattern reflected in these forms and the remnants of a declensional paradigm. He found in the items in *-a* the reflection of a specific case, the Old Hittite directive, and suggested that those in *-an*, along with *šer*, are derived from a "Locatif absolu", or endlingless locative case form (43-4). He also proposed a difference in function between the latter and the former set: "Opposition du statique au dynamique, de la position au mouvement; en grammaire: du locatif et des cas 'motionnels'" (44). Laroche further underscored the nominal origins of these items by citing their ability to combine with enclitic possessive pronouns (40,44), in contradiction with Zuntz' earlier classification of such phenomena as adverbial. Neu 1977 made this point even more explicit: "Als ehemalige Substantive können *peran, appan* oder *kattan*, wie jedes andere hethitische Substantiv auch, enklitische Possessivpronomina anhängen. Eine Form *peran-*

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šet unterscheidet sich dann strukturell in nichts von einer form wie *utne-šet* „sein Land"" (69).

1.1.4. Starke

Starke 1977 carried the implications of this similarity in shape to nominal case forms still further, noting that even in function these local adverbs, at least in Old Hittite, show nominal characteristics. As he observed of the forms in *-a:*

> Die ah. Adverbien sind nämlich Substantive mit starren Kasusmorphemen und werden auch als solche empfunden, weil sie weder formal noch syntaktisch... von Terminativen wie *parna* „ins Haus", *nepiša* „zum Himmel" etc. unterscheiden. Es wäre also eigentlich wörtlich zu übersetzen: *anda* „ins Innere", *appa* „zur Rück-/Hinter(seite)", *katta* „zur Unter(seite)", *parā* „zur Vorder(seite)", *šara* „zur Ober(seite)". (149)

As such then, Starke then lists the two sets of local adverbs as follows:

Gruppe I	*anda*	Gruppe II	*andan*
	appa		*appan*
	katta		*kattan*
	parā		*píran*
	šarā		*šēr* (133)

In another sense, however, Starke's work represents a departure from

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previous scholarship on the locative auxiliaries. Noting the difficulty faced by Zuntz, Goetze, and A. Kammenuber 1973 in attempting to delineate a precise distinction among postposition, preverb, and adverb based on the placement, he concludes that such analyses can only yield "Pseudokategorie[n]" (174), and that "Da es somit keine sichtighaltigen Argument gibt, die für das Vorhandensein von Postpositionen und Präverbien im Hethitischen sprechen, gehe ich davon aus, daß nur eine Kategorie „Adverb" besteht" (130-1). This of course results in a rather different understanding of these items. However, the disavowal of word order as a criterion for their classification has value according to F. Tjerkstra 1999, who credits Starke with offering "a turning point in the description of the function of the local adverbs, since it is the first to define the local adverbs in terms of the relation to the verb (6)."

1.1.5. Tjerkstra

Tjerkstra expands on the importance of the relationship between locative auxiliary and verb, although she finds a single blanket category as proposed by Starke insufficient. The crucial feature in her scheme is dependence on the verb: "Dependent local adverbs influence the meaning or the construction of the predicate and form a unity with it. Independent local adverbs have no relation with the predicate and may be omitted, leaving the predicate in the same sense and the remainder of the sentence grammatically complete" (131). These independent usages correspond with the traditional designation of postposition: "Independent local adverbs do not belong to the main construction but can

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govern a Complement in the dative-locative [in Middle and Later Hittite] on noun phrase level. These combinations of a local adverb and a Complement can be described as postpositional phrases" (138). One may further presume that independent local adverbs not taking a Complement may be considered strictly as adverbs. Nevertheless, Tjerkstra continues Starke's rejection of earlier syntactically based analysis of these forms:

> The unity of the [dependent] local adverb is of a semantic, not of a syntactic nature, since the local adverb and the verb can be separated from one another without effect on meaning. The traditional term preverb may therefore be a less appropriate description of the function of the local adverb in this connection. If one would need a separate designation, the best choice would probably be something like "converb" (136).

1.1.6. D. Salisbury: Conclusion

Still, the existence of a syntactic component to the functioning of the dependent local adverb cannot be completely denied. Salisbury (forthcoming) in her survey of the *anda/andan* dyad in Later Hittite finds that the placement of the preverb is limited to some specific points within the clause: "Syntactically, preverbs can only occur in certain positions: (1) directly before the verb, (2) in sentence-initial position (either absolute or after a conjunction-particle), or (3) separated from the verb by a single element of the predicate (a negative, direct object, or an allative/locative)" (4). Even if the word order of Later Hittite is crystallized somewhat from earlier stages of the language, the location of

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auxiliaries modifying or in nexus with the verb is never entirely random. The reliance of syntactic factors in differentiating between dependent and independent forms, or between preverbs on the one hand and postpositions and adverbs on the other, while not definitive in all cases, nevertheless remains a useful guide the great majority of the time. In the remaining instances, recourse to such questions as semantic relationships and similar contexts within the corpus of texts may be necessary, but, as will be discussed, even then a clear resolution may not always be possible.

2.0. Old Hittite Primary Texts Consulted

Unquestionably, then, the development of adpositions in Hittite cannot be considered independently of the usages of preverbs and adverbs. Accordingly, a major focus for the Anatolian chapter of the present undertaking will be the differences in function and frequency between postpositions and preverbs in Old Hittite, or more precisely, how the same lexical items serve both functions. And then, in the later portion of this effort, how these relations change in the Middle stage of the language will be examined. The initial exercise for this comparison of Hittite postpositions and preverbs is a tabulation of the items in several Old Hittite texts. The texts considered here are given, in estimated chronological order: *Der Anitta-Text* (Neu 1974), *Eine althethitische Erzählung um die Stadt Zalpa* (H. Otten 1973), *Die hethitisch-akkadische Bilingue des Hattusili I* (F. Sommer and A. Falkenstein 1974), *Ein althethitisches Gewitterritual* (Neu 1970), *Ein althethititisches Ritual für das Königspaar* (Otten and V. Souček 1969), the text of Hittite Laws

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(with editions both by J. Friedrich - *Die Hethitischen Gesetze,* 1959; and Harry A. Hoffner - *The Laws of the Hittites,* 1997) and *Der Erlaß Telipinus* (I. Hoffmann 1984).

3.0. Frequency of the Locative Auxiliaries in the Primary Texts

The distribution of items appearing as Friedrich's "locative auxiliaries" is given below, with tentative numbers of occurrences for each lexeme in the various roles as follows:

		An.	*Zal.*	*B.H.*	*Gwr.*	*Kp.*	*H.L.*	*Tel.*	*Total*
anda:	*preverb*	2	-	-	4	17	12	4	39
	postposition	-	-	1	-	-	2	-	3
	adverb	-	-	-	-	1	1	-	2
andan:	*preverb*	-	-	-	-	-	1	-	1
	postposition	-	-	-	-	-	1	-	1
	adverb	-	-	-	-	-	2	-	2
appa:	*preverb*	6	3	6	15	3	24	6	63
	postposition	1	-	-	-	-	-	-	1
	adverb	-	-	-	-	-	-	-	-
appan:	*preverb*	-	-	2	1	-	1	2	7
	postposition	5	-	1	1	-	-	2	9
	adverb	-	-	-	-	-	1	2	3
appanda:	*preverb*	-	-	-	-	-	1	-	1
	postposition	-	-	-	-	-	-	1	1
	adverb	-	-	-	-	1	1	-	2
arḫa:	*preverb*	1	1	-	-	-	4	5	11
	postposition	-	-	-	-	-	-	-	-
	adverb	-	-	-	-	-	-	-	-

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		An.	Zal.	B.H.	Gwr.	Kp.	H.L.	Tel.	Total
kaššaš:	*preverb*	-	-	-	-	-	-	-	-
	postposition	3	-	-	-	-	3	-	3
	adverb	-	-	-	-	-	-	-	-
katta:	*preverb*	1	4	4	4	-	10	-	23
	postposition	-	-	-	-	-	-	-	-
	adverb	-	1	-	-	-	-	-	1
kattan:	*preverb*	-	1	-	-	-	2	-	3
	postposition	-	-	-	-	-	-	1	1
	adverb	-	-	-	-	1	-	-	1
katte:	*preverb*	-	-	-	-	-	-	-	-
	postposition	2	6	-	-	1	-	-	9
	adverb	-	-	-	-	-	-	-	-
para:	*preverb*	1	-	4	21	10	5	4	45
	postposition	-	-	-	-	-	-	-	-
	adverb	-	-	-	-	-	1	-	1
píran:	*preverb*	-	-	1	-	-	3	-	4
	postposition	1	1	2	29	2	-	-	35
	adverb	-	-	-	-	-	-	-	-
pie:	*preverb*	-	-	-	-	-	-	1	1
	postposition	-	-	-	-	-	-	-	-
	adverb	-	-	-	-	-	-	-	-
šara:	*preverb*	-	1	5	6	5	2	1	20
	postposition	-	-	-	-	-	-	-	-
	adverb	-	-	-	-	-	-	-	-
šer:	*preverb*	-	-	-	-	-	1	-	1
	postposition	-	-	3	3	4	1	2	13
	adverb	-	-	-	2	4	1	-	7

3.1. Notes on the Figures

The present accounting has been content for the most part to follow the

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classifications of these words as preverb, postposition, and adverb offered by the editors of the texts examined where they are available; for example, Neu 1970 and Sommer and Falkenstein scrupulously itemize the occurrences of these lexemes according to whether they regard them as preverb, postposition, or adverb, while Otten and Souček as well as Neu 1974 simply list the appearances of the word without any more specific labeling. In most instances, this does not present a problem for classification. Preverbs in the two texts nearly always appear in Goetze's environments b) and d), immediately before the preverb or at the beginning of the clause, as set forth in Section 1.1.2. The exceptions are found in Salisbury's environment 3), with one intervening constituent between the auxiliary and the verb (Section 1.1.6). Postpositions, too, are clearly identifiable, with preceding genitives or suffixed enclitic pronouns, and they generally avoid potentially ambiguous contexts; one may speculate indeed as to a conscious design one the part of the scribes.

However, some ambiguous settings do occur in these texts, three in all. Two are in the *Ritual für das Königspaar*, within a few lines of one another, and both involve Salisbury's environment 3):

4) *ú-ga-aš-ma-aš-ša-an* ERÍN^{MEŠ}*-an še-e-er* III-*ŠU ya-aḫ-nu-ú-mi*
'and I turn the troops above them' (*Kp.* Vs. II. 31-2)

5) ERÍN^{MEŠ}*-ti-ma-aš-ša-an še-e-er* GÍR ZABAR *ki-it-ta*
'above the troops lies a bronze dagger' (*Kp.* Vs. II. 33-4)

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A postpositional usage for *šēr* in these sentences may immediately be ruled out as there is no complement in either the genitive or dative-locative case. However, this is an environment where preverbs commonly occur, and they may therefore reasonably be accepted as serving this function here. More complicated is one of the first few sentences in the *Anitta-Text*, where *katta* could arguably fall into any one of the three categories:

6) LUGAL ^{URU}*Ku-uš-ša-ra* URU-*az kat-ta pa-an-ga-ri-it ú]-*
'the king of Kussara [came] down from the city with great might'
(*An.* Vs. 5)

Once again, the auxiliary is in Salisbury's environment 3). Here, however, it immediately follows a noun in the ablative, which, as will be illustrated in sentence 12), is a possible adpositional complement where a sense of 'motion from' is present. But in this example, that sense is already made clear not only by the use of the ablative but by the directional prefix *ú-* on the verb. Moreover, *katta* also does not appear to color the meaning of the verb, assuming it is indeed *ú[it]*, so this, too, would appear to be an adverb.

Beyond such questions, these figures do not include items in the *Hittite Laws* attested only in textual variants. Friedrich takes the most complete tablets, listed as KBo VI 2, as canonical, and this, according to Hoffner, may also most safely be dated to the Old Hittite era (230), but some smaller fragments, most regarded due to orthographic practices as Later Hittite copies, have also been

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unearthed with variant readings. Even though these are copies of this Old Hittite text, the late date of their writing makes their inclusion in the above figures inappropriate[3]. Finally, instances where an enclitic possessive pronoun is suffixed to an item are counted as postpositions; all nine occurrences of *katte* are such cases, as are four of *píran* and six of *šer*.

4.0. The Two Sets of Locative Auxiliaries

The most striking result arising from this tabulation is a broad tendency for preverbial and postpositional functions to correlate with opposing forms reflected in the doublets appearing in these roles, as listed by Starke and set forth in Section 1.1.4. Starke's Group I, appearing overwhelmingly more often as preverbs, may be associated quite strongly with the Old Hittite directive case in *-a* (Laroche 1970:37-44; Starke 133), while Group II shows a similar correlation with the postpositional function. Such a consistent pattern of alternation strongly calls to mind the workings of an inflectional paradigm, as noted in particular by Laroche and Starke.

4.1. Their Derivations

Given the frequency, often noted by scholars on language change, with which adpositions are ultimately derived from nouns (see for example P. Hopper

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[3] Two variant readings of *kattan* for *katta* in these texts will be discussed below, in Section 11.2.

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and E. Traugott 1993:106), it comes as no surprise that the first group is commonly associated with the Old Hittite directive case in *-a* (Laroche 1970:37-44; Starke 133). The forms of the second group are also widely considered to be of nominal origin, but the question of the particular case reflected therein is considerably more controversial. At first glance, four of the five basically postpositional lexemes listed above appear quite readily to reflect the Hittite accusative singular in *-an*. J. Puhvel indeed derives the third of these items from "a root-noun **kat-* of which *kattan, katta,* and *katte-* would be petrifact accusative and dative-locative (including directive)" (1997:130; see also J. Tischler 1983:539). Starke draws on the ability of these postpositions to take enclitic possessive pronouns to form a similar conclusion as to their original case, noting that, as the enclitics are nominative/accusative neuters, the auxiliaries are presumably considered such as well (133). Of course the form may strictly speaking be termed nominative-accusative neuter singular, but their functioning in these contexts cannot in any sense be considered nominative. Sommer and Falkenstein also label *píran* accusative (1937:178).

On the other hand, the semantic senses of these postpositions are not easily reconciled with a derivation from the nominative/accusative case from. Such a difficulty is not encountered among the first group with their preverbial functions; compound verbs like *para úit* 'he goes forth, he goes to the fore' or *appa úit* 'he goes back, he goes to the back' clearly reflect the directive usages of

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para and *appa*. However, postpositional phrases such as *ammel appan* 'after me' or *šer-šmit* 'above them' reflect a directional orientation more appropriate to the locative case than to the nominative/accusative. Laroche exemplifies the distinction between the two sets as follows: "les adverbes hittites *peran* et *para*, qui signifient tous deux 'avant, devant', appartiennent le premier au système positionnel, le second au système motionnel: *peran* marque une position antérieure, dans l'espace et le temps, *para* marque un mouvement vers l'avant" (39). J. Boley, too, terming both preverbs and postpositions as Place Words, observes that they "have a *motion form* (in *-a*) and a corresponding *stative form* (generally in *-an*, but there are exceptions, the chief being *šer* corresponding to the motion *šara*)" (1994:130).

Adverbial uses of the accusative case among the Indo-European languages normally carry a sense of motion toward, across, or through an object. The usages of the Old Hittite postpositions to indicate position or state have driven some scholars to find an origin in the locative case of relational nouns for these forms. Laroche traces *píran*, as well as *šer*, to a "locatif absolu", or endingless locative (39, 40); *kattan*, *appan*, and *andan* may then be thought to have followed the same course. J. L. Garcia-Ramón offers a slightly different derivation, positing a compound of an endingless locative **pr̥* + **en*, familiar as

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the adposition represented, for example, in Latin *in* or Greek ἐν, and otherwise attested all over the Indo-European landscape.

Neither of these locative case-form hypotheses is entirely unproblematic, however. First, evidence for **en* as a discrete segment in Hittite is difficult to find. The place word *anda* is widely thought to be cognate with Old Latin *endo* and Greek ἔνδον, of which **en* is presumably an element (H. Kronasser 1966:351-2; Puhvel 1984:77)[4]. Kronasser also discusses the possibility that **en* may be reflected in such Hittite verbs as *immiya* 'mix' and *anda impai* 'be troubled' (159). Pokorny offers as possible inheritors of **en* Hittite *anturiyas* 'interior (adj.)' and *andurza* 'outside' (1969:314). Much of this is quite speculative, however, and it is noteworthy that all of these potential reflexes of **en* have that segment word-initially, while it would need to have been postposed to form part of *píran, andan,* and the rest. Perhaps more significantly, the very sketchiness of the evidence for **en* in Hittite indicates that it lost any existence as an independent unit relatively early in the proto-language. On the other hand, the heavily

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[4] Nussbaum reports that Greek χεῖμα 'cold, winter' could be derived from Proto-Indo-European **ĝʰ(i)i̯-ēm/*ĝʰi-m* 'winter' plus a locatival **-en* (1986:52n.11), thus suggesting a basis for postposed **-en* in Greek, and in a form with a Sanskrit cognate in *héman* 'winter' (on which see Chapter VII, section 3.3). However, while these forms evidently establish postposed **-en*, if not throughout Proto-Indo-European, then certainly in the subgroup containing Hellenic and Indo-Iranian, this fact does not bear directly on the Anatolian languages.

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nominal functioning of the postpositions within their phrases strongly suggests that *píran, appan*, and the others had begun to function as adpositions only comparatively recently. The final *-an* of these postpositions therefore appears unlikely to be a reflex of **en*.

As for the possibility of all five of these lexemes being endingless locatives in themselves without any compounding involved, Hittite does indeed present some remnants of such locative case forms. J. Friedrich offers several examples, such as É-*ir* 'in the house', UD-*at* 'by day', SÀ-*ir* 'in one's heart', *dagan* 'to the ground', *nepiš* 'to heaven', and *keššar-ta* 'in your hand' (1960:44), although this does not exhaust the inventory of such items. In particular, *šer* would appear to show some promise of fitting in quite comfortably with these items, and indeed Neu regards *šer* as deriving from just such a source (1980:36). However, given the posited roots for the other items, **pr̥, *e/op-, *kat-* (from **kom-ta*?- J. Pokorny 613), and **en-d-*, it is difficult to account for the appearance of the nasal on such a consistent basis in this set of endingless locatives. Indeed, both Neu (1980:35-6) and G. Dunkel (1992:159) note that, whatever the semantic difficulties of this set, morphologically they are accusative singular neuters. Indeed, Neu labels the derivation of final *-an* from the Proto-Indo-European **-m/*-m̥* "problemlos" (1979:192). Accordingly, the accusative case appears to be the more likely source for these forms on morphophonemic grounds.

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The possible exception, *šer*, may indeed be a locative. H. C. Melchert in fact proposes that it may in origin not even be a locative that is endingless: "[I]t seems most economical to derive Hitt. *šer* 'above' from the same **séri* which leads to CLuv. *šarri* and Lyc. *hri*. I marginally prefer to take the loss [of **i*] after /r/ in *šer* as regular and attribute the retention in *-ri* [in the present medial verb] to analogy with other present endings" (1994:183). This however seems too *ad hoc* an account to be entirely satisfying; *šer* may just as easily be a reflex of an inanimate accusative noun. With the evidence of the other four postpositions, the latter hypothesis is an inviting one. Of course, given the locatival senses of these five items (as will be explained in the next section), the possibility that an endingless locative *šer* served to influence a semantic reanalysis of the others is also not entirely unattractive (further discussion of *šer* and its possible relationship to Luwian *šarri* may be found in Chapter III, Section 3.1).

4.2. Their Semantic Senses

It remains, then, to explain the semantics of these purported accusative case nouns functioning as locational adpositions, stationary in sense. As mentioned, adverbial functions of the accusative are most commonly associated with senses of motion, above all motion toward, but also motion through, across, along, or around. Consequently, the accusative readily acquires the connotation of extent of space or duration of time. These latter two are fairly

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easily confused with simple location within space or time, as anyone who has had to explain to beginning Latin students that *paucās hōrās* may be translated as 'for a few hours' but not as 'in a few hours' can attest.

Indeed, even among native speakers of Latin there was confusion on such points in constructions of space or time. A. Ernout and F. Thomas note the use in Plautus and Terence of the accusative with *in* where the locative ablative would be expected: *esse in potestatem, esse in mentem* (1953:112). Ernout and Thomas also report vacillation by Caesar concerning distance expressions: *milibus passuum sex a castris caesaris sub monte consedit* (*De Bello Gallico* I. 48.1); *consedisse milia passuum ab ipsius castris octo* (*B. G.* I.21.1) (111). Certainly, then, from a sense of 'motion toward' to one of 'motion along', to 'motion within' a space, to 'extent' of space, to 'occupation' of a space, to 'simple location' in a space, is not a counter-intuitive leap. With an accusative origin for the Old Hittite postpositions clearly explicable on semantic grounds and far preferable in morphological terms, this hypothesis should therefore be considered more likely at present.

Of course, the preceding paragraphs assume a sense for the accusative similar to that found in such languages as Latin, Greek, and Vedic Sanskrit. It is possible, however, that these Old Hittite forms hearken back to an earlier stage of the accusative case in Proto-Indo-European. The ending *-an* on four of the five non-motional auxiliaries, as argued in Section 4.1, reflect the accusative in

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-om, or more precisely, the nominative/accusative neuter. It may further be argued, then, that this *-om* reflects the earlier Proto-Indo-European absolutive or inactive marker, contrasting with the ergative or active marker *-(o)s* (T. Gamkrelidze and V. Ivanov 1995:236-41). An origin in the "semantically inactive" class of nouns that eventually gave rise to the neuter gender, as well as giving its marker to the patient once the proto-language developed a transitive/intransitive distinction, is also an attractive means of accounting for the similarly inactive set of Old Hittite local items.

4.3. *kaššaš*

Perhaps at this point one other postposition, not elsewhere attested than in the *Laws*, should be discussed, *kaššaš* 'in lieu of, instead of', which may be found with the genitive singular possessive *-šaš*. Equally noteworthy, its object is also in the genitive case:

7) *ki-nu-na* 1 UDU LÚ-*na-aš ka-aš-ša-aš-ša-aš ḫu-it-ti-an-ta.*
'Now they adduce one sheep in lieu of the man' (*HG.* II.52.41);

8) 2 UDU^{ḪLA} GUD^{ḪLA} *ka-aš-ša-aš-ša-aš ḫu-u-it-ti-ia-an-ta.*
'They adduce two sheep in lieu of the cattle.' (*HG.* II.52.42);

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9) *ki-e-el* 1 UDU *ki-e-el-la* 1 UDU *ka-aš-ša-aš ḫu-u-it-ti-ia-an-ta.*

'In lieu of the one and in lieu of the other they adduce one sheep each.'

(*HG* II.82.4-5; all three translations from Puhvel 1997:119)

Unfortunately, a complicating matter is that the above three sentences contain the only three attestations of *kaššaš*. J. Friedrich in his edition of the *Laws* suggests that it may be derived "aus dem alten D.-L. Plur. eines Substantivs „Ersatz"" (111), and Puhvel proposes a derivation for this inflected form from a Proto-Indo-European **g^{h}end-* 'space, spot' (1997:119). This may well be a more inviting postulation than either an endingless locative singular or an inanimate accusative, given the ending of the form in *-aš*. Certainly the manifestly nominal features of *kaššaš* indicate a relatively recently-minted postposition, or even, in Josephson's phrasing, a postposition "*in nascendo*" (1981:101). However, this is also an eccentric form, attested nowhere else. It is therefore unlikely to be of use in generalizing about the derivations of the more common Old Hittlte local adverbs.

5.0. The Functioning of the Auxiliaries

5.1. *para/piran*

Turning to the functions of these items, Güterbock and Hoffner list both *parā* and *piran* as "adv., prev., and postpos." (1997:109 and 291, respectively). However, in the six Old Hittite texts consulted here, forty-five of forty-six

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instances of *parā* are designated as preverbs, with the remaining case a simple adverb. On the other hand, thirty-five of thirty-nine usages of *piran* are found to be postpositions, with the remaining items preverb. However, three of these preverbs are in the *Laws*, whose syntax of the local adverbs seems in some respects to foreshadow the changes that become prevalent in the Middle stage of the language. Still, with these figures it is necessary to bear in mind that arriving at such a determination as to the lexeme's category is not necessarily automatic. Of course, this disposal is not always difficult; where there is no noun governed by any type of grammatical or locational relationship, the designation of 'preverb' is clear:

10) *ta-aš-ta pa-ra-a pí-e-da-i.*
'And he brings them out.' (*Gwr.* Vs. I. 23)

Similarly, an oblique-case noun in no position to be so governed by the locative auxiliary presents no difficulties:

11) *ša-an ^{D}Ḫal-ma-šu-it-ti ^{D}Ši-i-uš-mi-iš pa-ra-a pa-iš.*
'And my god Siu delivers them out to the throne-god Halmasuit.'
(*An.* Rs. 47)

On the other hand, given that, where *parā* functions as a postposition, it generally takes an ablative complement (Güterbock and Hoffner 1997: 117), some instances, as already noted where sentence 1) was first discussed above,

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are difficult if not impossible to categorize:

12) LUGAL-*uš* SAL.LUGAL-*ša ḫal-ma-aš-šu-it-ta-az pa-ra-a ú-en-zi.*
'The king and queen come out from the throne.' (*Gwr.* Vs. I. 13)

In such *Mittelstellung* contexts as these, Güterbock and Hoffner is content to throw up its hands: "In the many cases where the order is *abl. - parā - verb*, a decision between preverb and postposition for *parā* is impossible and unnecessary" (1997:118). That *parā* is capable of serving as a postposition is ascertained because ablative + *parā* phrases appear to be in free syntactic distribution throughout the sentence (Güterbock and Hoffner 1997:118), while the placement of preverbs within a clause is more restricted, as Salisbury notes above.

The distribution of postpositions, on the other hand, is somewhat less restricted:

13) *ku-ut-ta-aš pí-ra-an kat-ta da-a-i.*
'Before the wall he puts it down.' (*Gwr.* Vs.II.16)

14) LÚ ^{GIŠ}BANSUR-*ša ku-ut-ta-aš pí-ra-an ša-aš-ki-it-ta.*
'And the table-setter sets it before the wall there for the god.'
(*Gwr.* Rs.III.46)

Note also that here *píran* governs the genitive as opposed to a more spatially-

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oriented case such as the ablative or dative-locative. Incidentally, the one occurrence in the *Bilingue des Hattusili I* of *píran* as a preverb is unambiguous:

15) *ka-a-ša-za-kán ku-u-un pí-ra-an ša-az-ki-it-ta.*
'And this one pushes this (other) one forward. (*BH.* II.58)

In the *Laws,* the three attestations occur in fairly quick succession, in sections 146-8, and all with the same root, *walḫ-;* Hoffner interprets the resulting compound as 'strike first' (207).

5.2. *appa/appan*

Unfortunately, few of the lexemes under consideration here are as straightforward to interpret as *para* and *píran.* Like *píran, appan* appears most often as a postposition, but unlike its counterpart, it is also found relatively frequently as a preverb or adverb. The *Erlaß Telepinus* offers examples of all three functions. It is found as a postposition with a genitive complement:

16) *ku-iš am-me-el* EGIR-*an* LUGAL-*uš ki-ša-ri.*
'...who becomes king after me...' (*Tel.* III.49)

And here incidentally a valuable contrast may be seen with subsequent Hittite adpositional usage. Hoffmann gives for her main text at II.40:

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17) *ku-iš am-mu-uk* EGIR-*an-da* LUGAL-*uš ki-ša-ri.* (*Tel.* II.40)

The meaning is the same as for the previous example, but the postposition *appanda* takes a dative/locative complement, as is more characteristic of Middle Hittite. However, she also offers for this locus the textual variant *am-me-el* EGIR-*an,* with the older genitive construction (32n.f, g).

One may also find *appan* listed as a preverb:

18) *ma-a-an* ^{m}*Ḫa-an-ti-i-li-iš* SAL.LUGAL ^{URU}*Šu-uk-zi-ia Ù* DUMU^{MEŠ}-*ŠU*
EGIR-*an ša-an-aḫ-ta.*
'When Hantili sought after the queen of Sukziya and her sons.' (I.58-9)

19) *nu-kán hal-ki-uš* EGIR-*an ma-ak-nu-nu-un.*
'And I multiplied again the grains.' (*Tel.* III.43)

Sentence 19) is unambiguous; *hal-ki-uš* is accusative plural and as such would be extremely difficult to construe as the object of a postposition for Hittite. Sentence 18), however, is far less clear. The position of *appan* is certainly appropriate for a preverb, as outlined by Salisbury, and Hoffmann evidently finds semantic grounds for regarding it as modifying the verb *šanahta* (172). Still, the use of Sumerograms and Akkadograms in sentences such as 18) to express the items preceding the word serves to obscure the case of these nouns,

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eclipsing any chance of determining whether they might be functioning as a postpositional complement.

And two instances of *appan* are listed by Hoffmann as adverbs:

20) *ki-i ut-tar šu-ma-a-aš* EGIR-*an še-ik-tin.*
'Know you (pl.) these words hereafter!' (*Tel.* II.68)

21) GIR-*an a-ap-pa-an ku-ut-ta-ni-iš-ši.*
'...the dagger (*acc.*) behind in the neck.' (*Tel.* IV.14/6')

In sentence 20), the placement of *appan* is once again very well-suited to a preverbial function; presumably Hoffmann sees no particular semantic connection between this item and the imperative verb. The second example is even less clear because the text is broken. It definitely is not a postposition, as it follows an accusative, but, if the missing word immediately following the dative/locative should happen to be a verb, this instance of *appan* might well be a preverb. It must above all be remembered that, for all these texts, the category classifications offered by their modern scholars can have varying degrees of certainty, given the context, but an absolutely firm determination is often impossible.

5.3. *šara/šēr*

Presenting still further ambiguity is *šer*, most often considered a

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postposition, but which is classified by Otten and Souček in their edition of the *Ritual für das Königspaar* as an adverb for six of its attestations in the text and as a postposition only twice. These eight instances are few enough yet complicated enough that all of them may be considered:

22) *še-er-še-me-ta* GÍR ZABAR *ki-it-ta.*
'And over them lies a bronze dagger.' (I.31)

While classified by Otten and Souček as an adverb, this occurrence of *šēr* might by its sentence-initial position be considered a preverb, as per Salisbury, or, more significantly, by its association with the enclitic inanimate accusative possessive pronoun, as a postposition. It may formally be regarded as a noun with the functions of a postposition, although its relational qualities that impute adpositional characteristics to it must at the same time not be discounted. These enclitic possessives, although normally expected in conjunction with nouns, are in Old Hittite also not at all uncommon with some postpositions, including *píran* and *katti* in addition to *šēr.* As observed by Laroche and Neu, and as will be elaborated further in Section 6.2.1, this tendency to affix enclitic possessives is an especially strong indication of the recent nominal derivation of these items. As such, in addition to the uncertainty as to whether these lexemes should on a given occasion be regarded as a postposition, a preverb, or an adverb, there is a nominal component to be borne in mind as well. Otten and Souček also note the

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"'nominal' konstruierten Wendung *šer-šmet*" (87).

Four instances of *šēr* occur in rapid succession and may therefore be considered together:

23) ^{MUŠEN}*ḫa-a-ra-na-an* LUGAL-*aš* SAL.LUGAL-*aš-ša še-e-er-ša-met* III-*ŠU* DUMU.E.GAL *ɥa-aḫ-nu-zi*.

'The courtiers turn the eagle thrice above the king and queen.' (II.30)

24) *u-ga-aš-ma-aš-ša-an* ERÍN^{MEŠ}-*an še-e-er* III-*ŠU ɥa-aḫ-nu-ú-mi*.

'And I turn the troops above them thrice.' (II.31)

25) ERÍN^{MEŠ}-*ti-ma-aš-ša-an še-e-er* GÍR ZABAR *ki-it-ta*.

'But above the troops lies a dagger of bronze.' (II.33)

26) ^{MUŠEN}*ḫa-a-ra-na-an* ERÍN^{MEŠ}-*an-na* LUGAL-*aš* SAL.LUGAL-*aš-ša še-e-er-še-me-et ɥa-aḫ-ne-me-ni*.

'We turn the eagle and the troops above the king and queen.' (II.34)

Otten and Souček label *šēr* in sentences 23) and 26) adverbs and those in 24) and 25) postpositions. However, in both sentences with *šēr* designated as adverbial, the word is preceded by the same genitive phrase, LUGAL-*aš* SAL.LUGAL-*aš-ša*, and followed by the same enclitic possessive pronoun, *-šmet*. On the other hand, 24) and 25) have purported postpositions preceded by an

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accusative singular collective noun and a dative/locative of the same noun, respectively. In the other Old Hittite texts considered here, postpositional *šēr*, where the case is not obscured by Sumerograms, generally follows a dative/locative, so the identification of *šēr* in 25) as a postposition appears secure. However, at least once, at *Tel.* IV.24/16', it takes a genitive and so, given the presence of the enclitic possessives, as well as the formulaic nature of the phrase LUGAL-*aš* SAL.LUGAL-*aš-ša, šēr* in 23) and 26) may well also be considered postpositions, or , in formal terms, as relational nouns[5]. The identification of 24) as a postpositional sentence remains puzzling. While the construction *šēr* + adverb (III-*ŠU*) + verb in 24) does not correspond exactly with Salisbury's specified alignment involving a preverb and verb separated by a single element, she herself observes that "Details of word-order restrictions need further investigation" (4). The inclination therefore presents itself to take *šēr* in 24) as a preverb, albeit in a slightly unusual context.

27) LUGAL-*aš* SAL.LUGAL-*aš-ša še-e-er-še-me-et ɣa-aḫ-nu-uz-zi.*
'He turns above the king and queen.' (II.46)

28) ...*še-e-er-ša-me-et* III-*ŠU ɣa-aḫ-nu-zi.*
'...he turns three times above them.' (III.35)

---

[5] The continuum between relational nouns, postpositions, adverbs, and preverbs in Old Hittite will be addressed in greater detail below, in Section 6.

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Sentences 27) and, insofar as far as we have it, 28) are virtually identical to 23). Sentence 27) lacks only the adverbial III-*šU* and 28) only the broken-off genitive complement. Despite Otten and Souček's designation of these instances of *šēr* as adverbs, they would thus appear to be postpositions as much as those in 23) and 26).

29) *še-e-ra-aš-ša-an* GAD-*an pí-eš-ši-e-mi.*
'And then I throw a cloth over.' (IV.22)

Here *šēr* begins the sentence, followed by the aspectual particle *-šan.* This may therefore confidently be regarded as a preverb. One thus finds in *šēr* syntactic qualities characteristic of nouns, adpositions, adverbs, and preverbs. The last of this group are sometimes considered as a subset of adverbs, but they are nevertheless much more closely bound semantically as well as syntactically to their associated verbs. As such, conventional classification of *šēr* and other such lexemes in Old Hittite as discrete and distinct parts of speech is not entirely adequate. A framework that takes into account the frequently muddy boundaries between these categories among the Old Hittite locational auxiliaries must therefore be consulted.

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6.0. Fuzzy Set Theory and its Applicability to Old Hittite Local Auxiliaries

6.1. Fuzzy Set Teory

The study of linguistics is accustomed to consider the presence or absence of a given feature or characteristic as a strictly yes-or-no, plus-or-minus proposition, be it [±*voiced*], [±*imperative*], or [±*verb*]. As observed, however, Old Hittite postpositions and related lexemes do not readily lend themselves to examination on such terms. The notion, aptly suited to this topic, that category boundaries are not necessarily so neatly delineated, is explored in Fuzzy Set Theory. This theory is based on the premise that "a fuzzy set is a class with unsharp boundaries, that is, a class in which the transition from membership to non-membership is gradual rather than abrupt" (L. Zadeh 1972:4). In the framework of this schema, a given item would not be assigned a label of plus or minus regarding its membership in a set, but is rather accorded a "grade of membership... between 0 and 1, with 0 and 1 corresponding to non-membership and full membership, respectively" (Zadeh 5). However, fuzzy-set theoreticians have tended to gravitate towards questions of semantics. One exception offers an approach to grammar with a slightly different terminology ("squishy" rather than "fuzzy"), but essentially the same basic premise, "reject[ing]... restrictive assumptions of the standard theory, such as that a constituent must either be a member of a certain category or not. Category membership is a matter of degree, and different syntactic transformations may differ in the degree of membership they demand" (J. Fodor 1977:203). Ross proposes

> that the traditional view of the categories *verb*, *adjective*, and *noun*, under

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which these three are distinct and unrelated, is incorrect. Instead,... these categories are (possibly cardinal) points in a linear *squish*, or quasi-continuous hierarchy:

> *Verb* > *Present participle* > *Perfect participle* > *Passive participle* > *Adjective* > *Preposition* > *Adjectival noun* (e.g., *fun*) > *Noun.* (1973:141; 1972:316)

Certainly in English the syntactic function and phonological shape of the verb and present participle overlap to some extent, as do the perfect participle and passive participle ('having broken' vs. 'broken'), the latter of which slides quite readily under the heading *Adjective.* J. Ross reports a number of syntactic tests for the category affiliation of a given semantic root as it affects syntax, for example:

> *Constituent deletion:* the nounier [the head word], the fewer constituents need appear in the surface structure:
>
> a. It is surprising (to me)/surprises *(me) that you know Tim.
> b. The surprise was considerable.
> c. I will attempt to leave, but Bill won't attempt *(to leave).
> d. Bill wants to leave, but the attempt (to leave) may tax him too much. (1972:325)

In these sentences, the degree of membership in the set of nouns for 'surprise' and 'attempt' in b) and d) would be closest to 1, that of 'surprises' and 'attempt' in a) and c) closest to 0, and that of 'surprising' in a) more middling. Such a system of categorization is of course more effective in a language such as English where clues afforded by morphology are missing or at least incomplete.

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6.2. The Situation in Old Hittite

As for Hittite morphology, one finds a greater level of distinctness between nouns and verbs, but, as in many of the ancient, heavily-inflected Indo-European languages, adjectives are very often employed as substantives and as such may be considered to possess a higher degree of "nouniness" than those of English. More pertinent to the subject at hand, for Old Hittite the following "squish" may be proposed:

Noun > Postposition > Adverb > Preverb.

That Old Hittite postpositions display a fairly high amount of "nouniness" may be observed from two striking syntactic features of these lexemes. First, there is a tendency, very regular in Old Hittite, for the nouns governed by these postpositions to appear in the genitive case, while in most Indo-European languages, and indeed in the later stages of Hittite, cases with values adverbially or spatially oriented, such as the locative or ablative, are employed with adpositions far more often than a more purely grammatical case such as the genitive. Constructions such as ***kur-ša-aš pí-ra-an*** 'before the shield' (*Gwr.* I.58) or ***at-ta-aš(-ma-aš) a-ap-pa-an*** 'after (my) father' (*An.* Vs. 10) thus display a high level of similarity to genitive + noun constructions. Indeed, these similarities to nouns are great enough that the question of whether these items might not simply be regarded as a relational subset of the inventory of nouns poses itself.

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However, these words also mark "case" relations[6] not indicated by the inflectional system of Old Hittite, and therefore already possess an essential adpositional quality. Admittedly, the fact that, over the history of the Hittite language, these relational qualities become more and more pronounced at the expense of their earlier nominal status makes it easier for one simply to think of them as adpositions rather than nouns even in Old Hittite. They must nevertheless be considered more noun-like (with a "nouniness" quotient closer to 1), than a postposition in a later construction such as ^{m}*Ḫa-an-ti-li kat-ta-an* 'with Hantili' (*Tel.* Vs. I.32). But even at this point, their role is sufficiently adpositional that the label is appropriate.

6.2.1 Old Hittite Adpositions with Enclitic Possessive Pronouns

Of perhaps even greater interest is the fairly occasional use of some of these adpositions with enclitic possessive pronouns. These particles are normally found with certain nouns, specifically those, like *at-ta-aš-ma-aš* 'of my father' quoted above, "which express concepts that are inherently relational. Examples... in English are *side, daughter,* and *face*... One doesn't say of someone that she is a daughter, only that she is somebody's daughter; and although it is possible to speak of having seen a face, the word is typically used when referring to 'his face' or 'your face' or the like" (C. Fillmore 1968:61). Such "inherently relational nouns" are considered to carry a feature of "inalienable possession", as

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6 For elaboration of this term, please see the Chapter I, Section 2.1.2.

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opposed to nouns that are not inherently possessed (Fillmore 61).

In Old Hittite, then, such inalienably possessed nouns are affixed with enclitic possessives rather than governing a noun in the genitive case. However, some postpositions also come under this relationship, such as the previously cited *še-e-er-še-me-et* 'above them', or perhaps, 'their above', as it were. Other such adpositions include *píran (pé-e-ra-am-mi-it* 'before me' - *An.* Vs. 79), and *katte (kat-ti-mi* 'with me' - *An.* Vs. 74). The difference in form between *-mit* and *-mi* is explicable as that of modifiers for an accusative and a dative/locative form, respectively. Marking for case is, of course, another highly noun-like characteristic, and, as noted previously, such constructions are marked for gender as well. To be sure, not all postpositions in Old Hittite reflect such inalienable possession, as may be noted from *am-me-el a-ap-pa-an* 'after me' (*An.* Vs.22). Presumably those that are marked with the possessive particles may be regarded, inasmuch as inalienably possessed nouns are as a rule those more intimately associated with the speaker or other narrative protagonist, as being higher on a scale of "nounish" character. The Old Hittite postpositional squish proposed above may thus be amended:

Inalienably possessed nouns > Alienably possessed nouns > postpositions.

As for the other end of the squish, adverbs should be considered intermediate between postpositions and preverbs due to their fairly free syntactic distribution throughout the sentence, similarly to adpositional phrases,

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and to their relatively loose modificational relationship to the sentence as a whole. Preverbs, on the other hand, are, as per Salisbury, more restricted syntactically, and semantically are far more closely bound to the clause's verb. As such, they are more distant on this continuum of decreasing noun-like quality than are postpositions and adverbs. Such a framework as Fuzzy Set Theory would thus appear to serve adequately to reflect that, with many category boundaries surrounding adpositions in Old Hittite not entirely clear, there are nevertheless some words fitting the description and filling the role of this part of speech to a greater or lesser degree.

7.0. Conclusions on Old Hittite

One point deserving of mention is that, at least at present, it is not at all clear whether, or at least to what extent, squishes such as that just introduced should be considered universal or language-specific. As touched on earlier, differing levels of morphological complexity from one language to another is a severely complicating factor. It is certainly difficult to imagine a squish encompassing both noun and verb, such as that proposed by Ross for English, in a language like Hittite where noun and verb are marked by widely divergent morphology.

With much greater confidence it can be asserted that these squishes, or at least the one offered above for Old Hittite, are in large measure reflective of diachronic developments. As will be elaborated in subsequent portions of this effort, many of the features of Old Hittite postpositions that are termed

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indicative of greater "nouniness", including elements of case morphology, government of genitive complements, and the capability of showing inalienable possession, also serve to demonstrate that, at an earlier stage of the language, these lexemes had been in fact nouns. After being fossilized into certain types of constructions and bleached of their primary semantic meaning, these words came to be reanalyzed as postpositions. In Middle and Later Hittite, on the other hand, postpositions are comparatively rare, and many Old Hittite postpositions resurface more often as adverbs and preverbs. In other words, they move further along this continuum over time and thus grow less noun-like.

8.0. Middle Hittite: Introduction

The span of some two centuries or so covering the transition from the Old to the Middle Hittite texts sees significant changes in the language's system of locational auxiliaries. Most immediately striking is the standardization of the tendency, already surfacing in such relatively late Old Hittite texts as the *Erlaß Telepinus*, for postpositions to govern the dative-locative case rather than the genitive. Moreover, the tidy distinction between the auxiliaries ending in *-a* and their counterparts, mostly ending in *-an*, has become obscured. Some Old Hittite postpositions appear more often as preverbs by the middle stage of the language, while others tend to lapse into disuse, and *anda* for one becomes prominent both as postposition and preverb. Taken together, these developments indicate that the nominal features of these items, reflecting their

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fairly recent derivation from nouns, have become almost entirely lost. Very few postpositions in Middle Hittite texts carry any trappings of an earlier nominal status, and the exceptions for the most part are not attested in the older stage of the language. In short, there is a rather clear movement further along the noun-adposition-adverb-preverb cline or "squish" described previously for these Hittite lexemes. These changes may be explained through an examination of the locational auxiliaries in a selection of Middle Hittite texts, similar to that conducted earlier for the older language.

9.0. Primary Texts Consulted: Frequency of Locative Auxiliaries in the Texts

The texts considered here are: *The Hittite Instruction for the Royal Bodyguard* (H. Güterbock and T. van den Hout 1991); *Die Militärischen Eide der Hethiter* (N. Oettinger 1976); *Das hurritische Epos der Freilassung I* (Neu 1996); and the *Hethitische Briefe aus Maşat-Höyük* (S. Alp 1991). As before, all lexemes with preverbial or postpositional functions are tabulated as follows according to how many times they appear as preverbs, adverbs, and postpositions in each text:

		RB.	*ME.*	*EF.*	*ML.*	*Total*
anda:	*preverb*	11	8	17	26 (+19)	81
	postposition	2	5	-	-	7
	adverb	-	-	2	4	6
andan:	*preverb*	-	-	2	-	2
	postposition	-	-	1	-	1
	adverb	1	-	-	-	1

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		RB.	ME.	EF.	ML.	Total
appa:	*preverb*	9	-	9	11 (+24)	53
	postposition	1	-	-	-	1
	adverb	11	-	5	-	16
appan:	*preverb*	16	-	-	12 (+5)	33
	postposition	-	-	-	-	-
	adverb	-	1	2	8	11
appanda:	*preverb*	8	-	-	- (+1)	9
	postposition	-	-	-	-	-
	adverb	5	5	-	-	10
aṛha:	*preverb*	13	13	28	31 (+6)	91
	postposition	-	-	-	-	-
	adverb	-	-	-	2	2
iṟar:	*preverb*	-	-	-	-	-
	postposition	-	7	10	-	17
	adverb	-	-	-	-	-
katta:	*preverb*	14	6	2	- (+7)	29
	postposition	-	-	-	-	-
	adverb	1	-	1	-	2
kattan:	*preverb*	-	1	-	2 (+4)	7
	postposition	1	-	1	4	6
	adverb	-7	-	-	1	8
katte:	*preverb*	-	-	-	-	-
	postposition	6	-	-	19	25
	adverb	-	-	-	-	-
menaḫḫanda:	*preverb*	-	-	-	1	1
	postposition	3	-	-	-	3
	adverb	1	-	-	-	1
parā:	*preverb*	22	3	34	58 (+8)	125
	postposition	-	-	-	-	-
	adverb	4	-	-	-	4

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		RB.	*ME.*	*EF.*	*ML.*	*Total*
píran:	*preverb*	13	4	1	4 (+9)	31
	postposition	-	1	8	2	11
	adverb	1	-	-	1	2
šarā:	*preverb*	10	5	5	16 (+7)	43
	postposition	-	1	-	-	1
	adverb	-	1	-	-	1
šēr:	*preverb*	-	1	1	1 (+2)	5
	postposition	-	1	2	-	3
	adverb	1	-	-	-	1

9.1. Notes on the Figures

Not included in the figures above are attestations of these lexemes in the extremely fragmentary portions of the *Epos der Freilassung*, where there is no context to determine in what part of speech they are functioning. Among these instances are *anda* (4 attestations); *andan* (2); *appa* (1); *appan* (1); *arḫa* (1); *katta* (1); *kattan* (3); *parā* (1); *píran* (4); and *šara* (2). Otherwise, as with the tabulations for Old Hittite, the classifications of the editors are followed where these are offered. Still, it must be borne in mind that, as Güterbock and Hoffner point out for *parā*, a determination is at times quite difficult and perhaps not necessary, as in:

30) 1 DUMU.É.GAL ^{É}*ḫa-le-en-tu-u-az pa-ra-a ú-iz-zi.*

'one palace attendant comes out of the palace.' (*RB*. I. 12.64-5).

With this qualification stipulated, however, the present effort remains content

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with Güterbock and van den Houk's classification of *anda* in this instance as a preverb.

9.1.1 Classification in the *Maṣat Letters*

However, in some texts classifications of these auxiliaries' roles may be problematic, a case in point being the *Maṣat Letters*. For the *Maṣat Letters*, two figures are given in this tabulation for the preverbs, the first for those items identified as such by Alp, with those not so classified but appearing at least possibly to serve that function added in parentheses. Where a context is at hand, Alp's classification between preverb and adverb is often quite puzzling. He does not, as it appears, discuss his criteria for distinguishing between the two in his work, and the reader encounters great difficulty in seeking any consistency in this determination. On several occasions, Alp posits opposing classifications for locative auxiliaries in close and identical contexts, or nearly so:

31) *nu-ya-kán ḫal-ki-uš ar-ḫa ya-ar-aš-ki-iz-zi.*
'And he continues to harvest the grains off.' (*ML.* 25 Vs. 10).

32) *na-aš-kán ar-ḫa ya-ar-aš-ten.*
'Cut them off!' (*ML.* 25 Rs. 17)

In these two sentences, *arḫa* appears with the same verbal root, *warš-*, and in

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both cases the aspectual particle *kán* is employed. However, Alp labels only the second instance as a preverb. Similarly:

33) MUŠEN^{ḪLA}-*ma-an-na-aš* EGIR-*pa ti-it-ta-nu-uš-kán-zi.*

'And the birds continue to set themselves back from us.' (*ML.* 47 Vs. 6)

34) *nu-un-na-aš* MUŠEN *ku-it* EGIR-*pa ti-it-ta-nu-ut.*

'Because the bird set itself back for us' (*ML.* 47 Vs. 9)

35) *nu-un-na-aš* ^{URU}*Ka-a-ša-ša-ia* MUŠEN^{ḪLA} EGIR-*pa ti-it-nu-us-ki-ua-an da-e-er.*

'And the birds were taken to set themselves back for us from Kasasa.' (*ML.* 47 Vs. 11)

Here, too, the same verbal root, *tittanu-*, is employed with *appa*, and in all three instances without an aspectual particle. Yet Alp classifies *appa* in 34) and 35) as a preverb, but as an adverb in 33).

10.0. Adposition and Preverb in Middle Hittite

Clearly the distinction between preverb and adverb is a problematic one. As mentioned earlier, some, such as Starke or P. Friedrich, prefer not to address it, identifying only a broader category also covering Hittite postpositions labeled "adverb" or "locative auxiliary". Nevertheless, in many contexts, one finds little

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difficulty in recognizing a given item as governing or modifying, or, to borrow Goetze's term, "in nexus with" (98), a noun, verb, or clause as a whole. Remaining necessary is a means of classifying those instances where the part of speech is not obvious, those falling somewhere along the fuzzy boundary between the categories. Most troublesome for any attempt at such a classification is Goetze's environment (c), or Zuntz' *Mittelstellung*, as described in Section 1.1.1. The *Maṣat Letters* include several sentences of this type:

36) KUR-*e an-da du-ud-du-uš-ki.*
'Always govern over the land graciously.' (*ML.* 45 Vs. 11);

37) *na-an-za-kán* URU-*ri ša-ra-a pe-e-da-ú.*
'Let him bring them up to the city.' (*ML.* 45 Rs. 22-23)

In such instances as these one finds Hittite locative auxiliaries at their fuzziest. It is in such situations that Güterbock and Hoffner call the distinction between postposition and preverb "impossible and unnecessary" (1997:118). However, when one's purpose is to analyze the language, an item falling neither into one category nor the other, but rather partly in both, may prove troublesome. It becomes desirable, as Zuntz attempted, to sharpen the boundary, even if only artificially, so that the object under investigation may be identified on one side of the border or the other. In a situation such as for *anda* in 31) or for *šarā* in 32), to borrow from the terminology of Fuzzy-Set Theory, by all appearances the grade

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of membership is .5 in both the set of preverbs and that of postpositions, at Zadeh's "crossover point" (1972:6). Here a device for nudging the word to one side or another is appropriate. Given the paramount importance of the verb in the syntax of Hittite as in the other ancient Indo-European languages, in that the subject must agree with the verb and other primary nominal constituents such as direct and indirect objects are governed by it, one may reasonably resolve the impasse by assigning to *anda* and *šarā* in the above sentences a value of .51 for association, or nexus, with the verb, and therefore label them more preverbs than not.

This mechanism may be extended as well to another frequently-occurring syntactic environment. This corresponds to Goetze's (d), except that the local adverb is immediately preceded by a noun in the dative/locative, or a "directional" case:

(d_1) directional case - local adverb ... - verb

38) *an-tu-uḫ-ša-ša-kán ut-tar ki-iš-ša-ri-i an-da ka-ru-ú da-iš.*
'He formerly placed the business of men in hand.' (38 Vs. 5-7)

39) URU-*ri-ma-kán an-da-ki-iš-ša-an me-mi-ia-u-en.*
'In the city we spoke as follows.' (46 Vs. 17)

Here too one finds a locative auxiliary in a position entirely normal for preverbs, but with a noun in a case appropriate to be governed by a postposition

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immediately preceding. Despite its greater proximity to the noun, nexus with the verb is regular and expected in this context, and *anda* in 38) and 39) should therefore be considered a preverb.

10.1. Preverbs and the verb *es-* 'to be'

An environment in which local adverbs have traditionally not been recognized as preverbs by Hittite scholarship is in conjunction with the verb *es-* 'to be'. Locative auxiliaries in nexus positions with *es-* are classified by Alp, as by most editors of Hittite texts, as simple adverbs. However, given that the copula takes the same paramount role in the sentence as any other Hittite verb, with the same positioning for the auxiliaries in relation to it, there is no evident basis for not classifying such constructions as typical compound verbs with preverbs. This is suggested even where the supporting context is unavailable, as in *kat-ta-an e-eš-du* 'let him be close by' (84.1 Rd. a. 4), or *an-da e-eš-ta* 'he was inside' (88 Rs. 19). Where the complete sentence is attested, the connection is all the more apparent:

40) *na-aš-ta ka-a-ša* ERIN$^{MEŠ.ḪLA}$ *pa-ra-a tu-uk-pát e-eš-zi.*
'And now look, the troops are out just with you.' (70 Vs. 5-7)

Here *parā* is clearly modifying *ešzi;* the spatial relationship of the troops to the dative/locative pronoun would be utterly misread with the plain copula. This

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compounding is especially well-supported in the other Indo-European languages, where combinations with the copula are quite common, as in Greek *πάρειμι* 'I am present', or Latin *abest* 'he is absent'.

Further evidence that the Hittite copula may be modified by a preverb is found in nominal sentences (terminology per F. Josephson 1972:397), where, as is also common among the Indo-European languages, the copula is omitted but understood. Although the verb in such instances need not be overtly stated, any associated preverb will still appear:

41) *na-aš-ta* NU.GAL *ku-iš-ki an-da.*
'And now no one is there at hand.' (58 Vs. 11)

42) *ma-aḫ-ḫa-an* ^{D}UTU*ŠI an-da.*
'As soon as his Majesty is inside.' (64 o. Rd. 27)

In each of these sentences, although the copula is missing, *anda* is still present to modify its sense. Indeed, an especially striking aspect of such constructions as those in 41) and 42) is that *anda*, placed immediately before the spot occupied by the verb were it stated, comes to occupy the final position of the clause. As such, taking the position normally held by the verb and modifying the underlying verb, these preverbs represent the closest items in the surface sentences to the verb. They have, in short, moved somewhat further along the *noun* > *postposition* > *adverb* > *preverb* "squish" proposed earlier, going so far as to take

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on additional verb-associated characteristics, as a result of its association as a preverb with the main verb of the clause.

11.0. The Collapse of the Old Hittite Distinction Between the Two Groups of Auxiliaries

11.1. The *anda/andan* dyad

With criteria for determining preverb, postposition, and adverb settled upon, an especially noteworthy feature of the Middle Hittite auxiliaries evident from these results is the collapse of the Old Hittite distinction between the two case forms evident in the groupings listed by Starke, mentioned previously. By Middle Hittite, the separation between the functions of these two sets has become rather badly eroded. In the case of the *anda(n)* pair, the directive form has essentially taken over the functions of both members. Indeed, Salisbury suggests that *andan,* not notably frequent even in Old Hittite, "dropped out of usage completely during or by the end of Middle Hittite" (forthcoming:14). It does appear in a few sentences in the *Laws,* all but once in variant portions of the text, and five times in the recently-published *Epos,* but it is significantly rarer than *anda.* On the other hand, *anda* remains common as a preverb, not only with an overt directional sense:

43) *na-aš an-da pa-iz-zi*

'Then they converge.' (*RB.* II.27.63; Güterbock and van den Hout trans.);

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or with such a sense more by implication:

44) *nu-uš-ši* ^{LÚ}ŠÀ.TAM ^{GIŠ}BAN *ḫu-it-ti-an* <u>*an-da*</u>*-ma-at-kán* ^{KUŠ}*pár-du-ug-ga-ni tar-na-an.*

'And the quartermaster gives him a stringed bow in a bow-case.'
(*RB.* II.22.39-40; Güterbock and van den Hout trans.);

or even in a rather abstract sense, adessive rather than the allative connotation expected with this item:

45) *a-aš-ka-az-ma ku-iš nu-uš-ša-an* LÚ.MEŠ.ŠUKUR.GUŠKIN <u>*an-da*</u> *a-ra-an-ta.*

'But at the wall which is on the outside, gold-spear-men are standing.'
(*RB.* I.3.16; Güterbock and van den Hout trans.).

In short, as Salisbury points out, the semantic range of *anda* as a preverb appears to have expanded (13).

This broadening of scope for *anda* in fact extends to the locatival or stative postpositional function vacated by the near-disappearance of *andan*:

46) *ki-i-ya-kán* ^{UZU}SA *ma-aḫ-ḫa-an ḫa-aš-ši-i an-da ḫu-ur-ša-ak-ni-e-it-ta.*

'Who braises these sinews on the hearth.' (*ME.* Vs. II.9-10)

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47) LIL-*ma-aš-si* A.SA *ḫar-sa-ú-na-aš ú-el-lu-aš an-da ú-el-ku-ua-an li-e ḫu-ua-a-i.*

'In fields, tilled soil, and meadows let food not grow for him.' (*ME.* Vs. II.39-40);

In Middle Hittite, *anda* is found as a postposition about as often as such familiar postposed lexemes as *píran* and *šēr.* Thus all traces of a demarcation in usage according to the desinences of the old nominal paradigm have been lost for *anda.* The word has progressed further along the cline or squish: *noun* > *postposition* > *adverb* > *preverb* by shedding this essential nominal characteristic. The fact that *anda* has become more prominent as a postposition, more to the left on this pathway, should not be misread: The more-common Old Hittite postpositions are more disposed than *anda* of this period to display nominal features, including governing genitive complements and, in some cases, enclitic possessive pronouns. Such noun-like qualities are missing in *anda* as seen in Middle Hittite.

11.2. *katta/kattan*

Developments in the *anda(n)* dyad, however, cannot be regarded as representative of the auxiliaries as a whole. As Salisbury observes, the eclipse of *andan* in Middle Hittite is not paralleled in the other pairs, and "This suggests

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that each pair of these adverbs has its own history and will have to be studied independently of the others" (16). Nevertheless, the evidence of the *Laws* texts shows a growing indistinctness in other pairs as well. While the *Laws* are for this effort considered an Old Hittite text, inasmuch as the principal text dates from the Older period of the language, many additional copies, usually fragmentary, dating from the Later period have also been discovered. As such, minor errors in transcription may sometimes illustrate differences in usage that have arisen during the intervening age of Middle Hittite. One such example maybe be found between *katta* and *kattan*. For example, where a verb occurs with a preverb *katta:*

48) *ták-ku ar-nu-ua-la-an ku-iš-ki kat-ta še-eš-ki-iz-zi.*
'If anyone is regularly cohabiting with an *arnuwalas*-woman...'
(*HL.* II.86a.25-6; trans. Hoffner 153)

a later variant (see Hoffner 297) reading with *kattan* is also attested, so that corresponding to 48) is:

49) *ták-ku ar-nu-ua-la-an kat-ta-an ku-is-ki se-es-ki-iz-zi.*
(*HG.* KUB XIII 14 II.1; Friedrich 86 n.7)

In both cases, the meaning of the preverb + verb complex is the same regardless of whether *katta* or *kattan* is employed. Such a change strongly

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suggests that *kattan* has become acceptable, either as an alternative or even as a replacement to *katta*, as an auxiliary to express this action during the intervening time. Here then, within the same body of laws, and expressing the same concepts, it is possible to find both *katta* and *kattan* employed in identical environments during different stages of the language.

11.3. *šara/šēr* and *parā/píran*

The most ubiquitous postpositions of the Old Hittite period, *šēr* and *píran*, still appear in these Middle Hittite texts in this same capacity somewhat frequently, but less often than they function as preverbs. In the older documents, *píran* is found as a preverb only once, and *šēr* not at all. In the more recent set, on the other hand, they are associated with verbs more frequently. A particularly striking attestation of *šēr* as a preverb is found in the *Laws;* at KBo II.47.20, the older text includes *šarā*:

50) *tá̀k-ku* PA_5*-an* EGIR*-iz-zi-az ku-is-ki sa-ra-a na-a-i.*

'If anyone upstream (?) partially (?) diverts an irrigation ditch...'

(*HL.* KBo II.47.20; trans. Hoffner 129)

There is also a Later Hittite variant reading, at KBo VI 15.7, offering rather: *še-ir da-a-ai* 'takes [it] upwards'. Thus *ír* may be observed conveying virtually the

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same sense as its erstwhile directive counterpart, and the textual variants dating from different stages of the language reflect a change in usage. Both *šēr* and *píran* in these later texts show an increasing distance from their original nominal moorings, modifying verbs either directly or indirectly far more often in Middle than in Old Hittite.

12.0. Further Losses of Nominal Characteristics

Of course, *šēr* and *píran* do, as mentioned, still serve fairly often as postpositions. Here, too, however, they have shed most of the characteristics of their earlier nominal status. Most significantly, they no longer govern nouns in the genitive case, taking a dative-locative complement instead:

51) *ki-nu-un-ma-aš* ^{D}IM-*aš* ^{M}*Me-e-ga-i tu-uk pí-ra-an ša-ra-a ar-ta-ri.*
'Now the Weather god stands upright before you, Megi.'
(*EF.* KBo XXXII 18 Vs. II.9-10; Neu 379);

52) URU-*ri še-er ḫa-as-túl*$^{ḪI.A}$ *pé-eš-ši-e-et.*
'He condemned the transgressions in favor of the city.'
(*EF.* KBo XXXII 15 Rs. III.20; Neu 297)

In 52) may be observed as well *šer* in a much more abstract sense than is normally expected in Old Hittite texts; as Neu remarks, "Dieser Umstand scheint dafür zu sprechen, daß man *šer* hier nicht konkret räumlich, sondern übertragen

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in der Bedeutung „für, zugunsten, um... willen, im Interesse von" zu verstehen hat." (367). Moreover, even where the form of the postpositional complement could be construed as a genitive:

53) *na-aš-kán i-na-na-aš še-ir ar-ḫa pár-ši-ia-ad-da-ru,* (*ME* Vs.I.44)

the text's editor, Oettinger, is at pains to point out, "*inanaš šer* wird man lieber mit dem (unüblichen) Plural (Dativ) von *inan-* rechnen als mit einem Genitiv (Sg.) dieses Wortes bei Postposition *šēr*" (29). As such, the sentence is to be rendered, 'Let him be completely broken as a result of sicknesses'. Indeed, the Sumerogram fortuitously serves to disambiguate the ending *-aš* in the following:

54) *nu* DINGIR^{MEŠ}*-aš pí-ra-an li-in-kat-ta.*
'Now he has sworn before the gods.' (*ME* Rs. III.15)

Thus, while in Old Hittite, the words functioning most often as postpositions regularly take a genitive object, one finds a transitional period exemplified in the *Erlaß Telepinus* where the noun in such constructions wavers between the genitive and the dative-locative; then, in Middle Hittite, the dative-locative has won out among these postpositions. Presumably, the original nominal affiliations of these items have by this time been entirely forgotten. Consequently, the genitive case, representing as it does a noun modifying another noun, would no longer appear appropriate to convey the relationships

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conveyed by these adpositions. Instead, such relationships have come to be seen primarily as those of direction and spatial orientation, and therefore within the sphere of the dative-locative. These lexemes are thus by the time of Middle Hittite truly locative auxiliaries, having been bleached of virtually all their noun-like properties.

12.1. *katte/kattan: iuar*

A small exception to this de-nominalizing tendency for the most prominent Hittite adpositions is found in the *Instructions for the Royal Bodyguard* and the *Maşat Letters*. Most striking in the *Maşat Letters* is the ubiquity of *katte* and even *kattan* as adpositions with enclitic possessive pronouns suffixed. *Katte* is employed in this fashion rather commonly in the older stage of the language, and in the *Maşat Letters*, *kat-ti-mi* 'with me' occurs seven times, with eleven attestations for *kat-ti-ti* 'with you'. The *Instructions for the Royal Bodyguard* includes six instances of *kat-ti-iš-mi* 'next to them, with them'. One thus finds in this corpus a remarkable archaism calling for additional investigation. Even more striking is the use of these enclitics with *kattan*, a construction listed by Puhvel as found only in these letters (1997:4.129): *kat-ta-an-ši* 'with him' (91 Rs. 15), *kat-ta-aš-ma-aš* 'with you (pl.)' (57 Vs. 6). With this particular lexeme, the more archaic construction is thus extended in its application in comparison with

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the Old Hittite texts. This unusual construction demands more detailed inquiry, as to whether the semantic properties of this item led to this seemingly reacquired nominal propery, or, as seems at first glance more likely, *kattan* was selected for these enclitics based on an analogy with other adpositionis in *-an* known from older texts.

One other postposition taking a genitive complement, somewhat widely attested in Middle Hittite, is *iyar* 'like, as'. It is found seven times in the *Eide,* including:

55) *na-as* ^{UZU}SA-*aš i-ya-ar ḫur-ša-ak-ni-ia-ad-da-ru* MUN-*aš-ma i-ya-ar pár-ši-it-ta-ru.*

'And let him be braised like the sinew, and let him be crushed like the salt.'
(*ME.* Vs. II.14-16)

It also occurs ten times in the *Epos der Freilassung,* as in:

56) ^{URU}*E-eb-la-aš* URU-*aš kat-te-e-ra-aš ya-aḫ-nu-eš-šar ar-ḫa* ^{DUG}*te-eš-šum-mi-ia-aš i-ya-ar du-ya-ar-na-aḫ-ḫi ša-ra-a-az-zi-ia-ša ya-aḫ-nu-e-eš-šar ar-ḫa ḫu-uš-ši-li-ia-aš i-ya-ar ša-ak-ku-ri-e-mi.*

'The walls of the lower city of Ebla I completely break like a drinking-glass; the walls of the upper city of Ebla I trample down like a trash heap.'
(*EF.* KBo XXXII 19 Vs. II.27-31; Neu 383)

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It even appears once in the old Hittite texts considered for this effort, in the relatively late *Erlaß Telepinus*:

57) *du-ud-du-mi-li-ma* ^{M}*Zu-ru-ḫa-aš* ^{M}*Da-a-nu-ḫa-aš* ^{M}*Ta-ḫur-ḫa-i-li-ḫa-aš*
^{M}*Ta-ru-uḫ-šu-uš-ša i-ḫa-ar li-e ku-na-an-zi.*
'But in secret, like Zuru, Danuwa, Tahurwaili, and Tarhusa they do not die.'
(*Tel.* Vs. II. 52-3)

Semantically, *iḫar* is a somewhat unusual adposition. Such lexemes generally convey some connotation of directional or spatial orientation, or at least they did so before evolving some more abstract meaning. Here, however, there is no indication of any such locational sense for *iḫar*, either as attested or in its past. Consequently, Neu prefers to term *iḫar* a "Vergleichspartikel" (266), although Oettinger, Tischler and Puhvel see it as a postposition (105; 452; 1991:499, respectively). Puhvel derives *iḫar* from the verb *iya* 'do, make': "the verbal noun *iya(u)war* is well attested, and one need not postulate a fossil, *iwar* being a 'frozen' reduction form of *iyawar*... The basic sense of gen. + *iwar* is 'in somebody's (manner of) doing' (1991:501). H. Kronasser, on the other hand, speculates on the root of *iḫar*, "könnte ursprünglich ,*Gang' (von **i-* ,gehen') gewesen sein" (1962:298). Tischler, too, sees the verbal root **i-* as a possible point of origin for this item, but also discusses the alternative that it should be

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connected with the **i-* of *pai-* 'give' (452-3). Puhvel sees the productivity of this genitive construction as analogous to Latin *aequoris instar* 'like the surface of the sea' or Greek δέμας πυρὸς 'like fire' (1991:502), and indeed Kronasser too remarks on the similarity with the Latin construction (273). Like *instar*, then, although *iyar* no longer declines, it would still have been recognizable as a noun.

Indeed, the distribution of *iyar* is substantial enough that it may be speculated to be of still recent derivation in Middle Hittite, given that it still retains many characteristics of its erstwhile nominal status. However, as a result of its unusual semantic sense, *iyar* shows an eccentric course of development for a Hittite adposition; it is not found additionally as an adverb or preverb, but as a conjunction. In this regard, according to Puhvel, "The postpositional usage clearly antedates the conjunctional one; the transition was effected by attraction to conjunctions such as *man* 'as, like'" (1991:500). This subsequent reanalysis into a more abstract item does nevertheless parallel that of the locational auxiliaries under discussion. In any event, *iyar* has clearly not progressed as far along its cline as have *šēr, píran*, and other postpositions prominent from the time of Old Hittite.

13.0. A Note on *Das hurritische Epos der Freilassung I*

A Middle Hittite text deserving of particular comment is *Das hurritische Epos der Freilassung I*, an anomalous work indeed. The Hittite texts here are very

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close translations of the Hurrian original, and as a result some rather basic constraints of Hittite word order are violated. Most jarringly, the verb may not necessarily appear sentence-finally, in contravention of all expected standards of the language:

58) *pa-ra-a-an-kán ḫu-it-ti-at* UDUN-*ni-ia-az.*
'He took it out from the oven.' (KBo XXXII 14 Rs. III.10; Neu 85).

59) *nu* SIG_7-*an* GU_4^{HLA}-*un ha-at-te-eš* ^{D}IM-*un-ni šal-la-i pí-ra-an.*
'And the thousand oxen lay slaughtered before the great Weather god.'
(KBo XXXII 13 Vs. II.15-16; Neu 223)

Given even the presence of such sentences in the Hittite translation of the *Epos*, let alone their frequency, it is highly dubious whether this text may be considered representative of the standard written language. At the same time, however, these occasional unexpected formations beg the question of how representative of the Hittite language as actually used among its speech communities the written language as it is known to us truly was. Neu, while hopeful that the bilingual text may shed some light on the extent to which the Hittite *Ortsbezugpartikeln* were authentically Indo-European (14-5), is primarily interested in what the documents reveal of the heretofore obscure Hurrian language (12-3). Nevertheless, the implications of the Hittite syntax in the *Epos* most certainly calls for more intensive study.

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14.0. *menaḫḫanda*

One additional item of note concerns an item occurring once in the *Maşat Letters* as a preverb:

60) *na-an-za-kán me-na-aḫ-ḫa-an-da a-ú.*
'Look towards them!' (2 u. Rd. 9)

And once in the *Instructions for the Royal Bodyguard* as a postposition:

61) 2 LÚ.MES.SUKUR.DUGUD-*ma-kán* LUGAL-*i me-na-aḫ-ḫa-an-da* ZAG-*az a-ra-an-ta.*
'But two high-ranking spearmen stand opposite the king on the right.' (*RB*.II.17.9)

This auxiliary, like *parā, anda*, and so on, may also function as a postposition or an adverb, as Güterbock and Hoffner list (1989:274). Its shape is of course not identical to those of the two-syllable items discussed over most of this chapter, but some similarities are present. Most striking is the desinence in *-a;* like *parā* and the others with this ending, *menaḫḫanda* is traced by Puhvel to a noun in the directive case (1991:94). Unlike its directive counterparts, however, the word is derived from a compound whose second member is *ḫanda* "'(to the) fore'" (Puhvel 1991:94). The first part, on the other hand, appears to be a fossilized form of *meni/a* 'face, cheek' (Güterbock and Hoffner 1989:289), and Starke

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reconstructs a phrase "**menan* (Pl. G. + **ḫanda*" (1977:191f; 1990:131 n.393). Taken as a whole, then, *menaḫḫanda* carries a sense of "'facing, towards, vis-à-vis' (lit. 'face-to-fore')" (Puhvel 1991:94). The compound thus attests to the antiquity of the Hittite postpositional word-order, representing as it does a frozen adpositional phrase.

15.0. Conclusion

By the Middle stage of the language, then the reanalysis of these former nouns as auxiliaries capable only of filling postpositional, adverbial, or preverbial roles is virtually complete. Apart from the relatively new *iuar*, genitive complements for postpositions are unknown in Middle Hittite, and the suffixation of pronominal enclitics for such items very nearly so. The traces of a nominal paradigm found in *parā* and *píran*, *šarā* and *šēr*, and so on, are largely obscured by this time. The features of the Old Hittite adpositional system are found at this stage only in lexemes whose development follows along fairly parallel paths but beginning later in time The overall course nevertheless appears remarkably similar for all these lexemes, and the cline *noun* > *postposition* > *adverb* > *preverb* appears consistent for Hittite.

As will be seen in subsequent chapters, however, most Indo-European languages follow a somwhat different path in the derivation of their adpositions. Luwian, to be discussed in Chapter III, also bears signs of an immediate source among its substatives for its locative auxiliaries. But outside the Anatolian group,

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these segments tend instread to be drawn from adverbs with a local sense. In Old Hittite, however, the nominal characteristics of these segments are unmistakable. There is moreover a clear distinction in this stage of the language between forms more likely to be employed as postpositions and those more likely to serve as preverbs, and this distinction is made along lines corresponding to nominal case morphology. Only later do the former set lose their nominal qualities and occur as preverbs more often than they do as adpositions. This, then, is the course of development that unfolds among the Hittite local words between its Old and Middle stages.

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Chapter III:
The Emergence of Prepositions in Southern Anatolian

1.0. Introduction

An examination of adpositional functions or indeed of syntactic operations in general for Later Hittite offers rather less insight than those of the Old and Middle stages of the language into the development of a vital, evolving speech. By the time the Late Hittite documents were set down, the language had ceased to be spoken among the population of the kingdom at large. Instead, Luwian, whose speakers had begun to spread across the Anatolian Peninsula quite early in the history of the Nesite, or Hittite, Kingdom, appears to have superseded Hittite except in official documents. As E. Laroche observes, "On a supposé que la langue hittite étant sortie d'usage dès le XIVe siècle, le louvite l'aurait remplacée sur toute l'étendue du territoire" (1959:13). Nevertheless, the Later period, that is the thirteenth century BC, presents a great volume of written Hittite addressing a wide variety of state and religious functions. Even here, however, the growing influence and increasing presence of the Luwian language is apparent, most obviously in ritual and magical texts, which often include extensive passages in Luwian:

> ... [D]ie k[eilschrift-luw[ischen] Texte erst im 14. und 13. Jh. abgefaßt worden sind... Sie konnte dies um so eher, als der Einfluß des K.-Luw. auf des Heth. gerade während dieser Zeit beachtlich zunimmt und deshalb einen unmittelbaren Zusammenhang mit dem Einsetzen einer k.-luw.

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Ritual- und Beschwörungsliteratur nahezulegen scheint (F. Starke 1985:30).

Therefore, even with these Luwian passages representing only a small fraction of the attested literature of that time and place[1], and appearing in extremely fragmentary and disconnected form, a far more fruitful avenue of investigation into the development of adpositional usages in the Anatolian languages presents itself here, the more so given that the surviving representatives of this linguistic group are much more closely affiliated with Luwian than with Hittite. And in these relatively few texts, some remarkable features new to the Anatolian languages appear. By far the most striking of these is the occurrence of prepositions, which are, as established, unknown in Hittite. These prepositions evidently correlate with an increasing flexibility of word order, whereby the sentence-final position of the verb, while still common, is no longer as obligatory as in Hittite.

These developments represent a precursor to the syntactic characteristics of the later Anatolian tongues, particularly Lycian. In Lycian, the successor to Luwian across Southern Anatolia, sentential word order is by no means rigid, but the verb almost never comes at the end of the sentence. More directly relevant to the present study, postpositions are evidently unknown; the syntax of such phrases has become entirely prepositional. However, with the earliest-attested step in this course of word-order change appearing in Luwian, this is the

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[1] Of course, our knowledge of Luwian is greatly supplemented by the words in Hittite texts marked by a *Glossenkeil*. Nevertheless, phrases at the least are necessary for the purposes of the present study.

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best starting-point for an investigation of such events in this sub-family.

1.1. Previous Scholarship

Research on the Southern Anatolian languages has in most respects not advanced to the extent that Hittitological studies have. Until quite recently, discussion of some questions, including those on the functions of the local adverbs, was sporadic and informed in large measure by what was already known of their Hittite counterparts. The study of Luwian and its descendants may be considered to have come into its own in the middle of the previous century, with Laroche's *Dictionnaire de la langue louvite* (1959), which includes texts and a brief grammatical sketch, along with his series concerning "Comparaison du louvite et du lycien"; the 1957 installment offers the first substantial examination of these auxiliaries. Not long after, P. Houwink ten Cate's monumental *The Luwian Population Groups of Lycia and Cilicia Aspera during the Hellenistic Period* (1965) presented the first modern treatment concentrating on Lycian grammar, along with thoroughly edited texts.

Perhaps the best overview of the locative auxiliaries throughout the Anatolian languages may be found in P. Meriggi's "Schizzo grammaticale dell'anatolico" (1980). Starke's *Die keilschrift-luwischen Texte in Umschrift* (1985) collated all the known Cuneiform Luwian texts into one volume. While much research in past decades has been occupied with laying the groundwork for an understanding of the phonological and morphological workings of these languages, a more recent and most welcome branching out is highlighted by A.

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Garrett's "Topics in Lycian Syntax" (1992). At about the same time, H. C. Melchert almost singlehandedly brought a quantum leap to the general understanding of the field with his dictionaries of Cuneiform Luwian (1993a) and Lycian (1993b), followed by his *Anatolian Historical Phonology* (1994).

Recent years have also seen great advances in our understanding of Hieroglyphic Luwian, underscored by M. Marazzi's *Il geroglifico anatolico - Problemi di analisi e prospettive di ricerca* (1990). Extensive work has also been published by T. Oshiro; most relevant to the present effort are "The Allative and Locative Adverbs in Hieroglyphic Luwian" (1988a), "*api* in Hieroglyphic Luwian" (1988b), and "The Spatio-Temporal Adverbs in Hieroglyphic Luwian" (1995). J. Hawkins, too, has made major contributions, culminating in his *Corpus of Hieroglyphic Luwian Inscriptions. Vol. 1: Inscriptions of the Iron Age* (2000). These, then, are the works that may productively be consulted in investigating adpositions and preverbs in Luwian and Lycian.

2.0. Frequency of the Locative Auxiliaries in the Primary Texts

The texts consulted for this investigation are those presented in Starke 1985, credited by Melchert (1993a:i) with establishing a definitive Cuneiform Luwian corpus; particular attention, however, is afforded the extended passages presented by Laroche due to their completeness and clarity. The Luwian texts in Starke's volume are more scattered and less substantial than the Old and Middle Hittite documents consulted in Chapter II, and the occurrences of lexemes that, as in Hittite, may function as adposition, preverb, or adverb are correspondingly

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few by comparison. Still, the number of attestations of each of these words in the aforementioned functions, as well as occasions where the item is attested but without adequate context to determine its role, are given below as classified by Melchert:

	Postp.	*Prep.*	*Prevb.*	*Advb.*	*Indet.*	*[Total]*
annan 'below, under'	-	2	5	4	-	11
anta 'in(to)'	-	2	38	3	-	41
andan 'inside (?)'	-	-	-	-	1	1
appa 'back, again'	-	-	16	-	-	16
appan 'behind, after'	2	-	5	-	-	7
appanda 'behind'	-	-	1	-	-	1
parran 'before, in front'	7	-	1	-	2	10
pari 'forth, away'	-	-	10	-	1	11
pariyan 'beyond'	1	-	-	3	-	4
šarra 'on, upon'	-	2	-	2	6	10
šarri 'above, up'	2	2	1	6	-	11
tawiyan 'facing, towards'	1	-	-	1	1	3

3.0. Luwian Locative Auxiliaries Compared with those of Hittite

To the observer familiar with Hittite, forms corresponding to *katta(n)* and *arḫa* are conspicuously absent from the above table. However, Meriggi notes that *arḫa* is common in Hieroglyphic Luwian (1980:369), and J. Puhvel lists *kata*

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as a preverb in the Hieroglyphic language (1984:4:129). It is possible, however, that these may be borrowings from Hittite. More precisely, the preverb is there represented as SUB-*ta*, corresponding in directionality to the postposition SUB-*nan*, or *annan* in Cuneiform Luwian (Laroche 1957:181; the situation in Hieroglyphic Luwian will be discussed at greater length in Section 6 below). In *annan* one therefore finds a lexical innovation on the part of Luwian which partly but not completely supplants *katta(n)*. This information, however, also underscores the incompleteness of the Cuneiform Luwian records as we have them, and consequently the necessarily tentative nature of any conclusions to be drawn from them.

As is the case with the corresponding lexemes in Hittite, a distinction may be observed between items functioning overwhelmingly often as preverbs but seldom or never as adpositions on the one hand, and on the other those occurring more regularly as adpositions while serving as adverbs and preverbs as well. This distinction, however, is not as neatly drawn as in Old Hittite. The Old Hittite predilection for preverbs derived from the directive case in *-a* appears to be reflected in *anta* and *appa*, but the pattern does not hold up so well with *šarra*, which, where its role is discernible, is as likely to be a preposition as a preverb. Postpositions drawn from adverbial accusative forms may be seen in *parran* and, to a lesser extent, *appan*, *annan*, and possibly *andan*, but there is no exact Luwian correspondent to Hittite *šēr*, as will be elaborated below. Both

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pariyan and *tawiyan* have the look of derived forms, the latter from *tawa/i-* 'eye', and indeed Melchert traces it to an adjective **tawi(ya)* 'of the face' (1993:225). In short, one finds traces of an earlier system based on a directive-accusative distinction obscured by subsequent developments, very similar to the situation in Middle Hittite.

3.1. Reflexes of Nominal Cases in Luwian Local Adverbs

Indeed, over ninety percent of the attested Cuneiform Luwian texts are dated from the fourteenth and thirteenth centuries BC, roughly contemporary with the later Hittite period (Starke 27-29). And, similarly to Hittite after the Older stage of that language, any demarcation between the former nominal cases of the local adverbs has become badly blurred. Moreover, Luwian does not in any event appear to have developed precisely the same sets of auxiliaries delineated for Hittite by Starke 1977 (131). One finds, as in Hittite, forms reflecting the directive in *-a* and the ending in *-an*, be it a reflex of the accusative, as argued in Chapter II., or of an endingless locative.

However, the forms *pari* and *šarri* indicate that the locative in *-i* is significantly more prominent in the Luwian auxiliary system than in that of Hittite, where it is limited to *katti*, found only with reflexive possessive pronouns. Some scholars, such as Melchert (1994:138), have been led in part by this *šarri* to postulate that Hittite *šēr* also reflects a locative form whose final **-i*

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was lost. It is, however, not a fruitful avenue of inquiry to rely on exact matches between the Hittite and Luwian systems. The form *pari* establishes that Luwian has drawn more extensively from locative case forms. Furthermore, the presence of directive *šarra* and locative *šarri* from one root; locative *pari* and accusative *parran* from another; and directive *appa* and accusative *appan* from a third, demonstrates that an inventory of local adverbs along two axes, as in Hittite, likely never prevailed in Luwian. The languages are both Anatolian, fairly contemporary with one another, and typologically very similar. But they are not so closely related that they should be expected to mirror one another exactly. And since we do not have any substantial evidence of a Luwian stage corresponding to that of Old Hittite, where adpositions still show significant nominal characteristics, we are able only to conclude based on the evidence known to us that the functioning of these local adverbs in such a putative stage of Luwian included some notable differences from that of Old Hittite.

3.2. Case Government by Postpositions

A further similarity of the adpositions in these documents with those of Middle Hittite is the case most commonly governed. As overwhelmingly most often holds in that stage of Hittite, these items too generally take complements with dative/locative desinences:

1) ^{D}UTU-*ya-az* ^{D}*Kam-ru-se-pa-i da-u-e-ja-an ma-am-ma-an-na-at-ta.*
'The god looked toward divine Kamrusepa with favor.' (KUB XXXV 107

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iii 8)

2) *a-ta a-ap-pa* DINGIR^{MEŠ}*-an-za par-ra-an ni-is a-ú-i-ti.*
'And it does not come again before the gods.' (KUB XXXV 54 iii 38)

3) *a-ti a-pí* EGIR-*an tar-si-is* AN.BAR[-.
'And the iron object on the [a-pí]...' (KBo IX 145.9)

Each of these postpositions, then, functions as do their counterparts in Middle Hittite. Starke for his part argues that an archaism is to be found where the Akkadogram *ŠA* is employed in conjunction with *parran:*

> im Ausdruck *ŠA* EN SISKUR.SISKUR *pár-ra-an*, vor dem Ritualherrn' KUB XXXV 54... ii 40, wo das Syntagma Sg. D. *malḫassassan* EN-*ya*... graphisch als Genitiv zum Adverb *parran* erscheint, was an die altheth. Konstruktion, lokativischer Adverbien mit dem Genitiv (z.B. LUGAL-*uas peran*, vor dem König') erinnert (1990:55).

However, he appears in this instance to misinterpret the phrase boundary so as to omit the preceding dative/locative DINGIR^{MEŠ}*-an-za* (in line 39), in contrast with the more accurate rendering offered by Laroche, "devant les dieux du maître du rituel" (1959:155). The government of the phrase is thus similar to an English construction such as, "a study of Los Angeles' languages", where the complement of the possessive preposition *of* is separated by an additional

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possessive form.

A rather more securely archaic construction, on the other hand, may be observed at KBo XIII 260 iii 12:

4) *pa-ri-ia-am-ša-at-ta tar-za-an-du.*
'And let them [do something] beyond it.'

Here, the enclitic possessive pronoun *-sa* is affixed to the adposition *pariyan*, as if to convey the idea 'over its beyond', analogously to the construction common in Old Hittite with adpositions recently derived from nouns and still retaining many nominal properties. This formation is much rarer in the middle stage of the language, but one may hypothesize that, as with Middle Hittite *kaššaš* 'in lieu of', where, in contrast with most other postpositions of the time in that language, the enclitic possessive is common, the reanalysis of *pariyan* as an adposition took place relatively recently.

3.2.1. *pariyan*

However, a terminological question presents itself with *pariyan*. Despite the clear similarity with the aforementioned Hittite forms, where all adpositional lexemes are considered postpositions even where an enclitic complement is suffixed, Melchert lists *pariyan* as found in 4) as a preposition (1993a:169). This is in fact the only known locus where *pariyan* is clearly functioning as an

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adposition; Melchert finds three other attestations, which he lists as adverbs. Two of these as it happens follow dative/locative substantives. One (KUB XXIX 43, 6) is in a fragmentary context, but the other is just a few lines below the passage cited in 4):

5) *a-at-ta a-ad-du-ɥa-an-za pa-ri-ɟa-an ad-du-ua-li-ia-an ɥa-at-ta-ni-ɟa-an up-pa-an-na-an-du.* (KBo XIII 260 iii 13-15)

A possible rendering might be, 'And they shall continue to bring a (?) of evil beyond evils', although the unknown sense of *wattani(ya)* is certainly a complicating factor. But a postpositional interpretation for this passage would appear to be at least possible. On the other hand, Melchert's reading has the advantage of being consistent with similar combinations of locative auxiliaries and proclitic particles that, as will be discussed below, are taken by Garrett (1992) as prepositions and their objects. In any case, while it is certain that *pariyan* has adpositional properties, the word is also illustrative of the difficulties of attempting a grammatical analysis based on such a limited corpus.

4.0. Prepositions in Luwian

Nevertheless, there remains an exceptionally striking contrast between the Luwian adpositional inventory and that of Middle Hittite: in the Cuneiform Luwian texts are also found some half-dozen prepositions. Like the postpositions just examined, these prepositions also take dative/locative

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complements:

6) *a-du-ya-an an-na-a-an pa-a-ta-an-za du-ú-ya-an-du.*

'And let them put it below the feet.' (KUB XXXV 39 iii 29)

7) *šar-ra za-a-ti-i.*

'on this' (fragmentary context, KUB XXXV 14 i 19; KUB XXXII 8 + 5 iv 22)

8) *a-ri-ya-ad-da-li-is* ^{D}ISKUR-*an-za šar-ri ka-ši-i ḫu-u-e-ḫu-u-i-ya*

tap-pa-aš-ša-i šar-ri ti-ya-am-mi ḫu-i-ḫu-i-ya.

'Storm-god of the mountain, run above [something],

Run in heaven above the earth.' (KUB VII 53 + XII 58 i 58-59)

Melchert's classification of these items as prepositions is not entirely without controversy. Meriggi for his part views such a typologically improbable phenomenon with a very skeptical eye. For the two instances given of *šarri* it would appear that he follows Rosenkranz and Otten in seeing an adverbial function (376), although in such an eventuality a reading of the locative *tiyami* as 'above the earth' rather than 'on the earth' becomes difficult. And although it is true that the two phrases *šarra zati* both occur "fra due lacune" (373), the identical joining of *šarra* with the dative-locative demonstrative, at the end of a line of text in the first example and at the beginning in the other, is too attractive to ignore not only for Melchert but for Laroche (1957:158). In any event,

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Meriggi, too, agrees that the phrase *annan patanza* is an "esempio decisivo" (372), so that there may be some dispute as to the frequency of prepositions in Luwian , but not the fact of their presence.

4.1. Ambiguous Instances

Further complicating matters, moreover, is the possible attestation of *šarri* as a postposition also, at least according to Melchert.

9) *ua-n]a-ti-ia-ti-ia šar-ri ma-am-ma-al-hu-un-ni*

]x-du ua-na-ti-ia-ti-ia har-ri

ma-am-ma-al-hu-un- ni.

'We break [them?] above the feminine one... [?] we break [them?] above the feminine one.' (KBo XXIX 16 ii 3-5)

Melchert lists the form *wanatiyatiya* as a nominative-accusative inanimate adjective, which would make a reading of *šarri* as a postposition highly unlikely on its face. However, it is difficult to be certain about any facet of a word when it is found only in a rather incomplete context such as this. If, on the other hand,*wanatiyatiya* is an *i*-stem noun or adjective, it then becomes at least possible to construe an ending in *-iya* as a dative-locative singular and as such a plausible adpositional object (see Meriggi 291).

Similarly ambiguous is the other purported attestation of *annan* as a

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preposition:

10) *za-an-ni-in* KASKAL-*an ku-i-in a-at-ti ku-i-iš ni-mi-ia an-ni an-na-an i-li du-du-ma-ni-ša-an.* (KBo IV 11.43-44)

The crux of the problem here is that the words on either side of *annan, nimiyanni* and *ili*, both look as if they could be dative-locative substantives but are otherwise unknown, as for that matter is *atti*. The final verb is a plural imperative with the sense, as suggested by Melchert, of 'have mercy on', and the first two words may be construed as 'this path' (accusative singular), but otherwise the sentence is quite obscure. With no means of determining whether *annan* in this situation governs *ili*, as indicated by Melchert, or *nimiyanni*, as proposed by Meriggi (372), it is impossible to state with certainty whether *annan* here is a preposition or a postposition.

In any event, one is left with at least the possibility that Luwian, in addition to possessing both postpositions and prepositions, may also feature adpositions that may be either postposed or preposed; if so, however, their status as adpositions is most likely not completely cemented. Looking ahead, one finds that the reflexes of *šarri* and *annan* appear in Lycian only as prepositions. More strikingly, of the Luwian adpositions tabulated above, only those found as prepositions offer reflexes in Lycian as (preposed) adpositions.

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4.2. Conclusions on Postpositions and Prepositions in Cuneiform Luwian

One finds in Luwian itself, then, an adpositional system quite similar to that of Middle Hittite, apart from a pronounced tendency towards preposing. The question of whether postpositions or prepositions are predominant in Luwian is less clear-cut. Five lexemes are identified by Melchert as prepositions, including *pariyan* as discussed above; and four are listed as postpositions, including the problematic *šarri*, which arguably takes either position. Twelve postpositional loci are found in the texts consulted here, as opposed to nine that are prepositional. However, the balance is tipped by *parran*, whose seven postposed attestations make it the most frequently found Luwian adposition. Nevertheless, even if one is hard-pressed to describe Cuneiform Luwian definitively as a postpositional or a prepositional language, its future direction is clear; the prepositions eventually prevail.

5.0. Luwian Sentential Word Order

Concurrent with this syntactic feature unexpected in the Anatolian languages that begins to appear in Luwian is another: in contrast with virtually all of Hittite literature, with exceptions occurring only in such heavily calqued texts as the *Epos der Freilassung I*, a sentence-final position for the verb in a declarative sentence is not entirely obligatory. This is so even beyond the occurrence of the imperative in initial position, found also in Hittite. As Laroche observes, the Anatolian syntactic "ordre <normal> est en réalité fréquemment

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troublé par des raisons obscures, peut-être en partie rhythmiques, surtout dans les formules de magie qui constituent l'essentiel de notre documentation actuelle" (1959:145). One common variation listed by Laroche is "le rejet emphatique du sujet après le verbe" (145). He cites KUB XXXV 88 iii 13-14 as an example:

11) *la-la-a-i-ya 9-un-za* ^{GIŠ}GAZUM-*za* ^{D}*Kam-ru-še-pa-aš.*
'Divine Kamrusipa takes nine combs.'

Laroche also mentions as a possible environment for such variation "l'asyndète: énumération de notions complémentaires ou de couples antithétiques" (145). Such an enumeration certainly appears to be a plausible explanation for the word order of KUB XXXV 43 ii 29-30, where the verb is *tappatta*:

12) *a-aḫ-ra-an ya-aḫ-ra-an tap-pa-a-at-ta ta-a-pa-ru*
ta-ta-ar-ri-ia-am-ma-an ḫi-i-ru-u-un ma-a-ia-aš-ši-in EME-*in.*
'Pain and woe he spit upon - the evil curse, the oath, the gossip of adults.'

Another term for this phenomenon is extraposition, defined by C. Watkins as the movement "of constituents to the right of the verb" in languages where the verb is expected at the end of the sentence. This process may include any sentential element; Watkins offers as an example the Homeric Οὖτιν ἐγὼ πύματον ἔδομαι μετὰ οἷς ἑτάροισιν 'Noman I will eat last among his comrades', in

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which a prepositional phrase is extraposed to follow the verb (1998:70). In Luwian, however, this syntactic feature, while common, tends rather to involve the verbal complement or direct object. Sentence 12) presents one such example; a quite lengthy extraposed direct object may be found in:

13) EN-*ia ti-ua-a-li-ia pí-ia-aš* ^{LU}KÚR^{MEŠ}-*in-zi*

kat-ta-ua-at-na-al-li-in-za ú-ut-na-aš-ši-in-za ḫi-iš-ḫi-ša-aš-ši-in-za

ta-pa-ru-ua-at-na-al-li-in-za ta-a-ta-ri-ia-am-ma-na-aš-ši-in-za ḫi-i-ru-ta-aš-ši-in-za

EN^{MEŠ}-*an-za.*

'Lord of the sun god, you have given over the enemy, the plaintiffs, the magical words, the binding spells, the evils, the curses, the oaths, the masters'. (KUB XXXV 45 ii 18-21)

In sentences such as those found in 12) and 13), the syntax calls to mind that of Lycian rather than of any stage of Hittite. It may therefore be speculated that this construction, where lengthy verbal complements are shifted to the lattermost part of the sentence, may have been either a contributing factor in the Southern Anatolian shift to verb-object (VO) sentence order and preposed adpositions, or at least a prominent example of this general drift. One thus finds that in Luwian, the change from a verb-final, postpositional syntactic structure to the pattern securely in place in Lycian is already well underway; clearly it had begun in the prehistory of the language. By the time Luwian makes itself known to us, although the verb still ordinarily takes final position, there is a flexibility

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unknown in Hittite. This flexibility correlates with the increasing tendency for locative auxiliaries, when reanalyzed as adpositions, to be preposed. It is severely tempting to suggest that the one feature created an environment favorable to the other, although Luwian at present does not offer us adequate information to make such a statement about this proposed sequence of events with great confidence.

5.1. Omission of the Verb

Another syntactic variation allowable in Luwian involving the verb is its omission, and not, as in many Indo-European languages, only in sentences with the copula:

14) *a-ta a-ap-pa za-aš-ta-an-za* DINGIR^{MEš}*-an-za*
pár-ra-an ni-iš im-ma-ra-aš-ša ^{D}ISKUR*-aš-ša-an-za*
ḫal-li-iš-ša. (KUB XXXV 54 iii 6-8)

Laroche notes that this sentence has a "syntaxe très obscure, dépendant de la nature et du sens de *ḫallissa*" (154). If one follows Melchert's suggested gloss of 'sickness, pain', one may , with the help of the preverb *appa*, infer a verb with a sense of 'bring back' : 'May the thunder-god of the open country not [bring] back sickness before these gods'.

In a sentence such as 14), the preverb, then, is the overtly stated item that comes closest to functioning as a verb. Indeed, bearing in mind the *noun* >

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adposition > *adverb* > *preverb* cline or squish proposed earlier for Hittite, *appa* comes remarkably close in this situation to moving still further along this continuum and taking on such verbal characteristics as governing a direct object. One may therefore speculate that a construction of this sort, where the verb is not overtly present but is nevertheless modified by a preverb, played a pivotal part in the increased flexibility of Luwian word order. After all, if the word functioning as a verb in many respects need not necessarily occur in a sentence-final position, then by extension, the verb itself may also migrate elsewhere in the clause.

5.2. Conclusions on Sentential Word Order

This increased variability in the ordering of such an essential element of the sentence as the verb is of interest to the present effort because, as has often been observed from Greenberg's landmark study of word-order universals and so on, the ordering of adpositions within the phrases they govern often correlates with that of verbs within the clause as a whole (see especially J. Greenberg 1966:62). And in the history of Luwian and the languages that followed it, it seems unlikely to be mere coincidence that a syntactic innovation allowing the verb to appear elsewhere in a clause than final position manifested itself more or less when prepositions also became an established feature of the language for the first time.

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6.0. Hieroglyphic Luwian

The cuneiform texts are of course not the only recorded variety of the Luwian language. Following the Hittite collapse at the beginning of the twelfth century BC, a number of successor states arose to the south-east of the former kingdom, in "the region of northern Syria" (Watkins 27). As Watkins notes, this dialect of Luwian is quite closely related to that of the Cuneiform script. The hieroglyphs suggest an adpositional system with both similarities to and differences from the cuneiform dialect that are quite intriguing. An inscription dating from the reign of Katuwas at Karkamish around the end of the second millennium BC includes the phrase POST + *RA/I-wa/i-sa-ti* 'in the future' (13+14.KARKAMIŠ A2+3.13); Hawkins renders POST + *RA/I* as *apara/i-*, which he describes as "a derived form... of uncertain morphology but clear contextual meaning" (2000:111). This *apara/i-* would appear to be a derivation from the root **appa-*, very familiar from Hittite as well as Cuneiform Luwian. Here then may appear evidence of the tendency for preposing observed in the cuneiform language, although Hawkins notes that the form "is of uncertain morphology", and V. Ivanov cites adjectives in Indo-Iranian and Gothic with parallel shapes (forthcoming:6). On the other hand, while *appan* occurs as a postposition in Cuneiform Luwian, the hieroglyphic language presents a different, derived form with a different syntax. Presumably its reanalysis as an adposition took place at another (and presumably later, given the overall history of the Anatolian

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languages) time than that of *appan*.

Other loci of the Hieroglyphic Luwian texts, however, present more extensive but contradictory evidence. Oshiro in his work finds no *apara/i-*; rather, he posits the form *api*, generally written *á-pi* and suggested in previous scholarship to be perhaps based on the inanimate singular of demonstrative pronoun *apa-* (1988b:246), as a preverb corresponding to Hittite *appa* (251-2; 1995:216). Other striking differences between the cuneiform and hieroglyphic languages appear in Oshiro's work. In the cuneiform texts, *anta* is attested twice as a preposition, but it is found as a postposition in the hieroglyphs (spelled *à-ta* as preconsonantal *n* is omitted in this variety of Luwian - Oshiro 1995:214):

15) ("MALUS") *ha-ní-ia-ta-pa-wa/i-ta-'* REL-*ia* (TERRA)*ta-sà*-REL+*ra/i* *à-ta*
á-sá-ta wá/í-ta (TERRA)*ta-sà*-REL+*ra/i ARHA [...]-ha-há.*
'and the evils, which were in the land, I drove out of the land'
(KARATEPE XII-XIII; translation by Oshiro 1995:215)

As noted in Section 3, ***kata*** evidently makes its presence felt in Hieroglyphic Luwian, but the locational form corresponding to this directive case reflex is, as in the cuneiform language, *annan*, rendered INFRA-*na-na*. But, as with *anta*, while *annan* in Cuneiform Luwian is classified by Melchert (1993a) as preposed when serving as an adposition, the hieroglyphic item is once again

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postposed:

16) ... ("PES")*pa-tà-za* SUB-*na-na* PONERE-*há.*

'I put them under my feet.' (KARATEPE XII; Oshiro 1988a:26, 1995:219)

Oshiro interprets the forms PRAE-*i* and PRAE-*na* as *pari* and *paran,* essentially the same in shape as in Cuneiform Luwian (1995:219-20), although he remarks that a difference in function between the allative and the locative is not easily discerned (221). Finally, Oshiro reads SUPER+*RA/I* as **sara* and SUPER+*RA/I-ta* as **saranta* (1988a:27-8). The latter form Oshiro derives with a suffix *-anda,* productive in Hittite as in such items as *appanda* but not otherwise known in Luwian (1988a:28-9). As such, it corresponds in function to Hittite *šēr* and Cuneiform Luwian *šarri.*

One finds, then, significant differences between the inventories of locative auxiliaries in Hieroglyphic and Cuneiform Luwian in terms of lexicon, with **kata* (assuming that this is in fact an indigenous Luwian term - see Section 3.0); derivational morphology, in **saranta* and perhaps in **apara/i-;* and in word order, with adpositional *anta* and *annan* postposed rather than preposed. These differences taken together demonstrate that Hieroglyphic Luwian, while undoubtedly related far more closely to its cuneiform counterpart than to Hittite, is nevertheless not simply the same language or a direct continuation of it.

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Rather, one may suppose that the hieroglyphic language represents an offshoot from the main course of Southern Anatolian linguistic development. In any event, the location of Karkamish and the other Hittite successor states employing Hieroglyphic Luwian in their official documents was well to the east of where Cuneiform Luwian and Lycian both flourished in southern Anatolia. As such, it cannot overshadow an examination of the course of language evolution shown by Cuneiform Luwian and Lycian.

7.0. Lycian: Introduction

While Hieroglyphic Luwian offers a continuation, more or less, of the language found slightly earlier in cuneiform texts, the Southern Anatolian linguistic branch is carried through the first millennium BC and beyond in the form of Lycian. Whether Lycian represents "a direct continuation of Luwian", as speculated by Houwink ten Cate (1965:86), or is rather descended from a dialect closely related to Luwian as it is known to us, as proposed, for example, by S. Luraghi (1998:173), it unquestionably displays the outcome of the striking syntactic developments found in Luwian. As such, an examination of Lycian is valuable in demonstrating where such incipient reanalyses as the prepositions in Luwian may lead.

8.0. Lycian Sentential Word Order

In Luwian, one finds a limited departure from the rigid Hittite requirement for postpositions and verb-final sentential order. Before Lycian

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prepositions are discussed in detail, the placement of the verb within the Lycian sentence may be briefly considered as well. While Laroche declares, "l'ordre des mots est *libre* en lycien" (1979:96), there are some clear tendencies, especially concerning the placement of the verb. In contrast with Luwian and especially with Hittite, the verb very seldom occurs in a sentence-final position. One example of such a formation is found at the end of the long Trilingual Inscription of Létôon:

17) *Pigesereye : meiyeserihhati : me-hriqla : asñne : pzzititi.*

'à Pigesere si l'on désobéit, alors l'acropole... fera respecter sa décision.'
(N.320.40-1; as rendered by Laroche 1979:76)

This is the only sentence in the extended text in which the verb is not followed by at least some enclitic particle, and it may be supposed that this placement is the result of discourse considerations, specifically its forming the conclusion of a long proclamation. It cannot be taken as a requirement for conditional sentences; this is not the normal position for the verb in such constructions:

18) *se ije ti edi tice mētē : me ne qasttu : ēni qlahi : ebijehi : se wedri : wehñtezi.*

'And whoever does harm to them, let the Mother of this precinct and the town Wehñtezi judge him'. (T.L. 56.3-4; Houwink ten Cate 92-3)

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19) *meijadi : tice tihe zuñm[ẽ me ne tulbeiti : muhãi : huwedri.*

'And if someone causes (makes) any damage, the assembled... Gods shall punish him'. (T.L. 59.2-3; Houwink ten Cate 94)

One may observe that, similarly to a Hittite or Luwian sentence, the Lycian clause generally begins with a series of cliticized particles. The verb, however, may come immediately thereafter, as in 18) or 19). In such sentences, as Laroche points out, the order verb-subject-object (VSO) predominates (1979:97). An extremely common variant places the object before the verb:

20) *ebẽñnẽ pr[ñ]nawã me ne prñnawatẽ klppasi kssẽñzijah tid[eimi.*

'This building, Klppasi, son of Kssenzijai, has built it'. (T.L. 91.1; Houwink ten Cate 96)

21) *ebñnẽ : kupa : mẽnadẽ: crehẽnube.*

'This tomb, Crehenube has made it'. (T.L. 52.1; Houwink ten Cate 92)

This formation occurs most often at the beginnings of epitaphs, suggesting that here too discourse factors play a role. A further variation on this type may place a locative element of the predicate in sentence-initial position:

22) *ebehi : isbazi : mi ije sijẽni ; padrñma : cumaza.*

On the couch of this [tomb] lies Padrñma, cumaza. (T.L. 49; Houwink ten Cate 92)

The subject, too, may fill this initial position:

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23) *mekisttẽnẽ: ep[ñ] tuwete : atli : eh- bi : skkulijah : ti- deimi.*

'Mehistte has set it up for his [own] person, the son of Skkulija-'. (T.L. 27.1-4; Houwink ten Cate 90)

Laroche, however, confirms the markedness of such positionings, labelling them "en position initiale emphatique", and stressing that this emphasis is a "trait stylistique plus que linguistique" (1979:97). In these emphatic variants, the complex of particles that normally begins an Anatolian sentence occurs between the emphatic preposed element and the verb. This conjunction of particles and verb suggests that in Southern Anatolian, the verb has been pulled leftward under conditions related to Wackernagel's Law. Still, such rules would appear to have been relaxed somewhat in attested Lycian, given the degree of variety allowable at both the beginning and the end of the sentence. Overall, however, the least-marked sentence-order is VSO, as noted by Laroche.

8.1. Lycian Adpositional Phrase Order

The significance of the placement of the verb in Luwian and Lycian sentences as regards adpositional ordering in these languages is further observed by Laroche, who remarks,

> Le glissement du verbe vers le début de la phrase ou de l'énoncé signifie l'éfacement de la fonction prédicative, fonction qui sera naturellement récupérée à l'aide d'autres procédés; la genèse de la préposition est une autre conséquence du même processus. Tout cela est bien connu des

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dialectes indo-européens, au cours de chacune de leur longue histoire. (1979:98)

This process, according to Laroche, may be speculated to have been set in motion by an accent shift: "La fixation de l'adverbe comme préposition a dû être motivée, comme en grec, par des conditions accentuelles" (1979:97). Unfortunately, accentual data for the Anatolian languages, and especially for those whose attested corpus is fragmentary, is necessarily quite difficult to extract, and consequently the effect of accent on syntax cannot be measured with any great precision.

It can nevertheless be stated with certainty that, whatever the nature of the process that drove both the verb and the adposition leftward in Southern Anatolian, it began in the prehistory of Luwian. As it happens, the clearest demonstration of this fact is found in Lycian, whose inventory of prepositions reflects the greater productivity of this formation over the postpositional construction throughout the history of the subgroup. The number of attested prepositional lexemes in Lycian is not great, but as a group they show a strong etymological affinity with the prepositions found in Luwian. The Luwian postpositions, on the other hand, have essentially disappeared in Lycian, with only scattered, highly dubious possible remnants. This outcome in Lycian strongly suggests that, by the time of Cuneiform Luwian, the set of postpositions was already moribund. Such a state of affairs does not arise instantaneously; it will have taken shape before the language was first attested.

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9.0. The Lycian Adpositional Inventory

9.1. *ẽnẽ*

Lycian, as mentioned, does not have a large inventory of known prepositions, and some of these are etymologically more or less opaque. However, because it is such a small set, each member can be examined. A preposition with an especially clear lineage, at least as far back as Luwian, is *ẽnẽ* 'under', from Luwian *annan*. As remarked above, the prepositional status of *annan* is at least somewhat controversial in Cuneiform Luwian, and, according to Laroche, SUB-*nan* (=*annan*) is a postposition in the hieroglyphic language (1959:181). However, given that the most unambiguous adpositional attestation of *annan* (KUB XXXV 39 iii 29) is an obvious preposition, and there is no evident necessity for the other (KBo IV 11.43-4) to be postposed rather than preposed, the hieroglyphic usage may be taken as a divergence from the cuneiform language remaining to be explained. Similarly, in Lycian, *ẽnẽ* in its clearest contexts appears as a preposition: *ẽnẽ χñtawata ker[i]χehe* 'under the rulership of Kerikhes (T.L. 43.2); *ẽnẽ periklehe χñtawata* 'under the rulership of Pericles'. Laroche argues that *ẽnẽ* in this construction is part of a nominal compound, so that *ẽnẽ Xñtawata* refers to a type of official: "'sous-officier de Péricles'", etc.; this would contrast with *hri χñtawatahi* "'l'officier supérieur'" (1957:182).

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However, the flexibility of word order in the examples offered here is indicative of a phrase rather than a formula frozen into a compound. As such, *ēnē* may be said to carry on the lexical innovation found in Luwian, where *annan* takes the place of *kattan*, as known from Hittite.

9.2. *hri*

The phrase *hri χñtawatahi* just mentioned includes a reflex of another Luwian locative auxiliary; *hri* is generally agreed to derive from Luwian *šarri* (Melchert 1993b:27; Laroche 1959:180; Meriggi 375-6). As discussed above, *šarri* appears to function as both a preposition and a postposition, although the prepositional contexts are somewhat less problematic. In any case, *hri* itself only offers the possibility of being interpreted as an adposition in the aforementioned phrase, at 44a51, which locus Meriggi describes as "oscuro" (376), with Melchert's agreement (1993b:27).

9.3. *hrppi*

Much more convincing as a preposition is the heavily-attested *hrppi* 'for, upon'. In contrast with *ēnē*, which is, according to Melchert (1993b:24) nowhere found clearly functioning as a preverb, and with *hri*, which normally fills that

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role, *hrppi* occurs quite frequently as both a preverb and a preposition. This word is very clearly both in sense and in the phonological shape of its initial syllable derived from *hri* and hence from *šarri*. Meriggi proposes that *hrppi* is a compound of *hri* and *epi*, which is a fairly frequent preverb in Lycian. This *epi*, according to Meriggi, "non ha corrispondenza in altre lingue anatoliche, ma soltanto nel greco epi" (370). Melchert, too, declares that *epi* is "hardly 're-, again' = Hitt. *appa*" (1993b:17). Whether *epi* is a borrowing from Greek, which had an extensive foothold in Asia Minor by the time of the greatest flourishing of Lycian in the mid-first millennium BC, or is a native Anatolian word that only surfaces at this stage, it would certainly reinforce the sense of 'on, upon, for' expressed by both *hri* and *hrppi*. And compounding of locative auxiliaries does appear to be productive in Lycian, both in fossilized forms, as in the preverb *ñtepi* (*ñte* + *epi*) 'in', and in compound preverbs, such as *ñte* + *epi*, *ñte* + *hrppi*, or *ese* + *hrppi*.

One thus finds particularly great productivity of compounding with *epi*. In its formation with *ñte* in *ñtepi*, it joins an item derived from Luwian *anta* which, while occurring as a preposition in that language, is overwhelmingly more likely to function as a preverb, as is also the case with its Lycian successor. On the other hand, *šarri* is attested in Cuneiform Luwian only once as a preverb; its adpositional function is far more prominent, relatively speaking. The

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combination in Proto-Lycian of *hri + epi* to form *hrppi*, with *hri* as the first, preposed member of the compound, indicates that, regardless of whether *šarri* was in fact allowable as a postposition in Luwian, it was the preposition that formed the basis of the Lycian lexeme. This newly-reformed preposed word serves to reinforce the similar item already found in Luwian.

With its sense of 'on' and by extension 'for', *hrppi* is especially common in Lycian epitaphs:

24) *ebēnne : kupu : me ti : prñnawatē : ida makazza : uherijeh tideimi : hrppi : ladi : ehbi : se tideime.*

'This tomb, Ida Makazza, the son of Uherija/i-, has built it for his wife and children.' (T.L. 57.1-4; Houwink ten Cate 93)

25) *ebēñnē prñnawu : me ti prñnawatē mñemruwi : kñtenubeh : tideimi : hrppi esedeñnewi : kñnahi ehbiehi : qurttāi lada.*

'This building, Memruwi, the son of Kñtenuba, has built it for his grandmother and the wives of the qurttas.' (T.L. 39.1-4; Houwink ten Cate 91)

The length of the phrase that may be governed by *hrppi*, especially as seen in 25), is rather striking.

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9.3.1. *hrppi* and *hri*: Preposition or Preverb?

Beyond its prepositional senses, *hrppi*, like *hri*, often serves as a preverb:

26) *me ije ne pemati tice : cbi hrppi ttãne.*
'And he does not permit to put anyone else upon him.' (T.L. 49; Houwink ten Cate 92)

In constructions similar to that found in 26), one finds evidence that in Lycian, as in Luwian and Hittite, the boundary between adposition and preverb remains quite fuzzy. As illustrated in 26), preverb and verb are not normally separated in Lycian. However, when this complex occurs sentence-initially, enclitic particles, including pronouns, may intrude; not only *hrppi* but also *hri* are cited by Garrett as occurring in such constructions:

27) *hrppije me=i : tadi : tike.*
'Thereon one puts someone.' (T.L. 57.8)

28) *hrije me=i alahadi tike terihe.*
'Thereon one 'buries' someone *terihe*'. (N309a.4)

In these two passages, Melchert (1993b) lists *hrppi* and *hri* as preverbs, but Garrett argues,

> The forms *hrppije* and *hrije* are presumably just the prepositions *hrppi*

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and *hri* respectively plus the locative clitic *ije* as prepositional object; in other words they are Prepositional Phrases. The pronoun *j* after *me* [a conjunction] in [27) and 28)] may be understood as doubling either the PP topic or its pronominal object. (206-7)

If *hrppi* and *hri* in these sentences are indeed to be read as prepositions, they would in a sense be following a long-standing Anatolian construction; adposition with enclitic pronouns suffixed are of course found in both Hittite and Luwian. Still, in Lycian, as in the later stages of Hittite, a reanalysis has taken place, as the locative pronoun is employed rather than the accusative as in Old Hittite. This change indicates that lexemes such as *hrppi* have long since shed any nominal properties; they are not nouns modified by possessives but adpositions governing complements. The only uncertainty as to the classification of these items concerns whether they constitute prepositions, preverbs, or adverbs in Lycian.

9.4. *ñtewe*

For a few other Lycian prepositions, a sure etymological link with Luwian has yet to be established. One of these is *ñtewe* 'facing, toward':

29) *sewene : χttadi : tike : ebineñtewe mahãnai : ebett/e : ebine : ñtewe : kumazi : ebehi*. (N.320.34-36)

'And let no one do harm either towards these gods or towards these

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priests.'

The sense of this preposition, then, is not difficult. There is, however, little consensus as to its derivation. Meriggi proposes that its origin lies "in gruppi o derivati oscuri: *ñte-j-ewẽ*" (367), thus representing a compound of the preverb *ñte* 'in, inside', and a particle of uncertain meaning. Laroche, on the other hand, suggests that "on restaure louv. *daweyan* > prélycien **dawan* > lyc. *ñtewe* = **dewen (ñt = d),*" while, contra Meriggi, "Il n'y a pas lieu de découper *ñte-we*". Although Laroche then speculates, "*ddewẽ, ddewe,* et *ddeu,* T.L. 44c 6-10, seraient-ils variantes graphiques de *ñtewe?*" (1979:90), this proposed graphic representation of a voiced stop seems more to anticipate Modern Greek than to correspond to known Anatolian orthographic practices. It also seems highly unlikely that the Luwian postposition *dawiyan* would subsequently have become preposed; there is no clear evidence for such a step in the Anatolian languages.

More plausibly, Melchert, following H. Eichner (1977:19n.26), does trace *ñtewe* to the Luwian root *tewe-* 'face', from which, the postposition *dawiyan* (or *tawiyan*) arises as well, as mentioned in Section 3.0 above. However, he offers that *ñtewe* derives from virtual **en + tewe* 'in(to) the face' (1993b:50). The problematic aspect of this posited etymology is the persistence of Proto-Indo-

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European **en*, which is found nowhere among the Anatolian languages as a discrete unit, as discussed in Chapter II, Section 4.1. Nevertheless, the connections with Luwian *tewe-* and *anta* both seem too promising to ignore; one is severely tempted to argue on behalf of a haplologized compound *ñte + tewe*.

9.5. *ese, ẽti*

Two other likely prepositions (but certainly preverbs) in Lycian are even more obscure in origin. In particular, the fullest hypothesis offered for *ese* 'with', by Carruba (1971:38), that its source is common with the Hittite particle *-asta*, is nevertheless termed "very dubious" by Melchert (1993b:20). Melchert does declare that the Lycian conjunction *se* is an aphaeresized form of *ese* (1994:321), but is unprepared to speculate on their source.

Regarding *ẽti* 'down, under', Melchert, in arguing that a rightward accentual shift in Lycian is indicated by a tendency toward initial vowel reduction, particularly "in several preverbs", such as *hri* and *ñte*, notes, "I concede that I cannot explain the different behavior of *ẽti*... < **ṇd*^{h}*i* 'in this respect' (1994:318). The auxiliary's anomalous phonology is however accounted for in Laroche's postulation of a Greek borrowing: "les lignes 22-33 [of the Trilingual Inscription] fournissent une équation parfaitement claire: gr. ἐν τῇ

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στήλη = lyc. *ẽti sttlai*" (1979:89). The shape of *ẽti* is as such the result of a misreading of the Greek definite article for a phrase borrowed wholesale into Lycian. There is however no evidence of any derivation of *ẽti* or *ese* from Luwian prepositions, but neither is there any reason to suppose that they derive from that language's postposed auxiliaries.

10.0. Possible Instances of Postpositions

One should note finally an anomalous construction involving *ñte* or *ẽti* occasionally presents itself in the Lycian texts that some scholars, particularly Meriggi and Houwink ten Cate, label postpositional. These generally involve the auxiliaries with demonstratives of various kinds, as in the following:

30) *ebi-d-alahadi-ti : ebei : ñte.*
'And who[ever] penetrates here into *this* [building]... (T.L. 57.9; Houwink ten Cate 93)

31) *me-ubu-hãtẽ: kbi-j-ẽti.* (44c4: Meriggi 372)
'And they made an offering under the other [?]'.

Houwink ten Cate (81) also cites the phrase *ebi ñte* in T.L. 134.7, although, unlike in 29), the position of *ñte* is not quite so odd:

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32) *tike : ebi ñte ni hrppi tãtu.*

Let them not place anything inside (nor) above.'

Melchert lists both of these occurrences of *ñte* as preverbs (1993b:49), and indeed there is no clear attestation of *ñte* as a preposition in Lycian, as its Luwian precursor *anta* occasionally is. Still, *ñte* in 28) certainly makes for a most unusual preverb. Laroche for his part also takes *eb(e)i ñte* in T.L. 57.9 and 134.2 as similar constructions, but as adverbial in nature (1960:183).

The passages involving *ẽti* are similarly controversial. Melchert reads for 31) not *kbi-j-ẽti* but *kbijẽti,* an accusative singular animate adjective with a meaning, perhaps, of 'double, paired' (1993b:33). Similarly, for another passage suggested by Meriggi as representing postpositional usages of *ẽti:*

33) *sẽ ne : tes-ẽti : qãñti : Trmmili-j-ẽti.* (T.L. 149.10)

Melchert reads a noun *tesÎti* and an adjective *Trmmilijeti* (1993b:71, 78). Indeed, a Lycian form *tes* appears not to be attested other than in Meriggi's proposed interpretation.

Nevertheless, the sentence-final *ñte* in T. L. 57.9 is quite bizarre, and Melchert's *kbijẽti,* if valid, is a hapax form. In short, a "postpositional"

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construction, as it has been termed, is at least a fairly plausible reading of a language that is still not very well understood. However, some caution should be exercised with the label "postposition". As mentioned, the most promising passages for such an interpretation involve the locative auxiliary preceded by a demonstrative of some sort. If these are in fact adpositional phrases (which, to be clear, remains to be established), it would appear that some sort of prosodic constraint, similar to that operating in such Latin forms as *mēcum, vōbīscum,* etc., is at work. One should therefore hesitate to term such instances of *ñte* or *ēti* in the same sense in which the word is used in discussing Luwian or Hittite.

11.0 Conclusion

One finds in Lycian, then, a syntax that comes as close to a perfect representative of a VO pattern according to Greenberg's framework as could realistically be hoped. In addition to unmarked sentences featuring the direct object following the verb, adjectives and genitives normally follow the noun, and, of course, adpositions are preposed. However, this syntactic structure did not arise overnight; its precursors are already quite apparent in Luwian. Sentences with a VO order are already fairly common, albeit marked. Prepositions also occur about as often as postpositions, and subsequent events as revealed to us by the Lycian texts indicate that postpositions by the time of Cuneiform Luwian were already moribund. The most prominent Luwian postposition, *parran,* leaves no traces in Lycian, while another, *appan,* does

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remain as *epñ*, but only as a preverb, a function it also served in the parent language. The prepositions in Luwian, such as *annan, anta*, and *šarri*, fare better over time, and this fact, coupled with other OV features in that language, strongly suggests that preposing was already the more productive means of deriving new adpositions by the time of attested Luwian. Moreover, the relative frequency of the prepositional constructions, in contrast with the highly-marked status of sentences with non-final verbs, suggests that, as discussed for Lithuanian and Greek (in Chapters V and I, respectively), in Southern Anatolian the shift to prepositions preceded that to a basic VO word order. The syntax of Lycian, then, is the result of processes set in motion in the prehistory of the Southern Anatolian languages.

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Chapter IV:
Word Order Change in Umbrian:
From Prepositions to Postpositions[1]

1.0. Introduction

One of the more striking syntactic features of the Umbrian language, especially as compared with Latin, is its adpositional system. Whereas nearly all Latin adpositions are preposed, Umbrian features a handful of frequently-used postpositions along with a larger number of prepositions. Most of these postpositions, moreover, are cognate with such Latin prepositions as *in, cum*[2], *ad,* and *prō.* The question of how these segments came to appear on opposite sides of their attendant phrases in the two related languages is a striking one for the investigator of historical syntax.

Indeed, the postpositions corresponding to these Latin words, *-en, -com* ('at, by, with' in Umbrian), *-ař,* and *-per*[3], also have cognates in Umbrian's more closely-related sister language, Oscan. Three of the Oscan adpositions, *com, az,* and *pru,* are, as in Latin, preposed, while *-en* is usually postposed. And the situation within Umbrian itself, as briefly described, is quite anomalous both among the Indo-European languages and in terms of general language

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[1] A preliminary version of this chapter appeared in 1999 in UCLA Working Papers in Indo-European Linguistics, Vol 1. V. V. Ivanov and Brent Vine, eds., pp. 85-101.

[2] On the origins of *cum* and Umbrian *com,* see Sections 4.1.1 and 5.2.1.

[3] Glosses for Umbrian words cognate with Latin will be given only where the meaning is believed to be different from that of the Latin cognate.

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typology. One normally expects to find a clear predilection either for prepositions, as in Latin, Greek, or the Germanic languages, or postpositions, as in Hittite or Classical Sanskrit. In this respect, Umbrian joins the Southern Anatolian and Baltic languages in atypically featuring both varieties of adposition. Consequently where typological classifications of Umbrian have been attempted, disagreement has arisen among some scholars as to whether it should be classified as a prepositional or a postpositional language (as will be discussed in Sections 1.1.2 and 1.1.3).

1.1. Previous Scholarship

1.1.1. The Late Nineteenth and Early Twentieth Centuries

Modern study of the Sabellian languages may be said to have begun in 1897 with the publication of two major surveys: R. von Planta's *Grammatik der oskisch-umbrischen Dialekte* and R. Conway's *Italic Dialects*. Both von Planta and Conway distinguish between postpositions and prepositions in their discussions of the adpositional inventories of these languages (II.439-457; 483-4, respectively)[4], although neither argues that one set is more basic or intrinsic to Umbrian, where postpositions are unquestionably more prominent than in the other Sabellian languages. It must be noted, however, that von Planta identified a couple of properties of Umbrian and, to a lesser extent, Oscan adpositional phrase formation. The first was the tendency for postpositions to be more phonologically slender than prepositions:

[4] The term "Präposition" is however evidently used by von Planta as a blanket term corresponding to 'adposition' in the present study.

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> Bei allen übrigen ausser den erwähnten kurzen einsilbigen Präpos. ist auch das O.-U. bereits zur Voranstellung übergegangen, nicht nur bei lang- und mehrsilbigen... sondern auch bei kurzsilbigen (*sup, op [eh]*)... Die postponierten kurzen Präp. *-en* etc. mögen infolge ihrer bequemen Sprechbarkeit in der Enklise und der Allgemeinheit ihrer Bedeutung der Tendenz zur Vorausstellung widerstanden haben (440-1).

This observation foreshadows later hypotheses, advanced most prominently by A. Nocentini (1992) that prosodic considerations play a deciding role in determining the placement of Umbrian adpositions (as will be elaborated below). Noted in passing by von Planta is the occasional occurrence of *en* twice in a phrase, as in **vapefem avieklufe**, *toteme Iiouinem, ocrem Fisiem* (441), the implications of which will also be examined below, in Section 3.3.

C. Buck (1928) essentially follows Conway in listing the adpositions found in the languages under discussion without attempting to draw any broader conclusions on their functioning. Moreover, while he does observe a distinction between preposed and postposed items, he refers to the latter as "postpositives" (203). The slightly different term is not explained. One intriguing remark of his concerns the double usage on *-en* mentioned by von Planta "as if it were a real case-ending" (114), presaging subsequent debate on whether it has, in fact, become reanalyzed as such.

1.1.2. The Mid-Twentieth Century

J. Poultney's edition of *The Bronze Tables of Iguvium* (1959) represents a

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significant advance in the study of Umbrian, offering an extensive and detailed commentary, as well as a more exact representation of the text. However, rather like Conway and Buck, Poultney lists the prepositions and postpositions of Umbrian, but does not offer any speculation as to how the language came to have both varieties of adposition.

None of the works mentioned thus far pay as much attention to syntax in the various Sabellian languages as they do to phonology and morphology. Within these examinations of syntax, as B. Konnecker remarks, "in no grammar - not even in von Planta's two volumes - does the section on word order occupy more than a page" (1972:38). Konnecker's dissertation indeed represents the first monograph-length treatment of Umbrian syntax, or, for that matter, of the syntax of any Sabellian language. Nevertheless, although she mentions what she calls the "predominance" of postpositions, while also noting the presence of prepositions (92; see also 44, 111), her work is more concerned with syntax at the clause level. She does however cite the diagnostic phrasal characteristics in J. Greenberg's recently-published study on language unversals in noting that a change in word order appears to be in progress in Umbrian (112). Most striking is her conclusion on the precipitating factors in this shift:

> While the later changes in Latin are focused on the movement of the verb, one of its earliest deviations in word order was just in the relative clause. The relation, if any, between that early innovation, the apparently equally-early shift to prepositions, and the subsequent displacement of the verb, should provide a rich field for investigation. For the present, I limit myself to the conclusion that a shift in dominant verb position is not the

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only, nor indeed the necessary, agent in precipitating syntactic order change. In Umbrian and possibly in Italic as a whole, the agent was the deep-structure relative clause. (112)

While the present investigation is not prepared to tender a postulated relative chronology regarding the shift in relative clause order as opposed to that of adpositional phrases, Konnecker's identification of the latter as a very early event in this syntactic change is most exciting, given similar evidence found in the Baltic and Southern Anatolian language (for which see Chapters III and V). Regardless of whether a change in the structure of the relative clause preceded or followed that for adpositional alignment, the additional support for early occurrence of this event serves to underscore its significance.

One other study of interest for its relating the Osco-Umbrian adpositional situation to broader questions of typology and universals of word order change is J. Kahr's article "The Renewal of Case Morphology: Sources and Constraints" (1976). Kahr's treatment of these Sabellian languages expands on Buck's observation of the resemblance of doubly-postposed **en* in phrases like *toteme Jiouinem* and *ocrem Fisiem* to case suffixes, and argues that, as often occurs with such postpositions, **en* in such instances is becoming morphologized in that role (128-9).

1.1.3. The Late Twentieth Century

By this time, awareness of Umbrian's postpositions had become fairly widespread. Baldi, in his general survey of the Indo-European languages, notes

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as Umbrian's only significant syntactic difference with its fellow Italic languages that it "has postpositions instead of the prepositions commonly found in Latin and Oscan" (1983:32)[5]. Nocentini's "Preposizioni e posposizioni in oscoumbro" (1992) represents the most substantial investigation of these adpositions, however. In this study, Nocentini alludes to "la convinzione unanimente espressa che l'umbro... sia una lingua posposizionale" (1992:200), and stresses that, aside from the four postpositions introduced thus far and a few others, *-to* 'from' being the most prominent, the majority of adpositional lexemes are preposed. Among these prepositions are *com,* with its more familiar comitative sense; *e(he)* = Latin *ex;* **hutra**/*hondra* = *infra; pert* 'beyond, on the other side of'; **pus**/*post;* **pusti**/*postin* 'at each'; *pre* = Latin *prae; subra* (+accusative), **super** (+locative), *superne* (+accusative), all 'above'; **supu** = Latin *sub;* and **tra**/*traf*/*traha(f)* = Latin *trans.* Seizing upon this numerical superiority, Nocentini declares, "Di consequenza la classificazione tipologica dell'umbro come lingua posposizionale non è sostenibile" (224). Of even greater note, moreover, Nocentini proposes a hypothesis to explain the presence of both varieties of adposition in Umbrian, certainly the most comprehensive and quite possibly the only one previously advanced. This schema will be analyzed in greater detail below, in Sections 4.1-4.1.1.

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[5] Of Baldi's statement that many of such Proto-Indo-European roots as **peri, *$h_1$epi/$h_1$opi* and **$h_1$en* "evolved into what are commonly known as postpositions, after passing through a stage of postpositional placement in the oldest stage of the language" (1999:87), it is hoped that the present effort will demonstrate such a summary of events to be rather oversimplified.

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One other work, published at the very end of the century, also deserves mention. J. Untermann's *Wörterbuch des Oskisch-Umbrischen* (2000) is far and away the most thorough dictionary for the Sabellian languages to date. Here one finds citations for all the loci in which a given word is found, in addition to summaries of all attempts at etymology. Untermann observes the preposition/postposition distinction in his listings, and his entries on individual items are of great use.

1.2. Umbrian Adpositions in Typological Terms

Nocentini appears to overstate his case. He downplays the greater frequency of the postpositions by arguing that *-to* and *-en* are morphologized beyond adpositional status (224), but even so postpositions are overwhelmingly easier to find in the Iguvine Tables. The 145 occurrences of *-per* alone dwarf all the attested prepositions combined, totaling approximately fifty[6]. No one

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[6] The exact number is difficult to determine due to ambiguous cases such as **sese** and *nesimei*. The latter occurs twice in VIa.9 and is analyzed as an adverb by both Poultney and Buck (144, 135-6, respectively), but it appears to function as a preposition, taking ablative complements: *nesimei asa* 'next to the altar', *nesimei uaspersus* 'next to the seats'. The former is also regarded as an adverb by Buck (212), while Poultney sees it as "probably" a preposition (323). It occurs three times in such phrases as **testru sese asa** 'from the right side of the altar' (III23, IV.15-16). However, their derivation is unlike that of the majority of heavily employed adpositions drawn from local adverbs: *nesimei* is a superlative form of an adjective (Poultney 89) in either the locative case (Untermann 493) or the adverbial ending **-ēd* (Poultney 115); **sese** is most likely an ablative form of an adjective (Meiser 1986:245; Untermann 671). As an added complication, Meiser and Poultney (207) both suggest that the complement of **sese** is in fact **testru**, not **asa**, and therefore functioning as a postposition. In any event, these are, as mentioned, not adpositions in the same sense that *-per, -en, post,* and *ehe* are. Nevertheless, to clarify the number of attested prepositions in the Iguvine Tables, there are forty-six if *nesimei* and **sese** are not to be considered as such and fifty-one if they are.

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preposition appears in more than eleven instances (*pre*). Even allowing for the ritualistic and extremely repetitive nature of the text in question and especially the contexts of such postpositions as *-per,* the postpositions are pressed into service more frequently. Poultney lists seventeen different phrase contexts for *-per,* still more than the total appearances of any of the prepositions, as well as nineteen contexts for *-cum,* and forty-six for *-en.* It may therefore clearly be seen that Umbrian's postpositions, enjoying far heavier usage than the prepositions, represent a much more central part of the language's vocabulary as expressed in the Iguvine Tables. On such grounds the typological classification of Umbrian as a postpositional language is certainly justified. Nevertheless, it remains true that the prepositions constitute a significant portion of the language's adpositional inventory, and even a minority of such prominence remains typologically most unusual. An accounting for this remarkable state of affairs is indeed of much greater interest than the determination of a typological label.

1.3. The Adpositional Situation in Umbrian

While on the strength of the numerical accounting described in the previous section Umbrian may be described as a postpositional language, it is rare among the Indo-European languages, and presumably universally as well, to find an adpositional system in which the ostensible minority formation appears in nearly one-sixth of all attested phrases, and where the majority of

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adpositional lexemes have the opposite orientation. Such scarcity would certainly indicate the great instability of such an arrangement; if after all it were stable and long-lasting, the comparative linguist would surely find more attested examples of such patterns than is in fact the case. It would likely not endure for an especially long time, and the situation in Umbrian will have developed relatively shortly before the Iguvine Tables were inscribed. The next question, then, is that of which subset of Umbrian's adpositional inventory arose more recently, the postposed items or the preposed ones. In other words, the crucial question is not, as is commonly asked, a synchronic one, in this case which variety of adposition is more basic, but rather a diachronic one, namely that of which variety prevailed earlier in the language's development, and in which direction this development appears headed.

2.0. Adpositions Elsewhere in Italic

A numerical counting of postpositions and prepositions cannot determine which formation is the relic and which, the innovation. Less pronounced mixings of types occur in Oscan, as mentioned, and in Latin as well. Latin traces of postposing include adverbs such as *eāpropter* 'on that account', and in pronominal phrases such as *mēcum* or *nōbīscum* 'with me, with us,' including also the sequence *tedendo (=tēd endo* 'toward you') in the extremely archaic Duenos Inscription (CIL I2 4).

Another Italic language, more closely related to Umbrian and Oscan than to Latin, is South Picene, which offers the great advantage to scholarship of

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attestations dating from the sixth or fifth century BC (A. Marinetti 1985:44), but the great disadvantage of appearing in only a small number of short, not entirely deciphered inscriptions. In these few, imperfectly understood inscriptions, such attested evidence as may be found appears to indicate a system similar to that of Umbrian. A total of seven instances of suffixed **-en* may be found, as in *mefiín* 'in the middle' (MC 1, AP 3)[7]. Inscription CH 1 contains the word *poioúeta*, which Marinetti interprets as including the postposition *-to* also found in Umbrian (112)[8]. One clear example of a preposition appears in TE 2: *postin viam*, where one thus finds an item known from both Umbrian and Oscan. One other possible prepositional phrase occurs in TE 1, *e súhúh*, for which Marinetti conjectures an interpretation of '*ē suō*' (99). These two loci however are the extent of the evidence for prepositions in South Picene. Despite the greater age of these inscriptions than of the Iguvine Tables, however, the South Picene texts offer no additional insight as to which set of adpositions is more recent in origin.

3.0. The Relative Ages of the Two Sets of Umbrian Adpositions

In evaluating the relative ages of the two sets of Umbrian adpositions, it is useful to recall the processes by which such parts of speech come into being,

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[7] Other attestations of **-en* in South Picene are in TE 2 (*vepelen*), CH 1 (*iepeten*), CH 2 (*ombrijen akren*), AQ 3 (*boúediín*).

[8] This interpretation is not universally held. Eichner reads the text in question as *poioúefa*, thus ruling out any such postposition (1993: 62-3).

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specifically the reanalysis of an item from a semantic word to a lexeme with a primarily grammatical function; that is, grammaticalization. In Umbrian, one may discern between the two sets of adpositions clear differences in degrees of grammaticalization. Some especially clear instances are to be found among the prepositions. As B. Heine, U. Claudi, and F. Hünnemeyer emphasize, an extremely natural and common early step in the process of grammaticalization is the transformation of a relational word into an adpositional or directional sense (1991; see especially 123-31). One such item in Umbrian is the preposition *hutra/* **hondra** 'below.' Cognate with Latin *humus* and *humilis*, its ending betrays its recent development incorporating the contrastive[9] marker **-tero-* (Poultney 91; Buck 1928:134). Derived with a different contrastive form, **-ero-*, is the complex *super/superne/subra*, all with the general meaning 'above' (Poultney 325). Synchronically, they also function as adverbs in some contexts. The sources of *hondra* and *subra* may be traced in further detail, with both reflecting an ablative feminine case ending (*-a* < **-ād*) (Poultney 115, Buck 136). Such etymological transparency further highlights the recent adposing of these prepositions, as they have suffered little phonetic erosion, and their adverbial usage indicates that they were derived from adjectives not long before. Clearly

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9 Poultney and Buck refer to this segment as a "comparative" marker, but, as Sihler points out, the term "contrastive" describes its senses more accurately (363ff.). The observation, however, does not originate with Sihler; E. Benveniste remarks, "Le suffixe sert donc à caractériser une position comme *distincte* d'une autre. Nous attribuerons à **-tero-* une fonction de suffixe «séparatif»" (1948:119).

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these items are among the most recently derived adpositions in Umbrian.

On the other hand, few of Umbrian's adpositions are as forthcoming about their lexical sources as the words just examined. Most, however, will prove recognizable to anyone familiar with the prepositions and preverbs frequently appearing in Latin. Such doubly-functioning items are extremely pervasive throughout the Indo-European languages, and may with seemingly equal facility be employed as adpositions, preverbs, or in the more archaic languages like Hittite, Vedic, and, to some extent, Homeric Greek, as adverbs. P. Friedrich therefore proposes for them the designation "locative auxiliaries" (1975:34-5). The distribution of these items is fairly homogeneous among the Italic languages, so it may reasonably be supposed that in Proto-Italic the antecedents of Latin *cum, ad, ex, in, de,* and so forth still functioned as these locative auxiliaries and only subsequently were solidified into their roles as adpositions and preverbs. However, there is no need to insist that they were so reanalyzed en masse; rather, some will inevitably have undergone reanalysis earlier than others. Umbrian's shift in adpositional order affords us the opportunity to determine a relative chronology for some of these adpositions. It is to be expected that, of the two groups of adpositions, the one derived earlier will have undergone a greater degree of grammaticalization.

3.1. Allative *-co(m)* and comitative *com*

One feature that is not a characteristic specifically of grammaticalization but which is frequently associated with the phenomenon due to the intensified

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usage of grammaticalized segments is accelerated phonological attrition. Unfortunately, Umbrian as a general system has undergone such widespread attrition as a result of the Proto-Italic word-initial accent, most significantly word-medially (G. Meiser 130ff.), but also to a striking degree word-finally (Poultney 30; Buck 101). Moreover, while the changes wrought by the accentual shift are more extensive in medial syllables, those at the end of a phonological word are of greater significance to an investigation of Umbrian adpositions; after all, it is at the end of the word that the boundary between word and suffix or clitic lies. As a result or this joining of word and clitic, it is difficult to pinpoint any exceptional degree of phonological loss. Nevertheless, one useful minimal pair appears to present itself. The comitative preposition **kom* 'with' is found in each of the three most prominent Italic languages. In Umbrian, however, this usage is far less common than its postposed allative cognate *-co(m)* 'at.' This postposition is attested thirty-one times in the Iguvine Tables; on all but six occasions, final *-m* is lost. This final nasal is lost in none of the five instances of comitative *com,* including one in which it is postposed to a pronoun (*eru-com* 'with him' - VIb.50). On all five occasions, the word immediately following *com* begins with a consonant (*p-* in every case). Phonological erosion is by no means remarkable in Umbrian, but its absence may well be. The continued integrity of preposed *com* may then be regarded as evidence of its more recent derivation.

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3.1.1. A note on **kom*

As mentioned, reflexes of **kom* appear in all three of the most substantially-attested Italic languages. It also has cognates farther afield, including the Old Irish preverb *con-* and the Old Church Slavic preposition *kŭ* 'to' (V. Ivanov 1999:109). Its derivation, however, appears to be significantly different from those of most other Indo-European adpositions and local adverbs, where very often a nominal source may be found in the more or less immediately preceding history of the languages in which they appear. No such historical antecedent is known for **kom*, however. Rather, Hittite and Vedic evidence especially suggests that this segment was previously an enclitic aspectual particle with a perfective sense. The Hittite sentence-initial enclitic construction *nu-kan* may thus be equated with Sanskrit *nú-kam* and probably the Greek *νύ κεν* (Ivanov 109), although the positional correspondences between Hittite and the other two languages are not exact (Watkins 1997:618).

Even in Vedic, however, one finds evidence of an adpositional reanalysis. A. A. Macdonell notes that *kam* has not only an unaccented manifestation, as just observed, but also an accented variant. In the latter instance, it may be an adverb meaning 'well', or, following a noun in the dative case, a postposition with the sense 'for the benefit of' (1916:225). A similar reanalysis along different semantic lines appears to be complete in Umbrian as well as Oscan. Despite its unusual origin, the functioning of *co(m)* appears no different from that of any

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other adposition or preverb in these languages. There appear to be no traces of its earlier status as an aspectual particle. The allative form of the adposition in Umbrian is postposed, but in this its placement is the same as those of several other adpositions whose derivations are more conventional.

3.2. Adpositional Clisis

As mentioned, adpositions are often grammaticalized so that they become clitics, conceived by a speech community as having no more than partial independence of the nouns with which they are associated. Some evidence is found for clisis of prepositions in Umbrian, at least in the native script:

1) **pusveres : Treplanes** 'Behind the Trebulan Gate' (Ia.7)

2) **preveres : Tesenakes** 'Before the Tesenacan Gate' (Ia.11)

However, in the Latin script, such prepositions are written separately:

3) *post . verir . Treblanir* 'Behind the Trebulan Gate' (VIa.58)

4) *pre . verir . Tesenocir* 'Before the Tesenacan Gate' (VIb.1)

On the other hand, postpositions are joined to the preceding word in either script:

5) **tutaper : Ikuvina** 'for the Iguvine state' (Ia.21)

6) **testruku : peri** 'at the right foot' (Ia.29)

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7) *totaper . Iouina* 'for the Iguvine state' (VIIa.53)

8) *uerisco . Treblanir* 'at the Trebulan Gate' (VIIa.53)

3.2.1. The "Second Position" Constraint

Further evidence of the clitic status of the postpositions may be found in a prosodic constraint on their placement: with very few exceptions, the postposition must immediately follow the first word in the adpositional phrase. Von Planta mentions the enclisis of these adpositions (1897:411), and Poultney states, "The arrangement of adj. + prepos. + noun, as in L[atin] *magnā cum cūrā*, is fairly common," referring to *-com* and *-per* as prepositions (156). However, von Planta does not specify the unvarying nature of this placement, and Poultney appears to overlook the great frequency of phrases in the pattern of noun + postposition + adjective. The constraint as such is unsurprising when the phrase consists of a single word:

9) **akrutu** 'from the field' (Va.9)

Examples 5) - 8) show that this is also the case in a two-word phrase, whether the first word is a noun (5, 7, 8), as in fact often occurs, or an adjective (6). The Iguvine Tables offer no adpositional phrases of four words or longer, but the constraint is present in a three-word phrase:

10) **uraku : ri : esuna** 'regarding that sacred matter' (Va.5)

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Example 10) also shows that this second-position rule applies even when the first word is a demonstrative. The only apparent exception is in phrases beginning with a genitive pronoun:

11) *erer . nomneper* 'for its (m.) name' (VIa.54)

12) *erar . nomneper* 'for its (f.) name' (VIa.54)

One explanation for this exception might be that possessives are somehow conceived as forming a single unit in a manner in which adjectives are not, so that, on an underlying level, the second-position rule for postpositions is not violated. A possible Latin parallel for such an analysis may be found in such compounds as *eiusmodi, huiusmodi,* both 'of this kind, such'.

3.2.2. Adpositions and Sandhi

Further significant evidence for the view that postpositions in Umbrian were regarded by the speech community as being attached unlike prepositions is that the presence of a postposition often blocks phonological rules normally operating at the end of a word. An excellent example is seen in 3) and 4), where final **-s* is rhotacized, in contrast with 8), where it remains unchanged. Similarly, final *-m,* as in the accusative case, is frequently dropped:

13) *esto . persclo . .... . enetu* 'He shall begin that sacrifice' (VIa.1)

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Final *-m* is retained, however, when followed by a postposition:

14) **persklum : ař : karitu** 'He shall call to the sacrifice' (III.21)

Similar phenomena do not occur with prepositions:

15) *posti . acnu* 'in each year' (Vb.8)

Here, one might expect perhaps a **postiŋ acnu*. After all, if the same sort of clisis were occurring among the prepositions as for the postpositions, the final consonant should remain, as it does in 14). Such evidence strongly suggests that the Umbrian postpositions have developed along this grammaticalization cline, as a consequence of their having had more time to do so.

3.3. From Postposition to Case Suffix?

Finally, evidence exists not only in Umbrian but in the other Sabellian languages as well for the first stirrings at least of the consummate step in the process of grammaticalization: the reanalysis of the cliticized adpositions as grammatical affixes. These cases involve the postposition *-en;* in several noun-adjective pairs, the adposition has been extended to appear on both members of the phrase rather than the first only:

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Umbrian:

16) **vapefem : aveiklufe** 'toward the augural seats' (Ib.14)

17) *ocrem . Fisiem* 'on the Fisian mount' (VIa.46)

18) *toteme . Iouinem* 'in the Iguvine state' (VIa.46)

Oscan:

19) **húrtín ./ kerríín** 'in the grove of Ceres' (E. Vetter 147 A.1-2)

Apparent examples are also present in South Picene:

20) *ombrijen : akren* 'in the Umbrian land' (CH.2)

21) *mefiín : ... : vepetí* 'in the middle of the stele' (MC.1)

Of particular interest is 18). Here, the postposition is so completely regarded as part of the noun that an additional *-en* must once again be postposed to provide a satisfactory locative sense. A similar example occurs earlier in VIa:

21) *toteme . Iouine* 'in the Iguvine state' (VIa.26)

It is surely no mere coincidence that this incipient case affix functioning is found in the one adposition to be postposed in both Umbrian and Oscan. The postpositions of Umbrian show a significantly greater degree of grammaticalization than do its prepositions, with their regular rules for enclisis and integration with their associated preceding words. It therefore stands to

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reason that they have evolved as adpositions over a longer period of time and as such were so analyzed some time earlier than the prepositions were. As for Oscan, comparable evidence is scanty: *com* appears only in its preposed, comitative manifestation, while *-to* is missing. The adposition cognate with Umbrian **-ař, az**, appears once, preposed:

22) **az . húrtúm** 'to the grove' (Vetter 147 A.20)

3.4. Relative Chronology

On this admittedly slender basis, a relative chronology may be proposed. The first of the attested adpositions to be derived was **-en*, along possibly with **-to*, given Nocentini's observation that "Il suo valore è quello di un caso locale secondario" (218). These items are, as far as can be discerned, suffixed throughout Sabellian, and show far and away the most-developed grammatical usage. Incidentally, *-en* also shows significant phonological attenuation. Subsequently, **ad(s)* was adposed; in the intervening period, word order for new adpositions was reversed in Oscan but not in Umbrian. For allative **kom* there is insufficient evidence to determine whether it was adposed closer in time to **-en* or **ad(s)*. It can however be stipulated that, much as *-e(n)* in 18) and 22) continued as a postposition while beginning to behave as a case affix, **kom*

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continued for a time to serve as a locative auxiliary, taking on in the meantime a comitative sense; Heine, Claudi, and Hünnemeyer argue that the comitative is a more abstract sense than the allative, and therefore reflective of greater grammatical development (159). The preposing in Umbrian of *post, pre, ehe, traf,* and *supu,* along with comitative *com,* indicates that Umbrian eventually followed Oscan in shifting its adpositional pattern after the reanalysis of **ad(s).* The sequence of events may be summarized as follows:

	Umbrian	**Oscan**
Stage I (Proto-Sabellian) (**-en* adposed)	postposing	postposing
Stage II (**ad(s)* allative **kom* adposed)	postposing	preposing
Stage III (**posti, *prai,* etc. adposed)	preposing	preposing

4.0. The Mechanism for the Shift in Adpositional Word Order

It should be explicitly stated that, at least in attested Umbrian, once a lexeme has been analyzed as an adposition, it manifestly does not shift from one side of its attendant noun phrase to the other. Possible evidence for such an event surfaces in Oscan, where the phrase *en eituas* 'for a fine' appears in the Tabula Bantina (2.9). This document, however, is fairly late and may show no small Latin influence. A. Harris and L. Campbell mention just such a shift in Pashto accomplished via the intermediate step of "circumfixes," in which the

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adposition is repeated on both sides of the noun (1995:139), but Umbrian offers no trace of such a mechanism. Nevertheless, evidence of a mechanism is present in Umbrian, but of a different sort.

4.1. Nocentini's Hypothesis.

One fully developed explanation for the adpositional alignment found in Umbrian is that of Nocentini. Nocentini ascribes the enclisis described above to the phonological lightness of the lexemes involved, intensified by the Sabellian shift "da un accento melodico... ad un accento ritmico." This shift "ha l'effeto di accrescere il contrasto fra il costituente accentato e l'elemento clitico seguente" (236). Further cementing this positioning is the sentence accent; here, Nocentini draws on the work of L. Dezső (1980), who, he declares,

> enuncia un universale in virtù del quale in una lingua SOV [*which, Nocentini and Konneker agree, describes Umbrian*] l'accento di frase cade sul costituente che precede immediatamente il verbo oppure, nel caso che il verbo sia preceduto da un elemento fisso, sul costituente che precede questo elemento, secondi gli schemi XV e XYV, dove X è accentato e Y non accentato. (235)

In short, the combination of rhythmic word-initial stress accent and SOV sentence accent provided a slot into which such phonologically slender adpositions as *-ař, -co(m),* and *-pe(r)* could readily fall. On the other hand, prepositions like *pre* and *post* would have been more "pesanti" and would not

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fit into such a schema, although Nocentini speculates that, with the strong cohesion between adposition and noun, the word-initial accent might have been retracted to the preposition (237).

4.1.1. Analysis

Several questions pose themselves in response to this hypothesis. One concerns general typology: instances of accentual considerations affecting adpositional placement are not widely known. A clearly documented parallel would certainly bolster Nocentini's position, but the few scattered Latin relics exemplified by *quoad* and *mēcum* cannot be taken to constitute such a system. As for the word accent, Buck attributes the word-initial stress accent to "the Italic period," but he points out that the accentual status of attested Oscan and Umbrian "cannot be determined" (101). Sihler moreover flatly declares that "Latin is the only Italic language for which we have information regarding accent" (1995:239), although he, too, holds out the possibility that the Sabellian languages underwent a period of initial stress accent (60). Poultney, on the other hand, is rather less diffident, arguing:

> On the quality of the accent in Umbrian and the other Italic dialects we have no direct information, but from the extensive syncope of unstressed vowels we must conclude that at least during a period in the history of these dialects, the accent was characterized by a considerable element of stress. Of the laws regulating the position of the accent in the word and phrase we know next to nothing, but it is generally assumed that the initial stress which was dominant in Germanic, Celtic, and prehistoric Latin

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must at one stage have characterized Oscan and Umbrian as well. (30)

It may reasonably be conjectured, then, that the development of word accent in the Sabellian languages proceeded as Poultney describes. However, the influence of word accent upon the Umbrian adpositions as we know them cannot be assessed with any certainty. Finally, sentence accent remains an elusive and poorly understood concept even in living languages, let alone a tongue such as Umbrian where no linguistic scholarship or data are offered by anyone who could have heard it spoken. Furthermore, Dezső is careful to point out that the "universal" cited by Nocentini "should only be considered tentative and temporary" (22), and that the accentual rules of individual languages will be complicated by the inevitable mixing of types (23), which is obvious in Umbrian's adpositional inventory. Therefore, while it is quite possible that the word-initial accent reliably postulated for Proto-Sabellian had some effect on the placement of clitics in Umbrian (of which, as discussed in section 4.2, the postpositions were undoubtedly such), it is not sufficient to tell the whole story. For example, this proposed schema does not explain *why* one finds in Umbrian a postposed allative *-co(m)* alongside a preposed comitative *com* of comparable phonological weight; the two are manifestly cognate, but how did it come about that there are two distinct forms, one of only occasionally greater phonological weight than the other? An alternative accounting for these adpositions is proposed here, drawing on the more familiar mechanisms of reanalysis and analogical extension.

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4.2. Another Hypothesis

A preposition found not only in Umbrian but also in Oscan and South Picene is *postin*. In Umbrian and Oscan at least this word has a distributive sense, 'for each' (Poultney 149) or 'according to' (Buck 206). Poultney and Buck both derive this preposition from **posti* + the postposition **en* (318, 206 respectively)[10]. Presumably, like most of the less recently derived adpositions in the Italic languages and indeed throughout the Indo-European family, **posti* was one of P. Friedrich's locative auxiliaries before undergoing adpositional reanalysis. In this instance, however, the lexeme could be modified by the adjunction of **-en*, which, as noted, became adposed extremely early in Proto-Sabellian. As a short, phonologically very light segment, **-en* would quite readily have lent itself to the clisis common to all postpositions in the Iguvine Tables. Consequently, with **posti* not yet analyzed as an adposition, or not fully so in any event, and **en* already required prosodically to appear second in its phrase, it follows that an expression such as

15) *posti . acnu* 'in each year' (Vb.8)

could not be produced with a postposition at this point in the history of the language; a phrase **acnu . posti-en* would clearly violate the second-place

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10 This **posti*, as Ernout and Meillet point out (1959:526), is cognate with the familiar Latin preposition *post*, where the final vowel has been lost by apocope (Sihler 68-9).

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constraint. Inevitably, as **posti-en* with its sense quite distinct from **posti* alone came to be regarded within the speech community as a single unit, it will have remained preposed while the reason for such a placement grew obscure. Its analysis as a preposition, moreover, would by strong analogy have prompted a similar placement for **posti* alone as well as its antonym, **prai*. Subsequent adposings of locative auxiliaries and other relational words then followed this pattern.

Thus, to return to the question posed in the previous section about *-co* and *com*, the most compelling explanation is that the former was reanalyzed from an earlier status as a local adverb to that of an adposition prior to the reanalysis of **posti* just described, while the reanalysis of the latter from the same local adverb, this time occurring with a comitative rather than an allative sense, took place after that of **posti*. The prosodic constraint described by Nocentini may well have served to reinforce the reanalysis set in motion by compounds such as **posti* by further discouraging the postposing of phonologically weightier newly-minted adpositions, but the **kom* doublet demonstrates that the constraint cannot have initiated the process. After all, if one were to accept that the uniform retention of final *-m* in preposed comitative *com*, as opposed to its general loss in the postposed allative cognate[11], is a

[11] But not completely so, as in Ib 1, 4; IIIa 28; Va 5; VIb 43, 45.

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consequence of the word-initial stress accent, it would still remain to explain how the two segments came to be on opposite sides of the adpositional object in the first place. They would have been of equal phonological weight before the phonological erosion of the postposed item, in which case this postposing of allative *-com* is unaccountable.

4.2.1. Applications Elsewhere in the Italic Languages

In Oscan, such a compound as *postin* likely had a similarly pivotal role, but the preposing of **az** indicates a slightly different sequence of events for this branch of Sabellian. South Picene simply offers no other prepositions than *postin* with which to test this hypothesis of reanalysis spurred by compounding. Somewhat further afield, while Latin offers some traces of such compound locative auxiliary-turned-adpositions, these compounds, such as *insuper* 'above' and *dēsuper* 'from above', tend further to suggest a quite earlier preposing of its core adpositions when contrasted with *postin*.

Certainly there are tantalizing hints in Latin of a long-lost postpositional word order. Most obvious and most commonly cited are the postposings of *cum* to personal pronouns, as in *tēcum, nōbīscum,* and *sēcum,* and to relative pronouns, as in *quōcum* and *quibuscum.* There is the temptation to view this enclisis as a reflection of the earlier status of **kom* as a clitic particle (as discussed

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in Section 4.1.1; see also Ivanov 1999:110), and it is true that no other adposition in Latin occurs in such an enclitic position as frequently or as regularly. Nevertheless, here, as in Umbrian, the reanalysis of the item as a locative auxiliary is complete in Latin, even if its origins are reflected in its positioning.

In any event, one finds beyond such cases intriguing adverbs formed of compounds such as *quoad* and *quaad* (or *quad*) 'to what extent, to what point in time', and *eāpropter* 'on that account'[12]. However, these adverbial compounds are not consistent in their formation. In addition to the forms in *-super* cited above, one finds *adeō* 'to such an extent', *praetereā* 'in addition to that, besides' as well as *proptereā*, with the same sense as *eāpropter*. Compounds reflecting the well-known prepositional syntax of Latin are thus at least as easy to find as those suggesting an earlier postpositional formation.

As mentioned in section 3.0, the *tedendo* of the Duenos Inscription is also generally regarded as evidence that postpositions were once more widespread in Latin. However, even in that same text, there also appears the phrase *enmanomeinom*, variously interpreted but invariably segmented with *en* as the

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12 The *-per* of *paulisper* 'for a brief while' and *topper* 'quickly, at once' is to some extent a red herring. According to Ernout and Meillet, this postposed item is "Sans étymologie claire. Un rapport avec *per* n'est pas bien visible..., non plus que celui auquel on pourrait penser avec *-pe* suivi d'une particlue *-r*" (497). The fact that this segment is postposed remains noteworthy to this study, of course. But it is not a case of the familiar adposition *per* appearing as a postpositional relic.

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first word[13], thus indicating the preposition of **en* from its earliest attestation in the language[14]. Indeed, postposed items are as rare in the Old Latin texts as presented by A. Ernout (1957) as they are in the Classical stage of the language.

4.2.1.1. The "Second-Place Constraint" in Latin

One allowable and frequently-employed variation in Latin adpositional phrase order is a formation resembling the "Second-place Constraint" found in Umbrian easily recognizable from such expressions as *magnā cum laude*. These may occur with prepositions "light" in phonological weight, as per Nocentini:

23) *magnō cum luctu et gemitu totius ciuitatis, multis cum lacrimis...*
'with great mourning and grieving of the whole state, with many tears...'
(Cicero, *Verr.* 2.4.76)

24) *tempore in omni* 'at every time' (Lucretius I.26)

However, prepositions need not be phonologically light to be so arrayed:

25) *magna ex parte* 'from a large part' (Caesar, *Gall.* I.16.6)

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13 H. Eichner lists the two most widely accepted readings as *en mano(m) meinom* 'zu guter absicht, zu gutem vorhaben'; and *en manom einom* 'zu gutem Zweck' (1988-90:237n99). B. Vine however proposes an interpretation of *en mano(m) meinom* or *en manom (m)einom)* 'as a fine gift' (1999:298-301).

14 As for *endo/indu*, it appears elsewhere as a preverb in the Twelve Tables, albeit sometimes in ambiguous positions that would surely draw comment from Goetze (see section 1.1.2, Chapter II): *manum endo iacitō* (Table I.2); *quis endo eō* (III.3); *librās farris endo diēs datō* (III.4); and as a preposition in metrically congenial positions, as in Ennius *Annales* 272 Sk: *Consilio indu foro lato sanctoque senatu.*

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26) *saxa inter et alia loca periculosa*
'among rocks and other perilous spots' (Caesar, *Civ.* III.6.3)

In striking contrast to the paradigmatic *magnā cum laude* construction, such second-place prepositions may occur at the end of an adpositional phrase, rather than in the middle of one:

27) *hunc post* 'after him' (Cicero, *Tusc.* II.15)

28) *hunc circum* 'concerning him'
(Cicero, *Nat. Deor.* II.105; examples 23)-28) drawn from C. Touratier 1994:719-20)

Of course, these are literary texts by educated authors. Prepositions in second position are essentially absent from Old Latin texts; one rare exception, from the *Lex Cornelia de XX Quaestoribus* (CIL. I^1 202, I^2 587), is in an environment where second position is mandatory in Latin, following a relative pronoun:

29) *quas in decurias* 'to these 10-man committees'.

Apart from such standard environments, however, prepositions are invariably phrase-initial in Old Latin prose. One may speculate that the expansion of the use of second-place positioning for prepositions in Classical Latin is to some extent driven by its frequency in the literary language. In this connection J.

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Adams' observation that Plautus' syntax is in some important respects less conservative than that of Cicero and Caesar is worth recalling (1976:94). Certainly its range in the Classical texts extends well beyond anything found in Old Latin or in Umbrian.

4.2.1.2 Relative Chronology of Latin Adpositional Word Order Change

The other Latin evidence for an earlier period of postposing, too, suggests a different set of circumstances than is found in Umbrian. Formations such as *tēdendo* and *eāpropter* certainly indicate that the prosodic constraints proposed by Nocentini did not operate in Proto-Latin (even if their precise significance for Umbrian remains unclear). The latter item in particular also suggests that at some point in the pre-history of Latin, the postposition of less central, derived (with **-ter*) adpositions was permissible. Mester notes that the accent of the pronominal member of this compound shifts to the second syllable (1994:49), reflecting an encliticizing operation undergone by *propter.*

As a consequence of such formations, it is difficult to postulate a Proto-Italic adpositional syntax neatly accounting for the available evidence of both Umbrian and Latin. Both show definite signs of an earlier postpositional period, but the Latin change to prepositions unquestionably took place far earlier to have left such sporadic traces in the attested language. Indeed, one must be mindful when presented with such evidence as Latin offers of Baldi's warning against circular reasoning in typologically-oriented syntactic reconstruction:

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> Constructions which are harmonic with the supposed reconstructed proto-language are considered archaic and demand no synchronic explanation. At the same time these very constructions are used to support the initial hypothesis concerning the basic typological character of the proto-language (1979:51).

In other words, one must beware in this case of simply assuming a postpositional stage in Proto-Latin, and then taking synchronically anomalous constructions as evidence of such a stage. Nevertheless, evidence outside Latin supports such a conjecture as well; even if one rejects an Italic unity encompassing both Latin and the Sabellian languages, the fact that or the most part earliest-attested the Indo-European languages indicate such a syntactic alignment generally in the proto-language, as discussed in the introduction, casts the supposition of such a pattern in the pre-history of Latin as well in a much more attractive light. This postpositional era in Proto-Latin may well have been in the distant past indeed, but there is insufficient syntactic uniformity in the attested language to postulate that it never existed.

In any event, the balance of this change clearly took place very early, relatively speaking, in the development of this language. Adams, in reporting that the shift in sentential word order from a verb-final to an object-final pattern was well underway but not yet complete in the time of Old Latin, notes, "we cannot assert categorically that it was a change in the position of the verb in early Latin which caused all the other adjustments" in word order, such as Genitive-Noun to Noun-Genitive, Adjective-Noun to Noun-Adjective, and so forth (98). Indeed, the shift to prepositions was all but completed by the time Latin first

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makes itself known to us, and therefore, in this language too, it occurred significantly earlier than that concerning verb and object. However, because these events unfolded at such a remote time, their cause is not visible. Whether this change resulted from a reanalysis based on compounding, as in Umbrian, or from the preposing of such modifiers as local adverbs, as in Lithuanian, or from some other impetus, cannot be determined from the evidence at hand.

5.0. Conclusion

Nevertheless, within Umbrian enough material is available to allow a chronological reconstruction of the events leading to its eccentric adpositional patterning. Consistent with the verb-final word order found throughout the Iguvine Tables, the oldest attested Umbrian adpositions are postposed. However, the development of compounds of locative auxiliaries with postpositions, bound by the prosodic requirement that they appear immediately after the first word in a given phrase, prompted a reanalysis. Thereafter, newly minted adpositions in Umbrian were preposed. Unfortunately, the Iguvine Tables do not span a sufficiently broad period of time to inform us what effects such a development might have had on sentence order as a whole. The brevity and restricted subject matter of the text further limit our understanding of the evolution of Umbrian syntax, along with potential insights into syntactic evolution in general. Such questions would have to be answered through an examination of shifts in the adpositional phrase order of other languages.

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Chapter V:
The Development of Lithuanian and Baltic Adpositions[1]

1.0. Introduction: Lithuanian Postpositions and Prepositions

The Lithuanian language as it first appears to us in written form in the late sixteenth century presents a number of syntactic anomalies in comparison with other Indo-European language branches, and especially with its closest relatives outside of its Baltic group, the Slavic and Germanic languages. Most strikingly, the adpositional inventory of Lithuanian includes some postpositions, along with the prepositions found in its Slavic (and also its Germanic) neighbors. The most prominent of these postpositions, *-na* 'into' and *-pi* 'to; by, near, at', have well-known prepositional cognates. These are found within Lithuanian, in the prepositions (and preverbs) *į* 'into' and *apiẽ* 'about, of'; cognates also appear in the Slavic languages, as in Russian *v* and *o*[2]. Indeed, reflexes of these items are common throughout the Indo-European languages.

The question of how the etymological sources of these items came to be derived in the prehistory of Lithuanian as both postpositions and prepositions is a most compelling one. An account of the prehistory of these segments becomes desirable, as well as a reconstruction of the sequence of events leading to the

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[1] Portions of this chapter were presented in February 2001 at the Conference on Language and Indentity: Linguistic Reality and Linguistic Consciousness in Eastern/Central Europe (14-17th Centuries) at UCLA.

[2] The etymology of *-na* and *-pi* will be discussed in greater detail in Sections 3.1 and 3.2.

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state of affairs found among the adpositional items of Old Lithuanian. Evidence suggests that the postpositions are the more archaic forms, while the prepositional word order took hold and became prevalent fairly shortly before Lithuanian as a written language came into use. The cause of this shift from a postposing to a preposing adpositional syntax may be postulated to lie in the tendency in syntactic orders favoring postpositions to place modifiers, such as some of the adverbs of time or place that often give rise to adpositions, before the modified item. Moreover, these dual adpositional reflexes of a common root offer some insight into the continuous processes by which grammatical markers are replenished. Finally, the influence of neighboring Slavic languages, particularly the East Slavic dialects with which the Baltic speech area shared a boundary through much of its prehistory, in tipping the balance in favor of the prepositional syntax must also be considered.

An adpositional system featuring both prepositions and postpositions is found not only in Lithuanian but also in its Baltic sister-language, Latvian. However, the Latvian adpositional situation is fairly distinct from that of Lithuanian, so that one cannot readily speak simply of "Baltic" characteristics concerning adpositional syntax. For the purposes of the present effort, therefore, features occurring in Lithuanian will command the great majority of attention, above all because Lithuanian is roundly acknowledged to be on balance the more conservative of these two most prominent members of the Baltic sub-family. However, features of the adpositional systems of Old Prussian and Latvian will be briefly considered in Sections 7.1 and 7.2, respectively.

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1.1. Previous Scholarship

1.1.1. D. Klein

Grammatical analysis of the Lithuanian language began fairly soon after it was first encoded in written form. The first comprehensive grammar of Lithuanian was published barely a century after the first attested text in that language by Daniel Klein in 1653. Klein's analysis of adpositions and preverbs falls under the blanket term "præpositio" but captures their basic functions with some accuracy. Klein does however divide these items into two classes: "*Præpositiones* sunt aliæ *separabiles*, aliæ *inseparabiles*. *Inseparabiles* sunt: **ap/api/at/par/pra/da/pi/na**" (140). The separable forms, on the other hand, are further distinguished as either monosyllables, such as *priẽ, be, del,* and *po,* or disyllabes, including *tarpei, apliñk,* or *apiẽ.* Of the inseparable items, he notes further: "*Inseparabiles* componuntur: vel in *principio* tantùm cum Nominibus & Verbis, **ap/api/at/par/pra/da/**...vel in fine tantùm cum Nomine & Pronimine, **pi & na/** quæ ad locum significant, ut: ... **manespi & manesp** ad me, **miestana** & ablato a **miestan** in civitatem" (140-1). He also notes the addition of *-m* to the genitive plural before a final *-pi.* As such, Klein observes what proves to be an important detail from a diachronic standpoint, as will be explained in Section 3.3.

A major shortcoming of Klein's analysis, however, is the lack of differentiation between the "præpositiones" that occur predominantly with nouns as opposed to those generally found with verbs, and those employed only

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with one or the other. For example, *-na* and *-pi* may be as inseparable as *ap-* and *at-*, but the fact that the former are always joined with nouns and the latter with verbs is a distinction that is overlooked here. That *-na* and *-pi* are after all not preposed also does not draw notice as a significant feature. Nevertheless, Klein is correct in finding commonalities between the adpositions and the preverbs, and his work, both on this point and generally, offers a valuable starting point to the Lithuanian grammatical tradition.

1.1.2. The Nineteenth and Early Twentieth Centuries

August Schleicher's *Handbuch der litauischen Sprache* (1856) was the first grammar of Lithuanian informed by the foundations of comparative linguistics laid by the first generations of Indo-Europeanists. Schleicher's exposition of the inventory of Lithuanian adpositions distinguishes prepositions from postpositions (276, 292-3), as becomes standard from that point forward. It was, however, superseded in many respects by August Leskien's work, especially *Litauisches Lesebuch mit Grammatik und Wörterbuch* (1919), benefitting from, among other assets, an additional two to three generations of scholarship. Leskien includes capsule sketches of each of the language's adpositions, although the prepositions are discussed in his section on syntax (219-222), but the postpositions, while described by that term, fall under the heading of "Anhang zur Deklination" (179-181), presaging later debate over whether these items should be considered adpositions or case suffixes.

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Not long after Leskien's work, however, was published an essential starting-point for any subsequent investigation of Baltic adpositions, Ernst Fraenkel's *Syntax der litauischen Postpositionen und Präpositionen* (1929). For every adposition in the language, Fraenkel includes sections on etymology and attested variant forms, along with cognates, especially in the Baltic and Slavic languages; cases governed; and the semantics of each item. This work offers a wealth of information covering Old Lithuanian as well as the various dialects. These adpositions are also included in Fraenkel's *Litauisches etymologisches Wörterbuch* (1962), although most of the content therein appears to be distilled from the earlier monograph. This monograph, however, remains the most complete on this part of speech in Lithuanian to date, although it lacks a broader discussion of the Lithuanian adpositional system as a whole, or an accounting for its anomalies, including the presence of both prepositions and postpositions.

1.1.3. The Mid-Twentieth Century: Endzelins and Stang

The early part of the second half of the twentieth century saw the publication of two major comparative studies of the Baltic language branch, Janis Endzelins' *Comparative Phonology and Morphology of the Baltic Languages* (published in 1971 in an English translation drawn from the Latvian original (1948)) and Christian Stang's *Vergleichende Grammatik der baltischen Sprachen* (1966). Neither work to be sure concerned itself to any significant degree with the Lithuanian adpositional inventory, but they did both advance the position that *-na* and *-pi* should be considered primarily not as postpositions but as case suffixes,

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representing the illative (*-na*), allative, and adessive (both *-pi*) cases (Stang 230; Endzelins 166-7). Both moreover reconstructed Lithuanian locative endings including the singular in *-e* and the plural in *-(u)ose* as arising from a suffixed postposition **en* 'in, into' (Stang 186; Endzelins 135, 137). The question of the status of *-na* and *-pi*, whether adpositional or suffixal, will be considered below. In any event, however, these works taken together represent the modern foundations for a diachronic perspective to the study of the Baltic languages, including their adpositional systems.

1.1.4. The Late Twentieth Century

While Endzelins and Stang proposed reconstructions of the Baltic proto-languages, investigations into developments unfolding over the course of their histories, such as a relative chronology of the agglutinations of *-na* and *-pi* described by those scholars, or indeed the evolution of word order in this sub-group, have only been undertaken in earnest fairly recently. However, both of these gaps in Baltic scholarship were addressed in one volume of the journal *Baltistica*. Zigmas Zinkevičius, in his "Lietuvių kalbos postpoziciniai vietininkai" argues that "postpositional locatives in Lithuanian (inessive [=locative case], illative, adessive, allative) are not formations of the same epoch, but rather they were formed at different times" (1982:38). According to this schema, the formation of the locative with **en* in the singular occurred earliest, while the

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stabilization of other constructions with the various postpositions, especially in the plural, was continuing even at the time of the introduction of writing to the Lithuanian language.

V. Ambrazas, in his "Žodzių tvarka ir baltų kalbų sakinio tipo rekonstrukcija" appearing in the next issue of *Baltistica* (1982), observes the greater frequency of verb-final (OV) sentence patterns, along with other modifier-modified constructions consistent with such patterns, in folkloric texts and more conservative dialects relative to the standard language. He therefore concludes that an OV syntax is the older variety in Lithuanian, and that the language has largely undergone a drift towards a VO model. He also remarks that the shift to prepositional constructions evidently took place rather earlier than other changes in Lithuanian word order (113-4; 118). Ambrazas has thus proposed a relative chronology of word-order change in Lithuanian, and one that underscores the importance of the shift in adpositional phrase ordering to word-order change in general.

Other important works published in recent years include William Schmalstieg's *A Lithuanian Historical Syntax* (1987), which offers a great deal of data throughout the attested history of the language, but does not attempt any reconstructions of its prehistoric antecedents. Zinkevičius' *The History of the Lithuanian Language* represents a translated and condensed version of his five-volume history of the language published in Lithuanian between 1984 and 1992 (1996:v). More comprehensive in scope, however, is Pietro Dini's *Le Lingue baltiche* (1997), with its history, thoroughly supported by linguistic data, of the

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entire Baltic sub-group.

1.1.5. Conclusion

With a solid foundation of Baltic diachronic studies now in place, the course of development followed by the syntax of these languages has begun to be considered in detail. Missing thus far, however, from an account of the shift in adpositional orientation in these languages, the significance of which was indicated by Ambrazas, is a chronology of this change, together with an examination of its mechanics, and the factors, within and without the language, that set it in motion. The present chapter undertakes to answer such questions, emphasizing above all that the process of this shift was as continuous and gradual as the agglutination of *-na* and *-pi* described by Stang and Zinkevičius.

An examination of Old Lithuanian texts selected by Leskien (102-121) as representative of the written language of the time, offering a substantial mixture of Biblical translation which, as for many older European languages, comprises the bulk of extant literature, and original compositions in the form of sermons, provides the starting point for this investigation.

2.0. The Adpositional situation in Old Lithuanian

While the postpositions *-na* and *-pi* occur very frequently in the earliest Old Lithuanian texts, it should be borne in mind that there is at the same time an extensive and heavily-used complement of prepositions at this stage of the language as well. Some of these prepositions will be quite recognizable after this

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effort's previous investigations of the Italic, Hittite, and Southern Anatolian languages, including *apiẽ* 'about', *į* 'into', *per* 'through', *prõ* 'by, through', and *priẽ* 'at, near'. But while Old Lithuanian features a broad array of prepositions, some manifestly ancient and others, as will be discussed, of relatively recent derivation, the presence of postpositions alongside these items is the attraction to the Baltic linguistic sub-family and to Lithuanian in particular for the present study.

2.1. *-na* vs. *į/ing*

Of these two most prominent Old Lithuanian postpositions, *-na* 'into' is simpler in function. This "illative" sense (Schmalstieg 1987:265), in which the postposition is suffixed to an accusative case form, is the only one it conveys:

1) *tada praschau tawe, tiewe, idant tu ghy siustumbei namuosna tiewa mana.*

'Then I ask you, Father, that you send him to the house of my father'
(Luke 16:27, Vilentas translation - 1579)

2) *ir eia wissi pasiraschidinti, kosznas miestana sawa.*

'And they all went to be counted, each to his own city.'
(Luke 2:3, Bretkunas translation - 1591)

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3) *teymi, meldziu, łaukąn ir tereku isz warpu paskuy to.*

'Let me go, please, into the field, and let me gather ears of grain after him.'

(Ruth 2:2, Chilinskis translation - 1662)

This *-na* performs essentially the same function that in modern Lithuanian is expressed by the preposition *į*, also 'into'. Indeed, more modern translations of the same passages, and therefore describing the same concepts, rely on *į* where *-na* is employed in the earlier versions:

1a) *taĩgi, tẽve, meldžù tavę̀, kàd jį siųstumbei į mãno tẽvo namùs.*

(Kurschat translation, 1865)

2a) *Ir ẽjo visì pasiraszýdint, kõżnas į sãvo mẽstą.*

(Kurschat translation)

3a) *leisk manę eiti į lauką ir varpų rinkti paskui tą.*

(1949 translation)

Clearly the function of *-na* has thus been entirely subsumed by *į* in modern Lithuanian, although the *Lietuvių kalbos gramatika* reports that it may still be found in the eastern Žemaitis or Samogitian dialect (K. Ulvydas et al. 1965:I.207). This process of replacement was in fact already underway even in

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the earliest period of written Lithuanian. In some of the same narratives from which some of the sentences cited above are drawn, the precursor to modern *į*, *ing*, occasionally appears:

4) *ir buwa neschtas nuog angelu ing prieglabisti Abrahama.*
'And he was carried by the angels to the lap of Abraham.'
(Luke 16:22, Vilentas translation)

Compare the more modern Kurschat version:

4a) *ir̃ tãpė angėlų nùnesztas į prẽglobstį Abracẽmo.*

The competing constructions may even be found side by side:

5) *Eia taipaieg ir Josephas isch Galileos isch miesto Nazareth*
ing Judawos szeme miestana Dawido.
'And so Joseph too went from Galilee, from the city of Nazareth, to the land of Judaea, to the city of David.'
(Luke 2:4, Bretkunas translation)

5a) *Taĩ pasikẽlė taipojaũ ir̃ Jozẽpas ìsz Galileos, ìsz mẽsto Nãzaret*
į žỹdų žẽmę, į mẽstą Dóvydo.
(Kurschat translation)

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2.2. *-pi* expressing the allative

The postpositional construction, while common, was thus not the only means of expressing the illative at any time in the attested history of Lithuanian. Such inconsistencies are perhaps not quite as striking but nonetheless present in the usage of *-pi* 'to; near, at'. The sense most often conveyed by *-pi* in Old Lithuanian texts is the allative; in this formation, *-pi* is suffixed to the genitive case form of a substantive. This usage is especially common with verbs of speaking to denote the person addressed:

6) *Tare tada Naomi anitosp sawo:*
'Then Naomi said to her mother-in-law:...[3]'
(Ruth 2:20, Chilinskis translation)

7) *bíloia motina Jesaus iopi: ie nebetur wino. Jesus biloia iospi:.*
'Jesus' mother said to him: They do not have wine. Jesus said to her.:...'
(John 2:3-4, Bretkunas translation)

Of course, the more literal sense of 'motion toward' is also frequently conveyed by *-pi*:

[3] Most Biblical texts, of course, have Naomi addressing her daughter-in-law. However, that is not the normal sense of Lithuanian *anyta*. Dini (personal communication, relayed by Aurelijus Vijunas of UCLA) speculates that this reading is the result of a misunderstanding on the part of Chilinskis.

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8) *tie, kurie noretu isch scha iusumpi ateiti, negali, ney taipaieg isch tenai musumpi ateiti.*

'They, who may wish to go from here to you, are not able, nor also can [you] go from there to us.'
(Luke 16:26, Vilentas translation)

9) *ieyk Faranop.*

'Go to Pharoah.' (Exodus 9:1, Chilinskis translation)

As with *-na, -pi* has fallen almost entirely out of use in modern Lithuanian, surviving to convey this allative function only in a few scattered and remote dialects (Ulvydas et al. 209). Its functions have been replaced by other constructions; the person addressed with verbs of speaking is normally indicated by the dative case:

6a) *O Naemi tarė savo marčiai:* (1949 translation)

7a) *tãre Jẽzaus mótyna jám: vỹno netùr. Jẽzus tãrė jeĩ:*
(Kurschat translation)

The more literally allative sense of 'motion toward' is represented in the corresponding modern texts with *pas* and an accusative complement:

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8a) *tẽ, kurẽ norẽtų ìsz czè żemỹn pàs jùs nużeñgti, negál, neīgi isz tẽn pàs mùs péreīt.* (Kurschat translation)

9a) *įeyk pas Paraoną.* (1949 translation)

With *-pi*, however, as with *-na*, the constructions that ultimately took its place may already be found anticipating their modern functions in the earliest texts:

10) *telasau, meldziu, ir rękų warpus pas pedus paskuy pjoweju.*
'Let me scatter, please, and let me gather ears of grain behind the reaper'
(Ruth 2:7, Chilinskis translation)

The Daukša translation, dated 1599, renders the passage cited in sentence 7) as follows:

7b) *táre motina Iésaus iop: wíno ne tur. Ir táre iey Iésus:*

It should be pointed out, however, that Daukša lived and wrote in the central part of the Lithuanian language zone, and may therefore be expected to show some dialectal differences from Bretkunas' western variety (Zinkevičius 238, 246). Even so, later in the same passage, at John 2:7 and 2:8, both translators use that dative plural pronoun with a verb of speaking: *Jesus biloia ghiemus;*

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ghis biloia ghiemus in Bretkunas' text; *Bilo iiemus Iėsus, taria iiemus Iėsus* according to Daukša, with all four phrases meaning 'Jesus/he said to them'. The Žemaitish dialect spoken in the west of the Lithuanian language area and so employed by Bretkunas is regarded as especially archaic due to the influence of West Baltic substrata (Zinkevičius 214), but both translators regard the dative construction as allowable. And while Zinkevičius argues that the use of *-na* and *-pi* was on the whole somewhat less stable with plural substantives, they were rather more settled with illative and allative functions respectively at the beginning of the written language (1982:24). Moreover, in 7b), an example of a dative singular with an allative sense is found. Thus, as these examples demonstrate, the use of these postpositions, while prevalent, was never in the attested history of the language uniform.

2.2.1. *-pi* expressing the adessive

While *-pi* suffixed to a genitive form of the substantive carries a clear allative sense of 'to, in the direction of', *-pi* is also more rarely found with a complement in the locative case, a case which is not governed by adpositions in modern Lithuanian. In such instances, *-pi* is considered to have a meaning of 'at, near, with', and therefore to express an "adessive" sense (Schmalstieg 268-270; see also Leskien 1919:180). This is by no means a common construction in the

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texts consulted for the present effort. One clear example appears in the first known Lithuanian publication, the Mažvydas Catechism of 1547:

11) *kurio prisakimus czia manip regit.*
'...whose commands you see here with me' (Leskien 103)

The distinction between the allative with genitive construction and that of adessive with locative is illustrated neatly in the Chilinskis translation, albeit over a century later and in a central dialect zone rather than the Königsberg area where Mažvydas published (Zinkevičius 1996:230):

12) *ir atliko oszweyp sawo.*
'And she remained with her mother-in-law' (Ruth 2:23)

13) *O ana atajo oszwesp sawo.*
'And she went to her mother-in-law.' (Ruth 3:17)

These are, however, the only examples in the texts examined here of *-pi* with the locative rather than the genitive. Schmalstieg, moreover, reports that this adessive use of *-pi* was apparently on the decline even in the sixteenth century (269), and Fraenkel notes that the locative/genitive distinction with *-pi* was not always observed:

Da auf die Frage 'wo' bei *-pi* sowohl Loc. als Gen. möglich ist, so tritt

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> mitunter der Fall ein daß die Postposition zuerst an einen Lokative, in der Apposition jedoch an einem Genetiv gehängt wird; daher *táwimpi tewop sawo* 'bei dir, seinem Vater' in den maldos krikscioniskos von 1653. (1929:23)

This locative/adessive construction, then, was certainly not as robust as the genitive/allative formation with *-pi*. The two usages are nevertheless obviously distinct, and must both be taken into account in any discussion of the origin and development of postpositional functioning in Lithuanian. The fact that the genitive is evidently the case employed overwhelmingly more often with *-pi* will, of course, also require an explanation.

2.3. *-na* and *-pi* in the context of the Old Lithuanian adpositional inventory

Although in the earliest Lithuanian texts *-pi* and *-na* are in competition with other syntactic formations that involve either prepositions or nominal case forms ungoverned by an adposition, they are all the same vital, productive elements of the language's lexicon at this stage of its history. Both, especially *-pi*, are among the most frequently-occurring adpositional elements in the texts consulted for this investigation. These texts, presented in Leskien (102-121), represent a selection of the earliest-attested examples of written Lithuanian, combining Biblical translations and original writings from the sixteenth and seventeenth centuries. The translations are by far the more copious portion of

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this corpus, and so the Mažvydas Catechism and the sermons by Bretkunas, Daukša, and Šyrvidas offer a valuable test of the authenticity of the syntax of the translations. As such, in assessing the frequency of the various prepositions and postpositions in these Old Lithuanian texts, a distinction must be observed between those appearing in translations and those in original compositions. The numbers of occurrences of the most-frequently employed adpositions thus appear as follows, with sub-totals given for attestations in translated passages and original work:

	translation	*original composition*	*total*
añt + gen. 'on'	58	25	83
gen. + *-pi* 'to'	47	18	65
iš + gen. 'from, out of'	32	19	51
nuo + gen. 'from, away from'	11	8	19
su + ins. 'with'	10	7	17
acc. + *-na* 'into'	17	0	17
ing + acc. 'into'	8	3	11

Given that, as mentioned, the original writings in this sample are less extensive than the translations, the proportions of occurrences of most of the adpositions between the two categories is roughly consistent. The exception is the postposition *-na*, not found in any of the original compositions. This absence may be an accident of the particular texts selected, of course. Still, rather striking

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is the observation of Klein, who declares, in comparing the competing acc. + *-na* and *ing* + acc. constructions, "Elegantius tamen cum inseperabilibus Præpositionibus juguntur" (161). Possibly this may be taken as an indication that the prepositional formation was by Klein's time the more common in mundane language use, even if an assumption that *-na* was therefore moribund at this point would probably be an exaggeration. These postpositional constructions are in any event among the most common adpositional usages in the texts examined here; in addition to those just listed, twenty-two other formations appear in fewer than ten instances each.

2.4. Morpho-syntactic differences between the Prepositions and Postpositions

2.4.1. Verbal Compounding

Despite this great frequency of use, however, some notable differences in morpho-syntactic behavior present themselves between these preposed and postposed items. One is that *-na* and *-pi* evidently do not form compounds with verbs. As in most Indo-European languages, the lexical items acting as adpositions may also serve as preverbs in Lithuanian. Here, in contrast with Hittite, Vedic, and Homeric Greek, but parallel with most of the more recent languages, the segment thus modifying the verb is prefixed to the stem. So, in addition to *eĩti* 'to go', one also finds such compounds as *apeĩti* 'to go around (with *apiẽ* 'about, round about'); *išeĩti* 'to go out of, leave'; *nueĩti* 'to go away,

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depart'; *péreiti* 'to cross, go through'; and *sueīti* 'to come together, meet', among many others.

There is in addition a compound *įeīti* 'to go in, enter', formed with *į* 'into', often found in older texts as *ing*. One may also seek a compound with the competing allative item *-na*, perhaps **naeīti* or simply **neīti*, but such forms do not present themselves, with *eīti* or any other verb. Of course, such forms, if they did exist, would readily be confused with negated verbs consisting of the element *ne-* prefixed to the stem. It is therefore not difficult to imagine that a compound with *į-* would be preferred over any with *na-*, assuming such ever existed. Similarly, *-pi* is the object of some confusion with the preposition and preverb *priẽ* 'near, at', owing to semantic overlap and similar phonetic shapes, to the extent that the form *namopri* 'to/at home' is occasionally found for the expected *namopi* (Fraenkel 17). As such, it would not be surprising if compounds like *prieīti* 'to approach' supplanted any forms as **pieīti*, assuming, once again, that they ever existed in the first place.

The absence of verbal compounds with *-na* and *-pi*, then may simply be the result of accident. One other adposition, *dẽl(ei)* 'through, because of', may, as will be discussed below, be either preposed or postposed, but it too appears not to compound with any verb. On the other hand, this gap may also be the

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product of a syntactic constraint, whereby the postpositional nature of these items precludes their prefixation, even to verbs, in Lithuanian. This is, however, a question outside the scope of the current effort, deserving as it is of further consideration.

2.4.2. Adpositions or case suffixes?

Another characteristic shared by *-na* and *-pi* that distinguishes them from their preposed counterparts is their tendency to take on the apparent qualities of case desinences. Of course, the fact that these postposed elements, unlike the prepositions, are "inseparable" from the substantives they govern was observed as long ago as Klein's work, and this juxtaposition certainly gives such elements the appearance of suffixal rather than adpositional status. Moreover, when a noun governed by *-na* or *-pi* is modified by an adjective or an appositional substantive, the latter will also appear with the same element affixed:

14) *Tassai maxlas roda tikra kiely dewa sunausp*
musu ischganitaiap Iesausp Christausp.
'This doctrine shows the true way to the son of God,
to our savior, Jesus Christ.'
(Mažvydas Catechism, Leskien 104; translation after Schmalstieg 143)

15) *Sugrĕszkime Iónop szwętóp.*
'Let us turn back to St. John.' (Daukša, sermon, Leskien 113)

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16) *O Naomi tare Ruthosp anitosp sawo.*

'And Naomi said to her mother-in-law Ruth.'

(Ruth 2:22, Chilinskis translation)

17) *Teypo nuejo ana anon kulamon aslon.*

'So she went to that threshing floor.' (Ruth 3:6, Chilinskis translation)

From such examples End:zelins concludes, "That these forms are felt as true cases, can be proven by the fact that not only does the noun, but also the modifiers and the words in apposition, take the... ending" (1971:166). Schmalstieg follows suit in regarding these constructions as substantive with case suffix rather than postposition with object. Leskien and Fraenkel, on the other hand, regard them as postpositional. Certainly to some extent this distinction, as with those of some of the categories discussed in the Anatolian languages, is an artificial one. Nevertheless, some features argue for a postpositional rather than a suffixal status for *-na* and *-pi*. These include their relatively discrete shape in relation to the words with which they are attached. In Lithuanian, as elsewhere among the Indo-European languages, case endings with their substantival root form a single word, of which, if the case suffix is detached, the stem cannot stand on its own. Thus, to take the noun *anýta* 'mother-in-law', if the genitive singular ending *-os* were separated from the form *anýtos*, the resulting **anýt* would be a nonsensical item. On the other hand, *-na* and *-pi* are pasted onto forms that are indeed capable of standing alone, as *anitos* is in contrast with *anitosp* in sentence

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16). Indeed, *-na* and *-pi* are affixed onto lexemes that are themselves inflected. This double inflection, while not uncommon among the agglutinating languages of central Eurasia, is most extraordinary among the inflectional morpho-syntactic grammars of the Indo-European family; the nature of the construction appears to support J. Kahr's suggestion of a Finnic influence (1976:126).

In addition, while syncretism is quite common among Indo-European case morphemes, syncretism of number, at least between singular and plural, is extremely rare. Nevertheless, Lithuanian *-na* and *-pi* are more or less equally capable of standing with a singular or a plural substantive, as in *jõpi* 'to him' and *jump* 'to them', or *miestana* 'to the city' and *namuosna* 'to the house'. There are some restrictions on these items in the plural (Endzelins 166, Zinkevičius 1982:24), but, as the preceding examples demonstrate, they are certainly allowable in some environments. In these respects, then, *-na* and *-pi* function more as postpositions than as case suffixes, even if the boundary between the two categories is much fuzzier than where prepositions are concerned. In any event, their usage is similar enough to that of adpositions, and, as will now be demonstrated, their etymological connection with certain Lithuanian prepositions is strong enough, that for the purposes of the present discussion they may be confidently regarded as postpositions rather than as case morphemes.

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3.0. Etymology of *-na* and *-pi*

3.1. *-pi*

Both *-na* and *-pi* have substantial etymological bonds with adpositional lexemes throughout the Indo-European languages as well as in the Baltic sisters of Lithuanian and, indeed, within Lithuanian itself. Very many of these cognates are, however, preposed. Of the two postpositions, the derivation of *-pi* and its relatives is rather less controversial. J. Pokorny posits as a Proto-Indo-European source for this particle the root **epi, *opi, *pi* (1959:323-324)[4]. As such, cognates for *-pi* include Greek ἐπί 'to, on', the occasional postposing of which in the Homeric texts provides testimony for such an alignment for this item in the proto-language (see Chapter I, Section 2.4.4), and Latin *ob* 'in front of, before'. These two prepositions (as well as preverbs) are of course readily recognizable to one familiar with the Classical languages. On the other hand, Vedic *api* 'by, with' is more familiar as a preverb than as an adposition. Not so far afield, the Slavic preposition *o* 'around, at' belongs to this family.

Within Baltic, Fraenkel declares that *-pi* "hängt mit der lett. Präposition *pie* zusammen" (18), and he notes that, while this is predominantly a Latvian

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[4] Katz notes that both the o-grade and the zero-grade of this root have been the subject of some skepticism (1994:157ff.). The o-grade variant, of course, is reliably confirmed most clearly by Mycenaean *o-pi-*, and thus its Lithuanian reflex (see just below) may be accepted with confidence. As for the zero-grade, Katz cites such compounds as Greek πιέζω 'press tight' as derivatives form the Indo-European root **sed* 'sit' (158). While, as such, the zero-grade form **pi* is with greatest certainty attested as a compound form (Katz 159f.), there is no logical reason to rule out its existence as an independent segment as well.

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preposition, even in Lithuanian, such phrases as *py músū, py tèwęs* 'to us, to you' occasionally appear. Still, Latvian, like Lithuanian, also contains relics of postposed *-pi* with adverbs ending in *-p: kaip, taip* 'how, so' in Lithuanian, and *kurp, turp* in Latvian. A more prominent cognate with *-pi* in Lithuanian is the preposition *apiẽ* 'around', also reflected in the preverb *ap-* or *api-* (Pokorny 324). In the Old Lithuanian texts examined for this chapter, *apiẽ* appears ten times, nine governing an accusative complement, as it does in the modern language, and once with a genitive. Its functioning is thus different from *-pi* and Latvian *pie/py,* both of which regularly govern the genitive and convey a somewhat different sense, allative rather than circumessive. Thus are found in Lithuanian several different syntactic and semantic variants derived from the same original root.

3.2. *-na*

Scholars are more divided on the immediate source of *-na.* Stang, drawing on the variant readings of the Old Lithuanian texts, gives two possible allomorphs for *-na*: "Was den Ursprung der Formen betrifft, so glaube ich, dass die Postposition des Illativs **-nā* oder **na* gelautet hat" (229). As for their source, Stang argues: "Man ist geneigt, die beiden auf **nô* zurückzuführen":

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> Diese Form würde im Lit-Lett. **nuo* ergeben. Es ist auch üblich, die Präposition *nuõ* (von) mit pr. *no*, sl. *na* etymologisch zu identifizieren, indem man annimmt, dass die Bedeutung der Präposition von dem nachfolgenden Kasus abhängig war: **nô* + Lok. bedeute *auf*, *nô* + Akk.: *auf* (Richtung), *no* +Ge. (Abl.): von (herab); vgl. gr. para + Dat.: »bei«, +Akk.: »zu«, »längs«, +Gen. »von«. (230)

A semantic shift along the lines of 'from' >> 'direction from' >> 'direction towards' would thus account for the derivation of *-na* from an adposition *nuo 'from, of'.

Endzelins, however, suggests that this illative formation, beginning in the singular, "was formed by the addition of some postposition directly to the accusative ending when it still ended in *-n*. This postposition has now largely disappeared;" in other words, the original shape of the postposition was not **-na* but simply **-a*, from Indo-European **o*. With the subsequently developed plural form in *-na*, "the *-n* could have been taken from the singular" (166).

Endzelins cites Fraenkel in support of his claim for a particle **-a* from **-o*, but his reading of the earlier scholar must be considered somewhat selective. Fraenkel does discuss this change in the postpositional vowel (1-2), but he also declares that such forms as "*vardan(a), vieton(a)* usw. enthalten... eigentlich den Akk. + Postpos. **on*, die mit *en* ... im Ablautsverhältnisse steht" (1). This **en* would of course be cognate with Greek ἐν, Latin *in*, and Umbrian *-en*, and

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Pokorny finds a possible reflex of an *o*-grade variant in Old Church Slavic *vŭ* and modern Russian *v*, from Proto-Indo-European **en* (312). The source of the final vowel of *-na* seems to be from other ablaut variants of **en/on* (Fraenkel 1-2). The final vowel was then reinforced by phonotactic constraints, partly from the need in the plural to avoid an ending in **-sn*, given its suffixation onto accusative forms ending in *-s*, and partly from its use in demonstrative pronouns: "Dies erklärt sich aus der Tendenz der Vermeidung einer einsilbigen Form im Vollbetonten Pronomen" (Fraenkel 4). Singular forms such as *miestana* may then be considered the product of back formation.

The great advantage of Fraenkel's hypothesis is that it relies on a known adpositional root with relatively straightforward phonological and syntactic developments. Endzelins must posit a rather shadowy postposition **o*, and while this does correspond extremely attractively both in sense and phonological shape with G. Dunkel's posited "directival local adverb **o* 'toward'" that he posits as the source of the Indo-European directive case attested most clearly in Hittite (1994:30ff.), it strains credulity to suppose that this tenuous item could have persisted as an independent form over so vast a stretch of time even in the relatively conservative Baltic branch. Dunkel's closing observation is worth recalling in this connection: "It is time, after *5000 years*, for [the directive ending **-o*] to take its rightful place" in comparative Indo-European grammars (35; emphasis added). It was thus in Dunkel's well-founded schema an affixed case

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morpheme even in the time of Old Hittite; it should not be expected to retain its Middle Indo-European functioning in Proto-Lithuanian.

As for the long vowel in Stang's **-nā/-na,* it is less likely to disappear completely as it would have to for illative singular forms ending in plain *-n.* As for an *o*-grade reflex of Proto-Indo-European **en,* one would like to see some clearer derivations in the various descendant languages. However, it remains the most plausible explanation proposed for this form. We are left, then, with a segment that not only competes in Old Lithuanian texts with the preposition *į, ing* in expressing the illative function but is also cognate with that preposition. The question of how these syntactically contrasting formations arose from a common source to present themselves in Old Lithuanian, then, is one commanding especially close attention here.

3.3. Relative chronology of the derivations

Any attempt to account for the presence of these syntactically different but etymologically cognate constructions must first determine their relative ages and then trace the course of their development over the history of the Lithuanian language. Kahr in her discussion of the Lithuanian inventory of morphological case markers refers to *-na* and *-pi* as representing "a new set of postpositionally derived locational cases" (126). However, while their taking on the functions of case endings as detailed above may be a fairly recent event,

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there are grounds for supposing that their status as postpositions is not new. Some of these factors are more or less circumstantial. One has for instance the opinion of Klein mentioned previously that the use of the "inseparable" adpositions is "more elegant"; it is after all a general pragmatic rule that older constructions tend to carry greater prestige than those more recently innovated. In addition, the emerging dialectal distribution of these postpositions resulting in their modern restriction to the isolated upland regions of eastern Lithuania and present-day Belarus reflects a classic pattern of an older feature supplanted by a newer one. Moreover, the nearly ubiquitous employment of *-pi* in particular in Old Lithuanian texts indicates a formation that has been in existence for some time, even if by the same token the great frequency of *iš* and *ant* shows that the prepositions as a class have likewise not only just come into being. Finally, the very fact that the postpositions have taken on characteristics of case markers attests to their heavy use over a long period of time.

However, the most significant basis for attributing a greater age to the postpositions as a class than to the prepositions in Lithuanian, is the relative chronology established by the sound changes within the language. Lithuanian by the time it is first attested has shed most of the word-final consonants it has inherited from Proto-Indo-European; the only exceptions are **-r* and **-s* (Endzelins 53-54). This includes nasal consonants, so that one regularly finds the accusative singular and genitive plural, distinctive in Proto-Indo-European with an ending **-m*, or vocalic **-m̥*, marked in Lithuanian by a vowel, usually

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supplemented by a diacritic to indicate earlier nasalization. A frequently-occurring exception in Old Lithuanian, however, is the retention of the final nasal in the genitive plural when *-pi* is suffixed: *musump* 'to us', *iusump* 'to you (pl.), *yumpi* 'to them'. This survival of an otherwise lost consonant suggests that the presence of *-pi* blocked the operation of this sound change, and therefore, the presence of *-pi* must have been regularized before the change took place. In Fraenkel's analysis: "Da *pi* schon sehr früh suffixal geworden ist, so erscheinen vor ihm noch die Nasale der Gen pl.; vgl. alit.: *piemenump, wissump, Letuwininkump.*" (19).

The final nasal left by *-na* in the accusative singular, of course, reflects a different sequence of events. As Endzelins points out, a consonant rendered word-final by the loss of a following vowel is usually retained in Old Lithuanian (54). There is unfortunately no phonological evidence demonstrating conclusively that *-na* became a regular suffix before or after the loss of the final nasal of the accusative singular; any sequence of **-nna* would have been reduced by the regular degemination of Proto-Lithuanian (Leskien 144). On the other hand, the preposition *į*, cognate to and synonymous with *-na*, clearly reflects the loss of word-final **-n*. A different set of circumstances, however, is reflected in *-na*. Certainly the vowel will have blocked deletion of the nasal, and it therefore

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developed before the rule for the dropping of final *-n arose.

The conclusion, then, that *-na* and *-pi* both assumed their adpositional function before the preposition *į* took shape is inescapable. And while there is no basis for assuming that all postpositions or all prepositions are the same age, there is also no evidence to suggest that in any given language the locational adverbs that give rise to most adpositions are derived into either prepositions or postpositions at random. A language may rather be expected to be syntactically disposed to generate one type of adposition or the other, but not both, at a given time. The change on the part of a language from deriving new adpositions as one variety to the other is diagnostic of a broader syntactic shift in progress. In Lithuanian specifically, postpositions as a class may be regarded as older than prepositions as a class.

3.3.1. Adpositional renewal

Further evidence that the affixation of postpositions was a long-standing means of supplementing, or, in Kahr's term, "renewing" case morphology in Proto-Lithuanian (126) may be seen in the language's locative forms. The locative singular ending in *-(j)e*, found in all declensional types, is generally accepted as the reflex of a former postposition **en* 'in' (Endzelins 135); this replenishment of the locative case marker is therefore exactly parallel to the process observed in Umbrian with the suffixation of **-en*. Here we also have still

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another Lithuanian reflex of the proto-form **en*, this last having become completely grammaticalized by the time of the first written attestation of the language. This postposition-turned-suffix was subsequently extended to the plural of the locative case as well, where the inherited final vowel **-u* was supplanted by variants in *-a* and *-e*. This *-e* was borrowed from the reformed singular (Zinkevičius 1996:115), and ultimately became the standard form.

As Zinkevičius 1982 points out, the process by which **-en* was grammaticalized as a locative case ending and extended to all Lithuanian declensional types and numbers took place over an extremely long period of time. He estimates that of the identifiable postpositional segments developing morphological characteristics, specifically **-en, -na,* and *-pi,* the suffixation of **-en* and *-na* to *a-* and *e*-stem nouns were the first steps towards the situation apparent in Old Lithuanian (21, 38). From this point a pattern developed across the nominal system. The attachment of *-pi* according to this model must have taken place relatively early as it is well-established by the time the language becomes visible to us. On the other hand, Zinkevičius argues that the consolidation of these segments, or rather of **-en* and *-pi,* with the locative plural was still underway at the advent of writing in the language (27, 31, 38).

By this time of course prepositions were also firmly entrenched in the language. It is worth pointing out that a syntactic shift in a language, such as one

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from favoring postpositions to prepositions, does not necessarily herald the imminent end for all representatives of the older word order type. Prominent, well-used constructions of the now-obsolescent variety may remain and, as in this instance, even spread, although any newly-developed formations will follow the newer pattern. In the meantime, the presence of competing constructions and different syntactic alignments, along with, in the case of the Baltic languages, many dialects within a relatively small geographic area, will result in contaminations and borrowings between the two varieties, as in the preposition *py* and postpositional constructions such as *namopri* mentioned in Section 2.4.1. There will always be smaller details that counter-exemplify the overall grand generality.

3.3.2. The role of grammaticalization in renewal

The cognate adpositions known in the various languages more or less closely related to Lithuanian cannot be considered direct reflexes of a corresponding pre- or postposition in Proto-Indo-European; such largely grammaticalized forms with rather slender phonological shapes would not be expected to retain their status over such a long period of time. Instead, these adpositions are continually renewed and replenished, with the same local adverb possibly giving rise to several adpositions. As P. Hopper and E. Traugott observe:

> Renewal results primarily in alternate ways of saying approximately the same thing, or alternate ways of organizing linguistic material. The

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> persistence of older forms and meanings alongside newer forms and meanings, whether derived by divergence from the same source or by renewal from different sources, leads to an effect that can be called "layering" or "variability" at any one synchronic moment in time... Within a broad functional domain, new layers are continually emerging; in the process the older layers are not necessarily discarded but may remain to coexist and interact with new layers. Layering is the synchronic result of successive grammaticalizations of forms which contribute to the same domain. (1993:123-4)

This model explains the seemingly redundant means of expressing illative and allative functions in Old Lithuanian exceptionally well. One point to be added, however, is that we have in Old Lithuanian a situation in which some of the layering under discussion arises from multiple renewals from, if not precisely the same source, then at least from sources that are in turn drawn from a common origin, specifically *-na* and *į/-ing*, both ultimately from **en*. Certainly the presence of multiple derivatives from single roots filling similar functions can best be explained by such a conception. In this schema, newly grammaticalized items repeatedly being pressed into service to replace, or perhaps supplement, forms that are becoming phonologically and semantically eroded towards nothingness. These newly-minted adpositions would be either preposed or postposed depending on the syntactic conditions at a given point in time, but in either case with the same semantic sense.

In this instance, *-na* was analyzed as an adposition while the orientation of the Proto-Lithuanian adpositional phrase was postposing , while *į/-ing* took

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on such a function after the change to prepositions (as will be elaborated in the following section). The case of *-pi* is not quite so symmetrical, since its semantic domain has been split between the preposition *pas* and the dative case; *pas,* then, represents an instance of Hopper and Traugott's renewal from a different source. But both sets represent excellent examples of older and newer "layers" existing side by side in Old Lithuanian.

4.0. Word order change and adpositional shift

On the whole, however, the emerging predominance of prepositions over postpositions in Lithuanian has attracted a certain amount of comment with regard to studies of language universals. As is often the case for a language with a rich inflectional system, word order throughout the attested history of Lithuanian has been quite flexible. Nevertheless, the general statistical preponderance found in Ambrazas' 1982 study drawn from dialects and folkloric texts, both of which may be expected to show a significant measure of syntactic conservatism, favors a verb-final word order. Consistent with such a sentence structure is "the majority of phrase constituent patterns in which the moderator (operator) precedes the modified (operandum): AdjNom, PronNom, NumNom, GenNom; AdvV, Adv'Adv, etc." (118). One glaring exception to this predilection for SOV syntax is the new ascendancy of prepositions, where postpositions are found to correlate overwhelmingly with an SOV pattern in J. Greenberg's 1966 landmark study of word order universals. Schmalstieg goes so far as to conclude:

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> Exactly why did the prepositional constructions gain the upper hand over the postpositional constructions? The theory of word order development [arising in the wake of Greenberg's work] predicts exactly the opposite of what did happen... [T]he facts adduced by Ambrazas must be considered evidence that the entire theory of word order universals is deficient. (327)

However, according to this work by Ambrazas, specifically another study from 1986, cited by Schmalstieg, this shift in favor of prepositions has its origins in an expected feature of SOV syntax. Given a phrase structure in which the modifier regularly precedes the modified item, as is expected of an SOV pattern, adverbs, too, may normally be found before the verbs, adjectives, or other adverbs with which they are associated (Schmalstieg 321-324). And, as Schmalstieg observes,

> In Lithuanian, as in other related languages, prepositions and prefixes had their origins in earlier adverbs or preverbs. In fact the change of adverbs into prepositions is in Lithuanian a live process: such words as *apliñk* 'around', *artì* 'near', *gretà* 'side by side', *išilgaĩ* 'along', *pāskui* 'after', *pirmà* 'before', *príešais* 'opposite', *skersaĩ* 'across', *šalià* 'beside', *viršuñ* 'above', *viršùj* 'above' are sometimes used like adverbs and other times like prepositions. (326)

It is, to be sure, not shocking by any means to find a language that fails to fit one of the canonical sentential word-order patterns with complete consistency. And since, as Ambrazas points out, modern standard Lithuanian

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favors SVO orderings to a greater degree than did the earlier language, this shift must have begun somewhere among the various phrase-structure rules. Most striking for this study is Ambrazas' observation that prepositional constructions emerged in Lithuanian along with other features of SVO patterning, such as in "the standard of comparison (AdjSt) and major sentence constituents (V, O)" (1982:118; see also 113-114). Indeed, this new tendency to reanalyze locational adverbs as prepositions rather than as postpositions based on the preferred placement of these adverbs was evidently among the first syntactic features to break from the typical SOV pattern, and as such it most likely helped precipitate the shift towards a VO syntax.

Nevertheless, the position that the reanalysis of preposed spatially-oriented adverbs as prepositions necessarily made the shift from a verb-final to a verb-medial syntax is by no means demanded by the evidence at hand. Ambrazas does not argue that Lithuanian has become a predominantly SVO language; he rather indicates only that SVO syntactic features have come to occur with greater frequency. Moreover, he notes a historical "ambivalence between (S)OV and (S)VO types in Baltic" (118), with neither variety ever truly eclipsing the other, despite an overall tendency to favor constructions of the SOV pattern. Even in modern Lithuanian, SOV-favoring formations such as genitives preceding head nouns are regularly encountered.

5.0. Possible external influences

In short, the appearance of prepositions in prehistoric Lithuanian was no guarantee that they would subsequently emerge as the normal adpositional

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formation. There has been speculation that external influences played a significant role in the ascendancy of a number of syntactic features in Lithuanian. Given the presence of postpositions in the process of agglutinating onto substantives to form case endings, Kahr suggests that the influence of the similarly agglutinating Finnic languages is at work in Lithuanian (126). Dini points out that, while evidence probably indicates a greater Finnic influence on Latvian than on Lithuanian (152), this is not to say that Finnic influence on the latter language is absent. However, as discussed in Section 3.3, the postpositions have as a group clearly been present in Lithuanian longer that the prepositions as a group. One may therefore reasonably suppose that Finnic influence may have led to the longer survival of these forms (although, as will be seen in Section 8.0, their use is rather less extensive in Old Latvian), but the position that such influence led to this presence in the first place is rather less tenable.

More intriguing, on the other hand, is the question of whether any external factor may have helped drive the change to prepositions. Dini posits the division in Proto-East Baltic leading to the emergence of separate Lithuanian and Latvian languages as taking place over the course of the first through the fifth to seventh centuries AD (171-2). The fact that Lithuanian *-pi* is postposed while Latvian *pie* is preposed thus poses a convenient *terminus ante quem* for the separate (although parallel) developments of their respective adpositional syntactic patterns. In short, the shift in adpositional word order will have taken hold sometime around or after the mid-first millennium AD, even if linguistic conditions were already becoming favorable for such an event while Common

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East Baltic was still a vital entity.

As for a *terminus post quem* for this change, by the time Old Lithuanian appears as a written language in the sixteenth century, postposing was not a productive means of grammaticalization. Even if, as Zinkevičius 1982 argues, postpositional constructions were still becoming crystallized (see especially Section 2.2), new adpositions by this point were being preposed, and the prepositions were well-established. External influences between the mid-first millennium and the mid-second millennium are therefore to be considered as possible factors in this development.

Indeed, at the end of the fourteenth century, a Lithuanian state had grown into a major power in Eastern Europe, and Polish influence in Lithuania was growing rapidly, both for dynastic reasons (Lithuania's Grand Duke Jogaila became king of Poland as well - Zinkevičius 1996:79); and religious (Poles dominated the clergy that established dioceses in Vilnius and in the western Žemaitisch region - 80). But by this time, prepositions were most likely already prevalent. This was after all shortly before the first attestation of written Lithuanian, and the many prepositions found in these early texts cannot have been derived as such overnight.

More likely candidates include the East and North Slavic dialects, or in the latter case more accurately Northwest Slavic, as typified by the Old Novgorodian birch-bark texts. Zinkevičius posits the first contact between Lithuanians and East Slavs as occurring in the ninth century AD (1996:54), and notes that "From

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the 10th c. the Slavic expansion into the Baltic lands was motivated by the mission of Christianizing the Baltic tribes. Christianity accelerated the Slavicization of the Baltic areas in the east and was apparently the most important determining factor in that process" (55). Two centuries later, a period of eastward Lithuanian expansion began (55-6). The period from the tenth to the twelfth or thirteenth centuries, then, was a time of especially intensive contact between the Lithuanians and East Slavs. Dini details extensive contacts as well with speakers of North Slavic dialects (143-4). This period of time certainly corresponds with that of adpositional word order change in Lithuanian, and the Slavic languages throughout their attested history have been solidly prepositional. Unfortunately, research on the linguistic effects of Baltic and Slavic contacts has concentrated overwhelmingly on the lexical sphere. The question of syntactic influences by the East and North Slavic dialects on the Baltic languages, however, would be a most fruitful avenue for future research. For the present, in this case, the debate on such influence as a factor in this clear instance of word order change in Lithuanian remains open.

6.0. Other Lithuanian postpositions

6.1. *dẽl* and its synonyms

While *-na* and *-pi* are the most prominent Old Lithuanian postpositions, and most regular in that they are found only postposed to the substantives they govern, several less-commonly employed adpositions may also occur postposed. Many of these are among the forms cited by Schmalstieg above as adverbs that

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have recently taken on adpositional functions. However, of this class of items with ambivalent word order in adpositional phrases, the only one found in a postposed position in the texts consulted here is *dẽl* 'through, for the sake of', listed by Schmalstieg simply as a preposition that may occasionally be postposed (277). This item is subject to no small amount of variation in form, appearing sometimes as *dẽlei,* and then a synonymous and similarly ambivalent form syntactically is *drin(ei),* which through contamination with *dẽl* may also appear as *dril(ei)* (Fraenkel 230). Fossilized conjunctions reflecting a pronoun with a postposition such as *kodẽl* 'why' and *todẽl* 'therefore' may also appear as *todrin, kodril(ei),* and so on (Ambrazas 1982:109).

In the texts examined for this effort, *dẽl, dẽlei* or *drin* occur in a total of seven instances, all in sermons by Bretkunas, Daukša, and Širvydas. Each of the three authors is consistent in his usage. Bretkunas has *dẽlei* postposed to a genitive: *Schito delei* 'because of this'; *Togi delei* 'on account of that' (Leskien 112). Daukša has *drin'* before a genitive: *drin' padérmés* 'through accomplishment'; *drin' bursós* 'though chance'; *drin' tụ* 'because of that'. Daukša also uses a conjunction *todrin'* for *todẽl* (Leskien 114). Širvydas uses *dẽl* before a genitive or an instrumental: *del rubu* 'on account of the garment'; *del ukanos*

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'because of the ocean' (Leskien 115-116).

The explanation that leaps to mind based on the admittedly tenuous evidence for this jumbled state of affairs surrounding *dẽl* and its synonyms is that several different derivations of the lexeme took place in different dialects. Bretkunas lived and wrote in what became Prussia, in the western part of the Lithuanian linguistic region, while Daukša and Širvydas were in the more central areas. The pervasive use of conjunctions such as *kodẽl* and *todẽl* show that at one time *dẽl* was a postposition in all, or at least nearly all, Lithuanian dialects. While the clearest cognates sometimes claimed for this adposition are in Slavic items such as Russian *dlja* (as by Fraenkel 224), even these are more likely borrowings, as will be discussed in Section 7.3. And related forms farther afield among the Indo-European languages are difficult to find. For *dẽl*, Fraenkel suggests as a source the Lithuanian verb *dẽti* 'to place, put' (1962:87), whence, after a series of intermediate steps, a derived adverb *dẽlei* ultimately became an adposition. However, the many different forms for this item strongly suggest multiple derivations, presumably at different times and places. Quite likely the fossilization of the postposition in the conjunctions mentioned above spurred these new reanalyses as a means of renewing the means for expressing the concept involved. Subsequently contact between the various dialects will have led to contaminations and inconsistencies in the emerging standard language;

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even in modern Lithuanian, such phrases as *idumumo dẽlei* 'for the sake of interest' are allowable[5], although preposed *dẽl* is the most common form. But this inconsistency is the product of separate reanalyses to full adpositional status at various points among the various dialects.

6.2. Other items

In addition to *dẽl*, some other less common adpositions listed by Ambrazas as occasionally postposed include *liñk(ui)* 'toward', *pagal* 'along', and *pãskui* 'after' (1982:109). These are also representatives of Schmalstieg's set of adverbs recently turned prepositions, which Leskien classes "Von den sogenannten uneigentlichen oder unechten Präpositionen, d.h. Kasus von Nomina, die sonst noch in der Sprache gebräuchlich oder als Nominalkasus erkennbar sind" (222). For forms like *liñkui* and *pãskui,* inflectional endings are indeed plainly evident. Typically, as a lexeme becomes more firmly analyzed as an adposition, as the form *liñk* indicates, the case ending is dropped. The clearest illustration of this process may be observed in the preposition *tarp* 'between', derived from the noun *tárpas* 'space between'. In Vilentas' translation of Luke 16:26, however, the instrumental case of *tárpas* is used as a preposition governing the genitive: *tarpu musus ir yusu* 'between us and you'.

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[5] I am grateful to Aurelijus Vijunas of UCLA for bringing this information to my attention.

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The ambivalent syntax of these items may be explained by their ambiguous, or fuzzy, status as parts of speech. In many instances, they function to some extent as inflected substantives and, perhaps more often, to some extant as adverbs in addition to showing some adpositional characteristics. And while in Lithuanian a fair amount of flexibility in word order is acceptable, it is generally the case that adverbs precede the word or phrase they govern, while a substantive has a strong tendency to follow another noun, usually in the genitive, by which it is modified. Presumably, however, as the lexeme in question comes to be regarded more purely as an adposition, the word order of its phrase will become more firmly fixed.

7.0. Adpositions in Closely Related Languages

7.1. Old Prussian adpositions

While the combination found in Old Lithuanian of profound grammatical archaism and a substantial corpus of texts make that language the best member of the Baltic subfamily to examine for any topic in comparative Indo-European syntax, some striking forms appear in its sister languages as well. The evidence by which Old Prussian is known to us is too fragmentary to expand on our understanding of Baltic adpositional functioning in any significant sense. The known texts yield prepositions, many already familiar from Lithuanian: *en, an* 'in', *bhe* 'without', *ìs* 'out of', *no, nà* 'on, after, according to', *prei* 'at, to' (Schmalstieg 1974:144-145).

One astounding exception to this familiarity, however, occurs in a

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translation of the *Pater Noster* prayer analyzed by E. Mikalauskaite 1938, where the form *andangonsun* 'in heaven' appears. Endzelins sees the final *-un* as an illative suffix cognate with *-na* (167). The initial *an-*, on the other hand, is the Old Prussian preposition just mentioned, corresponding with Lithuanian *į*, serving to "strengthen" the "illative or locative" (Schmalstieg 1976:163). This double adposition is certainly a departure from the Lithuanian data. One may speculate that the former postposition had become so bleached of its sense as no longer to register as a meaningful segment; as such, the reinforcement of the preposition would be required. Still, one must restrain oneself from reading too much into a single form, and all the more caution is demanded given that Stang expresses severe doubt about this reading of the text (1954:16). Moreover, Old Prussian, unlike Lithuanian, has not dropped word-final nasal consonants; the accusative singular in *-n* was consequently quite common in Old Prussian. With the tenuous data we have on Old Prussian, especially, for the purposes of the present question, concerning its lexicon, the possibility that *dangonsun* is a noun in the accusative simply cannot be ruled out; indeed, V. Toporov lists it as such (1975:297).

Of note, however, is the fact that in many respects, Old Prussian bears greater affinity with Slavic than with the East Baltic languages; this is the case in the Old Prussian adpositional inventory as well. Especially striking is the small set of items ending in *-dau*, including *pirsdau* 'before' and *sirsdau* 'among'.

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Stang proposes a correspondence between *-dau* and the Slavic derivational morpheme *-dŭ* found on several local prepositions, although he admits that the phonological particulars are somewhat difficult (1957:238). The Slavic prepositions include **perdŭ*, attested as Old Church Slavic *prědŭ* and Russian *pered* 'before, in front of'. The root, according to Stang, is clearly cognate with Old Prussian *pir-š-*, with the sibilant representing a form of *š*-lengthening common in Baltic, as in Lithuanian *priẽš*, or Old Prussian *preisiks* 'enemy, adversary' (237). Stang points out that some of the Old Church Slavic items, including *prědŭ*, may also function as substantives (236), thus offering additional evidence for the nominal origins (and not necessarily dating back to Proto-Indo-European times) for most of these adpositions.

7.2. Latvian adpositions

At the other end of the Baltic linguistic area, Latvian, like Lithuanian, offers an extensive literary corpus dating from the sixteenth century as well as remarkable morpho-syntactic developments continuing to the present. On the other hand, as a result of an accentual shift preventing the morphological conservatism for which Lithuanian is well-known, along with substantial linguistic interference from Finnic neighbors, Latvian is a less fruitful field for comparative studies. Nevertheless, some questions of Latvian historical grammar deserve comment, shrouded in controversy as they are.

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The first such question is the fate of *-na* and *-pi* in Latvian. There is general agreement that *-pi* remains in a few relic forms, specifically directional adverbs ending in *-p* (Fraenkel 1929:18; Endzelins 1971:167); the presence of *-na* is less easily ascertained. Endzelins declares, "Illative [in *-n*], as productive case endings, are found (in the singular!) in the very first Latvian texts" (166-167). T. Nilsson 1997, however, takes note of the extremely difficult quality of the early Latvian manuscript tradition, observing, "Old Latvian texts are, as is well known, at times almost incomprehensible due to the admixture of Latin, German, and perhaps even of other source languages both in vocabulary and syntax" (91-92). A problem of particular relevance to the question of the illative in *-n* in Latvian is a persistent vacillation in the manuscripts between *n* and *m* where *m* should be expected on etymological grounds (92). For example, Nilsson cites:

> *isschim laikan* 'until the present time', cf. Modern Latvian *lidz sim:* here, the lack of congruence between *sim* and *laikan* can ONLY be viewed as a misprint for **laikam*. Whether these types of errors are due to the printers or to the scribes, can of course only be determined after collation with the original manuscripts, which unfortunately are no longer extant except in very few cases. (93)

On the basis of this textual evidence, Nilsson argues that these forms should be considered substantives in the locative case or, in some instances, the dative. As

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for the purported illative, however, "to set up a postpositional *-*en* here, just on the basis of its attested evidence in Lithuanian would be simply unjustified" (99).

Like Lithuanian, Latvian has some adpositions that may be either preposed or postposed, including *dẽl'* as well as *pêc* 'after' (Endzelins 1922:480, 523). In the modern Latvian language, moreover, one finds an array of lexemes exhibiting adpositional characteristics even more elaborate than those discussed for Lithuanian by Ambrazas and Schmalstieg. A. Holvoet 1993 argues for the presence in Latvian not only of postpositions (*durvju prieksa*) but even of circumpositions (*pa saknes, siekstas apaksu*) (131, 133). As is often the case in Lithuanian, these postposed items are drawn from inflected nominal forms. Indeed, partly on this basis, S. Lagzdina 1998 argues against labelling such words as adpositions. Listing some strict criteria for determining just what is an adposition (for example, prepositions, unlike nouns and adverbs, are function words and cannot stand alone; prepositions are grammatically obligatory, while adverbs are merely "grammatisch fakultativ"), she argues for considering such items as adverbs, specifically local adverbs, instead (165). Still, as such words in Lithuanian may demonstrate, these are lexemes that could perhaps take on more explicitly adpositional qualities in the future, eventually becoming reanalyzed as unambiguous adpositions. Nevertheless, it is apparent from the topics introduced here that the morphosyntactic structure of Latvian is on balance of greater interest from the standpoint of the universals of language evolution, with its development of new postpositions and even circumpositions, than that

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of comparative Indo-European studies. It is complex and deserves to be analyzed in depth, but such analysis is outside the scope of the present project.

7.3. Slavic Adpositions

While the Slavic languages are of course not so closely related to Lithuanian as Old Prussian and especially Latvian are, and indeed some questions about the extent of the relationship of Baltic and Slavic remain to be settled, nevertheless commonalities between the adpositional inventories of the two language groups are sufficient that this is a natural place to discuss the latter set. The Slavic languages are far more resolutely prepositional than their Baltic neighbors. There is great vacillation over the basic word order of Common Slavic and the earliest-attested Slavic languages, with H. Andersen listing it as SVO but subject to variation for pragmatic reasons (1998:451), while J. Friedrich declares that in these older languages and in Slavic folk literatures, "VSO is virtually uniform in narrative passages," while "[v]erb-final is the main alternate," again for pragmatic purposes (1975:62). Then too, A. Vaillant declares that the order of accent-bearing words "est en principe libre" (1977:255). However, the order of the prepositional phrase has never been in doubt.

Vaillant, however, identifies some exceptions. One, *dělja* 'because of', may be reliably connected to Lithuanian *dẽl*, examined in Section 6.1 above. The short vowel in the Slavic form indicates that it is borrowed from Baltic rather than a true cognate, with the adverbial morpheme *-ja* added (134). Another, *radi* 'because of' as well, is also borrowed, but from an Iranian language, where

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it also occurs as a postposition[6] (134). As such, the syntax of these items is not difficult to explain. One other postposed form, which too has the sense of 'because of', is *cěšta*. This Vaillant traces to a root **kait-*, which also manifests itself in the verb *čite-* 'to suppose, calculate' (135). Beyond that, its derivation is murky, but it unquestionably does not reflect a local adverb. All three of these items, however, survive into modern Slavic languages (V. Maldjieva 1995:49).

There is, however, a hint of a postpositional alignment in the prehistory of Slavic. The locative singular for masculine and neuter nouns of consonant stems in **-n-* and **-s-* is **-e*, where the more common pattern throughout the Indo-European languages and indeed in most Slavic stem types shows a locative in **-i* (H. Bräuer 1969:54). This **-e* is, like the same ending in Lithuanian locatives, traceable to a postposition, which was suffixed to a previously endingless form (J. Mikkola 1950:47). Here, then, we likely have yet another manifestation of **en*. It may be supposed, then, that Common Slavic had a postpositional syntax long before its descendants were first documented. Moreover, given that attested reflexes for **en* in the Slavic languages include such prepositions as Old Church Slavic *vŭ*, Russian *v(o)*, Polish *w(e)*, and Serbo-Croatian *u*, there is additional evidence for repeated adpositional reanalyses of this old locative auxiliary over the course of the histories of several tongues.

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[6] The development of this item over the history of Persian is discussed in Chapter I, Section 2.4.5.

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8.0. Conclusions

The evidence from Lithuanian, however, offers an excellent basis both for comparative study and discussion of universals of language development. Unlike most known Indo-European languages, Old Lithuanian has an adpositional inventory that includes a substantial mixture of both prepositions and postpositions. While there are more prepositional lexemes in the language, the most prominent of the postpositions, *-na* and *-pi*, are as heavily used as the most common prepositions. Moreover, as has also been observed in Umbrian and Oscan, these postpositions have also begun to take on the characteristics of case desinences. Usage of these postpositions is quite productive, suggesting that they have not filled this role for so long as to have become fossilized forms devoid of specific meaning. At the same time, however, the retention of the *-m* of the genitive plural before *-pi*, in contrast with the general deletion of final nasals in Old Lithuanian, shows that the postpositional formation is at least older than this sound change. This change is evident in such prepositions as *į*, and given the cognate status of *-na* and *į*, the presence of */n/* in the former indicates its earlier derivation as well. This set of circumstances allows us to establish a relative chronology whereby the postpositional class in Lithuanian is older than the prepositions; at some point in the past, Proto-Lithuanian had a postpositional syntax.

The shift to prepositions was evidently precipitated by a feature of the same SOV word order that normally favors postpositions, specifically the

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tendency to place modifiers, including the local adverbs that generally give rise to adpositions, before the modified items. Certainly the most remarkable fact in the evolution of Lithuanian adpositional phrase structure is the cognacy of *-na* and *-pi* with not only prepositions in other Indo-European languages but within Lithuanian as well. That *-na* and *į*, like *-pi* and *apiẽ*, are cognate within the language indicates the presence of a common source, presumably still a local adverb with fairly flexible word order requirements, at a relatively late stage in the proto-language. These cognate adpositions in Old Lithuanian may be considered to have arisen at different times over the history of the language by similar processes of grammaticalization; the fact that these constructions are found coexisting in the earliest Lithuanian documents, however, should not obscure this difference in age. Processes along fairly parallel lines took place in the other Baltic languages, but after Proto-Baltic and even Proto-East Baltic had begun to split up, resulting in somewhat different adpositional inventories. Such a course of development thus described offers the most comprehensive explanation for these unusual adpositional situations, especially that in Old Lithuanian.

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Chapter VI:
Adpositions in
the Other Indo-European Branches

1.0. Introduction

The present undertaking has concentrated on Hittite, Luwian and Lycian, Umbrian and the rest of Italic, and Lithuanian and the rest of Balto-Slavic as they offer the most substantial bodies of evidence for the development of adpositions, including their origins and changes in their syntactic usage, among all the attested Indo-European languages. This is not to say, however, that the other languages or branches of the Indo-European family are devoid of such evidence. There are in many other Indo-European languages varying amounts of traces of such events as well. These traces, insofar as they are discernible, are remarkably consistent in what they reveal about Proto-Indo-European syntax and the development of adpositional syntax in the individual language branches. Evidence supports the proposition that the proto-language was for all intents and purposes postposing, but the drift towards prepositional patterns began to take shape in each of the daughter branches apart from Hittite, suggesting it was underway before the general dispersal of the various dialect communities. While this phenomenon has already been observed in Chapters II-V, the remaining language groups offer further corroboration. A brief survey of the other branches of the Indo-European family, then, is in order, to mention the residues in these groups of earlier steps in their adpositional history, as well as to suggest fruitful avenues of future study.

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2.0. Tocharian

The adpositional syntax of Tocharian bears the hint of a most remarkable past. Tocharian is generally described as having a rigidly, virtually canonical SOV syntax, the expected postpositions included. Certainly there are many postpositions readily recognizable to one familiar with other Indo-European languages:

1) *Śrāvasti riyā* ***pre***

'outside of the city Śravasti' (Toch. A 395 b2 - O. Hackstein 1997:45)

2) *saryat=ompĕ poyśintse asāṃ* ***spe*** *kenne witskaṃ*

'he planted there by the Buddha's seat in the ground roots'

(Toch. B 388a2 - D. Adams 1999:718-9)

3) *śeśuwer* ***postäṃ***

'after the dinner' (Toch. B HMR 2a1 - Hackstein 50)

4) *tā͡u ytarimeṃ* ***parna*** *lantsi*

'to go out beyond this road' (Toch. B 330a3 - Adams 359)

The *spe* of 2) is cognate with Latin *sub*, Greek *ὑπό*, and Sanskrit *úpa* (Hackstein 40; Adams 719). In *parna* is reflected "PTch **päränā-i* from an earlier **pärärā-*, exactly matching in stem-formation Proto-Germanic **ferera/ō* seen in

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Gothic *fairra*, Old Norse *fjarri*, OHG *ferro*, Old English *feor(r)*. Both Germanic and Tocharian reflect a (late) PIE **perero-* 'beyond, far'" (Adams 359). Cognates of the other two postpositions are self-explanatory, with Adams finding in the final syllable of *postam* a reflex of the Proto-Indo-European particle **nu* (405).

2.1. Prepositions in Tocharian

However, in addition to these postpositions and others, there are in Tocharian some prepositions as well. A prominent example is Toch. A *sne*, *snai* in B, both with the sense 'without':

5) ***snai*** *śawaṃñe*
'without arrogance' (Toch. B 20a3 - Adams 712)

6) *wnolmi [tan]e* ***snai*** *spelke*
'beings here [are] without zeal' (Toch. B 3a6 - Adams 712)

In 6), *snai* thus governs *spelke* 'zeal'. The derivation of this item is not secure, although A. Ernout and A. Meillet connect it most closely with Latin *sine* and Old Irish *sain* 'different', along with, less directly, Greek *ἄνευ*, Sanskrit *sanitúr* 'besides, without', and Gathic Avestan *hanarə* 'without' (1959:628). Opinions differ, moreover, as to the implications of the unusual syntactic requirements of this item. Hackstein argues that, because it is not a "Richtungswort" like the majority of core adpositions in Tocharian and elsewhere (55), and, unlike such

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directional forms, cannot serve as a preverb, which he regards as the source of these Indo-European adpositions (51), it does not represent a true inconsistency within the Tocharian adpositional system. It is likely an archaism (54), but it is outside the inventory of locative auxiliaries.

On the other hand, Hackstein mentions another Tocharian preposition, A *śla*, B *śle/śale* 'with', but does not discuss it at length (54). This item has few likely etymological connections in the other Indo-European language, although V. Ivanov suggests that cognacy with Russian *dlja*, and therefore Lithuanian *dẽl*, is possible (1995:24). Within Tocharian, however, is the comitative case suffix for A, *-(a)śśäl* (J. Penney 1989:58, G.-J. Pinault 1989:76)[1]. One may thus recognize a situation similar to that of Lithuanian **en*, with different morpho-syntactic rules for reflexes of the same root, presumably derived at different times.

2.2. Reflexes of **en*

Even more striking, reflexes of this same **en* also surface in Tocharian in various positions. These include the locative case desinence A *-aṃ*, B *-ne* (Pinault 75; Adams 341); an intensifying prefix A *a(n)-*, B *e(n)-* (Adams 82-3)[2]; and another

---

1 This item in both A and B serves also as a conjunction, with the sense 'and' (Thomas 1979:155). This is an occasional further step in the grammaticalization of some local adverbs, as with Sanskrit *api* (see Gonda 1968:189ff.).

2 Adams writes that *e(n)-* "was originally a locative prefix meaning '± in' (and certain forms which contain it still have traces of than meaning remaining, cf. *eṣe* 'together', i.e., 'in one')".

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prefix, at least in B, *y(n)-* 'in, among' (Adams 516; J. Pokorny 1959:312). Both *e(n)-* and *y(n)-* occur in many compounds, by no means limited to adpositions or local adverbs (see Adams 82, 516). However, these are the parts of speech of interest here. The adpositional compounds are mostly postposed, such as *ekalymi* 'in the power of, subjected to', from **en* plus *kalymi* 'direction'[3]:

7) *pelaiknetse ekälymi nestsi preke*
'[It is] time to be subject to the law' (B - 281a4/5 - Adams 76)

Some compounds include as second members adpositions already discussed:

8) *śama=nepre poyśintse*
'she stood before the Buddha' (B 18b8 - Adams 84)

9) *päls[k]osa erwts=etrenkätte [sic] cimpā=neśle takoy mñrta*
'may he be broad of spirit and unencumbered like thee'
(B 248a3 - Adams 84)

Some of these compounds, however, are prepositions:

10) *walo kauṣale ese wertsyaimpa*
'the king of Kauṣale, together with his retinue' (B 18b7 - Adams 98)

---

3 In the following examples, the adpositional phrase is underlined in order to indicate the word governed.

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Items listed by Adams as prepositions may nevertheless be postposed, though:

11) *triwoṣ ... kattākemmp=ese*
'mixed together with householders' (B 31b6=32a8 - Adams 98)

Of course, Penney notes in passing that postpositions in Tocharian too may "occasionally precede the noun" (62).

2.3. Discussion and Speculation

Despite the occasional contradictory example, there remains the fact that Tocharian is largely postpositional but with some prepositions and, as often holds for an SOV syntax, is for the most part suffixing (J. Greenberg 1966:73) but has some prefixes with solid Indo-European roots. On this basis, Penney arrives at the remarkable conclusion that, in Tocharian, the prepositions, like the postpositions in Umbrian and Old Lithuanian, "represent a survival from an earlier stage". In support of this position, he declares "this impression is reinforced by the agreement between the A and B forms of the prepositions, by contrast with the greater diversity to observe amongst the postpositions" (56). This indicates that the prepositions took shape during the time of common Tocharian, while the postposed items did not arise until A and B had begun to go their separate ways. Moreover, Penney points out that "A number of these postpositions are obviously of recent origin, some being recognisable as Tocharian case-forms: so B *ṣesa* 'together', for instance, is simply the perlative form of *ṣe* 'one'" (64). The fact that in Tocharian, like Hittite and Vedic Sanskrit

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but unlike Latin, Mycenaean, and Classical Greek, preverbs do not form inseparable compounds with their verbs further suggests the relatively young age of the local adverbs also functioning as postpositions. In short, Tocharian bears the residue of a shift in adpositional alignment.

Assuming, then, a postpositional syntax for Proto-Indo-European, this evidence would indicate that Proto-Tocharian had undergone a shift to prepositions, but that a change back to postpositions was underway as Proto-A and Proto-B were beginning to go their separate ways. This would of course be most extraordinary but plausible, given the time depth before the second half of the first millennium AD when the languages as we know them were committed to writing. After all, Mycenaean, dating from nearly two thousand years before Tocharian A and B, is already solidly prepositional. In any event, Proto-Tocharian is acknowledged to have diverged from the principal dialect of Proto-Indo-European quite early, probably earlier than any other groups save the Anatolian communities (see especially T. Gamkrelidze and Ivanov 1995:790ff, 828), so that it will have had an especially long time to develop independently.

A second change in word order remains a reasonable conjecture as well in this period of time, particularly when one recalls the status of the Tarim Basin, where the speakers of the Tocharian languages most likely lived, as a busy station along the silk road and other trade routes (Mallory 59). Contact with other languages, then, will have been frequent and intense, with outside influence a significant factor in the acceleration of language change. The syntax of Proto-Tocharian need not necessarily have changed entirely to an SVO pattern and then back to SOV, either, if, as suggested by the cases of Umbrian and Old

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Lithuanian, adpositional word order change is a very early step in a wholesale syntactic reversal. A cause for this chain of events, unfortunately, has not as yet made itself discernible to us. Certainly compound prepositions like *eneṣle* and *enepre* present a tantalizing mirror image to Umbrian *postin*, but there is no trace in Tocharian of the operation of any sort of prepositional second-place constraint in Proto-Tocharian, and so an explanation remains to be found.

3.0. Indo-Iranian

The earliest-attested Indo-Iranian languages, Vedic Sanskrit foremost among them, are traditionally characterized as having an SOV syntax with the generally-expected correlates, including postpositions (see, for example, Lehmann 30-4, especially 31f.). This consensus dates from B. Delbrück's pioneering studies of the syntax of Vedic and the other Indo-European languages, with his declaration, "Im Allgemeinen gilt nun die Regel, dass die echten Praepositionen traditionell hinter ihrem Casus stehen" (1888:21). Even a cursory examination of the Vedic texts, however, will reveal this statement to be at best an oversimplification, and indeed Delbrück, after naming *ā* and *purā* as exceptions that "regelmässig voranstehen," observes, "Wird aber die Praeposition stark betont, so rückt sie nach vorn" (21).

3.1. Classification of Indo-Iranian Local Adverbs

In addition to this poetic artificiality, there is also a sharp contrast between the functioning of the locative auxiliaries in Vedic (and Avestan), and in the

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Classical Greek and Latin languages, and even Classical Sanskrit. As in Old Hittite, category boundaries are very vaguely defined, although in Indo-Iranian the core auxiliaries show few signs if any of being derived directly from nouns. Rather, there is a great lack of clarity as to what environments may be considered appropriate for the classification of a given item as an adposition, a preverb, or an adverb. This fuzziness was observed long before the term "Fuzzy Set" was coined; as early as 1892, A. Jackson argues: "Prepositions in the sense of words that 'govern' oblique cases do not strictly exist in Avesta, any more than in Sanskrit. There are, however, a number of adverbial words which are used with the oblique cases and which define such cases more precisely" (203).

More recently, H. Hettrich, following G. Dunkel's proposed division of all Indo-European morphemes into four classes ("Wurzeln, Suffixe, Endungen," and "Partikeln" - 1992:155), offers the term "Lokalpartikeln" for these items (1991:27). He also notes that the syntactic binding of these forms is much looser than in later languages, whether associated with a verb, as a preverb, or with a noun, as an adposition (27). Moreover, Hettrich argues that this looseness is the consequence of the early stage of development of these local particles:

> Diachronisch betrachtet dürfte ein solches Stadium den Beginn einer Entwicklung anziegen, die von einem Lokaladverb in voller Verwendungsbreite ausgeht und einerseits zu Präverbien, andererseits meist zu Adpositionen führt... Adpositive... und präverbale... Wortstellung signalisieren - dies sei deutlich betont - in dieser Phase noch nicht Adpositionen bzw. Präverbien, sondern zeigen lediglich die

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In other words, these local adverbs are in Vedic and Avestan only beginning to develop their functions as adpositions and preverbs. They still retain many of the characteristics of independent adverbs, just as their Old Hittite counterparts display many nominal features. Certainly in a sentence such as the following:

12) *rājeva jer avr̥ké kṣeṣy antáḥ*
'Like a king you triumph and you live in security'
(RV VI.4.4; Hettrich 1993:148)

one is hard-pressed to determine the function of *antāḥ*, whether it may be considered to modify the verb *kṣeṣy*, govern the locative *avr̥kē*, or serve as an adverb to the entire phrase. Of course, Dunkel points out that there are enough "strictly adnominal" and "specifically preverbal" constructions involving the local adverbs to be able to take the position with confidence that "the tripartition adverb-preverb-adnominal pre/postposition had already begun in the protolanguage" (1990:169). But clearly this tripartition was not as fully established in Old Hittite, Vedic, Avestan, or even Homeric Greek as in the later languages. And accompanying this flexibility of categorization is a similar flexibility of word order.

3.2. Word Order

Indeed, as a counterweight to the statements by Delbrück cited above, G.

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Holland lists several "traditional grammarians" adhering to the position regarding the locative auxiliaries that "the original placement of these items was free" (156-7) in Proto-Indo-European. H. Reichelt, in his grammar of Avestan, comments, "Die aw. Präpositionen finden sich zum großern Teil sowohl vor als nach dem Kasus" (1967:266), and while T. Burrow describes their Sanskrit counterparts as "mainly postpositional" (1955:284), exceptions are not difficult to find.

No study of adpositional word order, or for that matter of the frequency of occurrence in the Vedas or in Avestan of the local adverbs as adpositions, preverbs, or adverbs is currently known to the author of the present undertaking. Such an effort would indeed be a possible fruitful avenue for further investigation into the development of Indo-European adpositions. Some more general examinations of Vedic word order, focusing mainly on the placement of the primary constituents, above all the verb, have been presented, most recently by J. Klein (1991, 1994). Most notable of Klein's findings is that, for a purportedly SOV language, relatively few Vedic sentences are in fact verb final: Of a sample drawn from RigVeda I.1-34, 34.5% of the sentences are verb-final, 15.7% verb-initial, and a plurality of 46.8% verb-medial (1994:99-100).

Another noteworthy study on pre-classical Sanskrit word order is that of J.- M. Verpoorten (1977), on the *Aitareya-Brāhmaṇa*. Verpoorten offers a date of 600 BC for the composition of this commentary on Vedic ritual, following Mylius 1970:421. Verpoorten devotes a substantial amount of attention to adpositions in the *Aitareya-Brāhmaṇa*. His conclusion concerning "Prépositions qui précèdent

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habituellement leur régime" is that "Elles sont plus nombreuses et variées que les autres" (285). And indeed, he lists some eighteen words occurring principally or exclusively in a preposed position, compared with eight postpositions. And while the former group includes the same *ā* and *purā* cited by Delbrück as generally prepositional, many of these are clearly of recent derivation from adverbs or declined substantive forms, such as *purastāt, antareṇa,* or *madhye.* On the other hand, the postpositions identified include *anu, abhi, adhi,* and *prati.* In short, the words that are (more often) postposed in the *Aitareya-Brāhmaṇa* continue to be those clearly derived from the set of local adverbs. Most of the preposed items are transparently of recent adpositional coinage.

While such evidence upholds the proposition that Vedic Sanskrit is not a canonically SOV language, it also shows that placement of these items is not simply "free". Only *purastāt* of the items listed by Verpoorten is fairly ambidextrous in its attested positioning (it is preposed in six instances and postposed in four - 286). All the others have a clear preference for one alignment or the other, albeit with exceptions. Moreover, the much closer identity of the set of postpositions in the *Aitareya-Brāhmaṇa* with the local particles is strongly indicative once again of an earlier word order that was more strongly postpositional.

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3.3. Reflexes of **en* in Indo-Iranian

One additional argument offers the possibility of further confirmation that the word order of the Proto-Sanskrit adpositional phrase was more pronouncedly postpositional than its attested successor. S. Vanséveren 2000 finds in the Sanskrit adverbs *jmán* and *héman* a remnant of a formerly postposed **en*, cognate with the inessive **en* found in a great many of the languages discussed in the present effort. These lexemes are derived by Vanséveren from locatives in **dʰĝʰ-ém* and **ĝʰey-em* plus a postposed, and ultimately suffixed, **en* (121, 124). The derivation of *jmán* is sufficiently distinct from cognates in Latin, Lithuanian, and Gothic (*hemō, žmuo, guma,* respectively) not to be an Indo-European inheritance (121). It may be seen, then, that at least one postposition was perhaps carried into Proto-Indic, even if by the time of attestation, it had already fossilized beyond grammatical usage. Another reflex of **en* in Proto-Indo-Iranian presents itself in Sanskrit *antár* and Avestan *antarə*, where it is compounded with the Indo-European contrastive suffix **-ter* in a formation parallel to that found in Italic with Latin *inter* (J. Gonda 1938:453-4). In this latter example, of course, it is impossible to draw any conclusions about syntactic requirements, but the forms cited by Vanséveren make these clear.

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3.4. Subsequent Developments

In contrast with this postulated state of affairs for Proto-Indic and, by extension, Proto-Iranian, one meets with increased flexibility for adpositional placement in Vedic and Avestan and, as Klein demonstrates, of sentential constituents in general at least in Vedic. Evidently here, too, a drift is in progress from a verb-final, postpositional syntax to a verb-medial, prepositional one. In Iranian, this drift continued: prepositions are common (though not universal) in Old Persian:

13) *pasāva adam kāram pārsam frāišayam* ***abiy*** *vištāspam* ***hacā*** *ragāyā*
'then I sent a Persian army to Vishtapa from Raga' (DB 3.1)

14) *pasāva* ***hadā*** *kārā adam ašiyavam* ***abiy*** *sakām* ***pasā*** *sakā*
'then I went forth with an army, to Saka against the Scythians'
(DB 5.21; both examples from M. Hale 1988:37)

It cannot be said, of course, that this syntactic change continued inexorably throughout the Iranian languages; modern Pashto, for example, has both prepositions and postpositions (D. Shafeev 1964:51-2), and indeed, postpositions remain in Old Persian, as the case of *radiy* described by J. Kahr 1976 and P. Hopper and E. Traugott 1993 illustrates (see Chapter I, Section 2.4.5). In Indic, on the other hand, this direction of drift was arrested and reversed. Postpositions are the expected forms in Classical Sanskrit and on to the modern languages; C. Masica cites the influence of several language groups, including Dravidian as well as Munda, Tibeto-Burman, Turkic, and Iranian also, in forming a regional

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syntactic *Sprachbund* favoring such syntactic features (1991:333). This sequence of events may offer some insight into the turbulent history of Tocharian syntax; the Indic languages too show a postpositional prehistory and perhaps the beginnings of a prepositional early history before reverting once again to postpositions.

4.0. Germanic

One would at first expect to find very little of interest in the Germanic languages to an investigation on this subject. All of the attested Germanic languages are very solidly prepositional, and any clues as to the origin of these segments, such as Gothic *fra-* as a reflex of the directive case of **per/*pr̥*, are as clear if not more so in other branches. Nevertheless, here too are some traces of earlier syntactic structures worthy of mention.

4.1. The Runic Inscriptions

The Runic Inscriptions represent the earliest attestations of any Germanic languages, with artifacts dating back to the second or third century AD. Even in these texts, only prepositions are found[4]:

[4] These texts are of course often only provisionally understood. However, the presence of prepositions recognizable from better-known Germanic languages is clear, and certainly the phrase in example 15) is indisputably prepositional.

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15) *anahahaislaginaz / frawaradaz*

= segmented by E. Antonsen 1975:33 as ***ana hahai*** *slaginaz frawaradaz*

'slain on [his] steed, Frawaradaz [= "advisor of lords"]'
(Möjbro Stone, Uppland, Sweden, AD 300)

16) *ekwiwazafter . woduri / dewitahalaiban : worahto*

= *ek wiwaz* ***after woduride witadahalaiban*** *worahto*

'I, Wiwaz, after [in memory of] Woduriaz the lord, wrought [this]'
(Tune stone A, Østfold, Norway, AD 400; Antonsen 44)

However, other facets of the syntax of these inscriptions indicate a great deal of upheaval in the Germanic languages during this time. Of earlier inscriptions such as those just cited in a Northwest Germanic dialect, Antonsen remarks, "With the exception of gen. constructions with inanimate heads and the single instances of quant. + N and of deictic adv. after an adj., the phrase structure rules point to a language with an S+O+V order" (25). In this assessment Antonsen is supported by R. Lass (1994:221). H. Nielsen, however, elaborates on and qualifies this conclusion:

> The word-order of Early Runic would thus appear to be in a transitional state, in which an older OV layer is being replaced by a younger one with features characteristic of VO typology: indications of the latter are the use of especially adjectives and possessive pronouns as postmodifiers and the preference of prepositions to postpositions... The most significant, inherited OV features are (besides thed position of the verb) the place of genitives before and that of appositions (titles) after the head nouns plus the occurrence of OV-type compounds. (2000:174)

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This postulation of a syntactic transition is further supported in later documents. Of the North Germanic texts from the early second half of the millennium, Antonsen declares, "The position of the verb has undergone a dramatic change: fully 83% of all sentences have the verb in medial position... From the scanty evidence available, there has been a breakdown in the old ordering. No longer do demonstratives and attributes appear exclusively after the head [as in an SOV pattern]" (25). M. Bean finds additional traces of an earlier verb-final syntax for Germanic in the compounds of these inscriptions: "The evidence of compounds in the older fuþark would also indicate a prior OV ordering within the NP itself, e.g. earlier Gen-N, Adj-N, etc." (1976:92-3). Bean further points out that "indirect evidence is provided by the internal order of compounds and in the use of V-F [verb-final] ordering in subordinate clauses in Old Saxon and Old High German" (92) for such an alignment throughout Germanic.

4.2. Old English

Rather surprisingly, the residue of postpositional structures present themselves in the Old English texts dating from rather later than the Runic inscriptions, beginning in the late first millennium. Indeed, Lass notes that the transition in word order noted by Antonsen is not yet complete here: "The bulk of the OE material in fact presents a picture either consistent with or easily derivable from an ancestor that looked like NWGmc. Both OV and VO orders

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are common in transitive clauses (the latter increasing later on...)... It is basically prepositional, though postpositions do appear" (222). Reminiscent of Latin *cum* in their syntactic usage, the great majority of these postpositions occurs after pronouns:

17) *ond hine ymb monig snellic særingc seleriste gebēah*
'and about him many a brave sea-warrior lay down to his hall rest'
(*Beowulf* 681; J. McLaughlin 1983:28)

However, it is possible to find an item postposed to a noun:

18) *he hine forwæc mancynne fram*
'he banished him from mankind'
(*Beowulf* 109-10; cited by F. Colman 1991:58; glosses from F. Klaeber 1950)

However, the majority of Old English adpositions, as noted, are preposed. Moreover, Colman cites several factors in questioning "the extent to which Old English is a postpositional language" (56). First, Colman notes the permissibility of separation of constituents (73), in some of these postpositional phrases:

19) *sua us unnytte geðohtas to cumað*
'so to us a useless thought came' (Colman 67)

Second, Colman draws attention to "the uniformity with which supposed postpositions co-occur with the dative case - even where the same formative

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functioning as a preposition normally functions as another case" (77). At the same time, according to Colman, the dative case in Old English is commonly used with "(ungoverned) circumstantial locatives" (79) or in other adverbial senses (84). Thus Colman is led to conclude that a great many of the presumed dative postpositional complements are in fact filling just such functions, and that most of the ostensible postpositions are in fact non-affixed preverbs (82). Furthermore, given that most postpositional objects are pronouns, Colman, much as A. Nocentini proposes for Umbrian, suggests a tendency to postpose "light-weight arguments" (75), but that the items governing them are not true prepositions any more than *cum* in *mēcum* is.

Colman admits, however, that not all of these postposed locative auxiliaries can be accounted for in this fashion, identifying the sentence in 18) as an example of a noun that can be explained only as a postpositional object (86). Besides, McLaughlin, as noted, has *hine* in 17) as an accusative. Unquestionably, then, there are occasional postpositions in Old English. Given the observed drift mentioned by Lass towards a verb-medial syntax in this stage of the language, it is highly unlikely that this is a new or variant formation that failed to take hold. Rather, it is a remnant of an earlier word order, which as such must have survived at least to some extent into Proto-Germanic. However, given the established status of prepositions in the Runic inscriptions, where the rest of the syntax is undergoing a systemic change, the Germanic languages may be supposed to offer additional testimony for the proposition that a shift in adpositional word order is one of the first steps in such a transformation.

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Various hypotheses have been offered for the change in Old English syntax, from external influences (A. Danchev 1991) to the collapse of the nominal case-marking system (D. Kastovsky 1992), but in actuality this process began long before the development of English as a distinct language.

5.0. The Celtic Languages

The Celtic languages display a number of syntactic features that are unique among the Indo-European family. Most striking, especially for the specific topic of word order, is the sentence-initial position of the verb, firmly entrenched in Old Irish and its Goidelic successors and common in the Brythonic sub-branch as well. Indeed, as J. Fife and E. Poppe point out in the introduction to their collection on Brythonic word order, this uniqueness in particular has made the Celtic languages an attractive arena for syntactic studies (1991:vii). Unfortunately, however, the prominence of this feature has eclipsed other aspects of these languages for many investigators of Celtic syntax, and studies of other aspects of this subject are difficult to find. There is, on the other hand, testimony that the Celtic languages did not always have a VSO word order or, for that matter, the prepositions that, as J. Greenberg has shown, necessarily accompany such an alignment (62).

5.1. Evidence from Continental Celtic

The earliest texts in a language thought to be Celtic date from the mid-first millennium BC. These represent the scraps known to us of Lepontic, thought to be a precursor to Gaulish. The texts that have been discovered

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include some forms suggesting Indo-European verbal morphology, most likely imperatives in these instances:

20) *uvamokozis plialeθu uvltiauiopos ariuonpos siteś tepu*

21) *pelkui pruiam teu karite iśos kalite palam*

(K. McCone 1997:381-2; W. Meid 1997:402)

McCone and Meid both interpret the second text as consisting of two clauses. As such, we have two verb-final sentences and one verb-medial.

A somewhat more substantial corpus exists of Gaulish, appearing some half-millennium later, and considered by Meid to display an essentially SVO syntax (403), while McCone prefers to reserve judgement (382). However, among the largest single Continental Celtic texts is the Celtiberian Botorrita I inscription, dated to the second or first century BC, and generally agreed to have an SOV syntax (McCone 379-80; Meid 400). More intriguing for the present investigation, the adpositional inventory shows a pattern remarkably similar to that of Oscan and Umbrian. Most of the ten locational segments listed by Meid (1995:154-6) either occur only as preverbs or are prepositions. The exception, however, is *eni*, a preverb but also a postposition with the sense 'in'. A short passage of the inscription illustrates the different adpositional formations quite neatly:

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22) *śtena eś uertai entara tiriś matuś tinpitus; neito tirí cantam eni oiśatus*

'Let him bring out these items from the enclosure within three days; let him bring them into the district of Neitos'

(Meid 1995:137; Botorrita I.A. 53-63)

As in several languages already discussed over the course of this effort, here is a language with a largely SOV pattern, but with prepositions for the most part. The form *entara* would appear to represent another, preposed reflex of **en*, here with the contrastive suffix as with Latin *inter* or Sanskrit *antar*. The fact that **en* (or **eni*) keeps appearing among the exceptional forms is surely deserving of further investigation.

5.2. Old Irish

These few Continental Celtic texts are sufficient to demonstrate that an overall shift to the VSO syntax found in Old Irish did not occur in Proto-Celtic. On the other hand, such texts as the Celtiberian Botorrita I Inscription leave open the possibility that the adpositional shift was in progress by that point. There has been speculation that some of the earliest Old Irish documents, a series of mostly poetic legal texts, reflect a word order for the recent prehistory of that language more in keeping with the Continental Celtic, although McCone reports that this pattern instead reflects the closeness of these works to Latin antecedents (376).

Old Irish, then, appears to us with its word order firmly established. In a sense, this is unfortunate, as there remains a lacuna between Gaulish and Insular Celtic, not only in time but in the lack of a direct genetic derivation from the

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former to the latter. Consequently, we find with nothing to account for the radical syntactic transformation that presents itself in the latter. Of course, this "missing link" would likely have shed little light on the shift in adpositional orientation, which was already as advanced in Celtiberian as in the Sabellian languages. In any event, prepositions in Old Irish, like the verbs, carry features not found elsewhere in the Indo-European family. Most prominent is their ability to be conjugated for person and number, as with *di* 'from', cognate with Latin *dē*, and which takes a complement in the dative case:

		singular	plural
1st person		*dím*	*dín(n)*
2nd person		*dít*	*díb*
3rd. person	(m./n.)	*de, dé*	*diib, díb, díb, díbsem*
	(f.)	*di/dí*	(as above, all genders)

(R. Thurneysen 1949:274)

At first glance, one is reminded of the enclitic pronoun suffixes in Hittite, especially in the Older stage of the language. However, it is necessary to recall that the Hittite formations consist of a possessive pronoun in the same case as the locative auxiliary, thus suggesting their substantival origins and characteristics. On the other hand, the Old Irish enclitic pronouns are the substantives. They, and not the prepositions, bear markings for gender, number, and case; there are somewhat distinct forms suffixed to prepositions taking the accusative, as in *fort, fort-su,* the second person singular forms of *for*

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'on' (Thurneysen 275). The pronominal roots are of course the same, but clearly the Hittite and Old Irish constructions have evolved along different lines.

The cliticized pronouns also occur with Old Irish preverbs, which, as elsewhere among the Indo-European languages, are drawn from the same set of local adverbs from which the adpositions are derived. Here, however, the preverbs have the added advantage of preserving the same sentence-second positions for these pronominal clitics as are held by their Hittite counterparts. An exception, however, is the semantically empty *no-*, derived from **nūḫ* 'now', and as such cognate with the Hittite sentence-initial conjunction *nu-*, employed with verbs not otherwise requiring a preverb so that the second position for these pronouns is preserved.

In some respects, the Old Irish locative auxiliaries reflect their Indo-European origins rather faithfully. The preverb *ne/ni* 'down' represents the only occurrence of that item as a distinct segment outside of Indo-Iranian, and *tar* 'across' corresponds to Sanskrit *tiraḥ*, Avestan *tarō* (Thurneysen 523, 531). However, while it is possible to observe from the Continental evidence that over its earlier history, Celtic syntax followed much the same course of development as many other Indo-European language branches, with a reverse in adpositional phrase orientation preceding a more comprehensive change, clues for the more radical Old Irish change in word order do not readily present themselves, among the local formatives or elsewhere.

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6.0. Armenian and Albanian

These two languages may be treated together not due to any particularly close relationship between them, but because of their similar circumstances. Both make their earliest appearances already heavily and thoroughly influenced at all levels of the language by their neighbors, both Indo-European and, in the case of Armenian, non-Indo-European. Consequently, their value to a comparative study of any aspect of Indo-European syntax is limited. Nevertheless, there are formations in these tongues deserving of comment.

6.1. Classical Armenian

B. Vaux reports scholarly vacillation over whether the word order of Classical Armenian should be considered SVO or free (1994/5:18-9) before adopting the former position on the basis of the behavior of clitics derived from demonstratives that follow their head constituents (20-1). The adpositional system is overwhelmingly preposing. S. Mann lists one item, *zat* 'except, besides', as postpositional, which he derives from the preposition *z-*, and the widely-attested Indo-European local adverb **at(i)* 'back, away' (1968:39-40). This would appear to be a mirror-image of the Umbrian compound *posti(n),* with a preposition reanalyzed as a prefix to an erstwhile substantive, but this compound of course did not spark a corresponding syntactic reanalysis.

Other than *zat'*, while there has not been a great deal of interest in the

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Classical Armenian adpositional system from a comparative perspective, one finds some remarkable constructions from the stadpoint of universals of language change. One such item is this preposition z-. This is not a segment that occurs independently but is rather a proclitic. It may carry such senses as 'concerning' with the ablative case or 'around' with the instrumental, but it may also be a semantically empty marker of a definite accusative:

23) *ew tanǰec'in z-na tasn ayr*
'and ten men tortured him'

24) *or arar z-erkin-s ew z-erkir*
'who made the heavens and the earth' (both from Vaux 19)

Typologically, this formation is strikingly similar to the 'personal *a*' construction that has emerged in Spanish, wherein the preposition *a*, derived from Latin *ad*, has become obligatory with animate direct objects[5]. In both instances, a new object marker is springing up in a preposing syntax, where a new case suffix drawn from an adposition would not be expected. Such newly grammaticalized markers moreover commonly begin with a restricted application, be it for the category *+definite* or *+animate*, before becoming more generalized (see, for example, Hopper and Traugott 1993:100-3).

However, some of the core Armenian prepositions may take on even more of the appearance of case prefixes, with the prepositions affixing to both a

[5] Romanian *pe*, derived for Latin *per*, has a similar function.

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noun and its modifying adjective, forming a preposing counterpart to the incipient case desinences in Sabellian and Lithuanian:

25) *patmeac'... z-amenayn z-bansn*
'he tells all the tales' (Est. 4.12)

26) *y-ahagin y-anapati*
'in the vast desert' (Tom. II, 423; both examples from M. Minassian 1980:33)

27) *z-noyn z-hogewor z-kerakurn keran*
'they will eat the same celestial nourishment' (1 Cor. 10.3; Minassian 34)

This construction is not obligatory, and Minassian reports that it occurs "plus facilement si le determinant antéposé est séparé de son déterminé":

28) *z-noyn usoyc' noc'a z-hambrerut'iwn*
'he signaled to them the same thing, patience' (Ag. 356.11; Minassian 33)

One may speculate that this tendency arose from sentences such as 28), where the repetition of the preposition is helpful in reinforcing the agreement of an adjective separated from its noun, before its extension to more conventional noun phrases. The frequency of such proclitics in Armenian may be borrowed from Georgian and other Kartvelian languages, where they are widespread. In any event, such formations, even if they do not take root in a language, offer exciting evidence for the cyclical nature of morphosyntactic development, whereby a morphological system, eroded by inevitable phonological attrition,

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may be replenished by the grammaticalization of formerly independent words. Examples of such phenomena are of course very easy to find in languages with an agglutinating morphology and a synthetic, verb-final, postpositional syntax. Cases such as that of Classical Armenian demonstrate that they may also be found in inflecting or isolating, analytic, and prepositional grammar.

6.2. Albanian

Albanian does not appear to us in written form until the fifteenth century AD, and its grammar shows many of the *Sprachbund* characteristics shared with the other languages of the Balkans. As such, discernible traces of the development of its word order from the time of Proto-Indo-European are not evident. Adpositional phrases are preposed, and the expression of case relations in the sense advanced by C. Fillmore is thoroughly analytical. As Sh.Demiraj notes, "Die Präpositionen werden im Albanischen obligatorisch vor ein Substantiv (oder ein Pronomen) in adverbialen oder adnominalen gestellt" (1993:225).

The most remarkable feature of the Albanian prepositional system is the capability of a small number of items, *nga* 'from, away', *tek* 'to', and *ke* also 'to', that govern complements in the nominative case, which, as Demiraj points out, is quite unique among the Indo-European languages (1998:497). These are, however, not core prepositions derived from local adverbs, or "unechten" as Demiraj terms them (1993:228). Rather, they are derived from conjunctions; Demiraj proposes that their reanalysis stems from sentences of the type *Vajta*

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tek shoku është 'I'm going there, where my friend is'. The deletion of the verb *është* in the subordiate clause would then leave a phrase *Vajta tek shoku* 'I'm going to the friend' (230)[6].

7.0. Conclusion

As may be observed, then, all of the branches of the Indo-European family considered in this chapter describe a general drift towards a prepositional word order. Germanic, Celtic, Armenian, and Albanian are all solidly prepositional, although Germanic and Celtic both present possible indications of a postpositional prehistory. Classical Armenian on the other hand appears to be grammaticalizing some of its prepositions more purely into affixed markers. Indo-Iranian is more ambiguous, but these languages likely were gradually drifting towards prepositional patterns before the influences of neighboring languages had the effect of arresting and, in the case of Indic, completely reversing this tendency. The passage of time between the postulated breakup of the Proto-Indo-European speech community and the attestation of the Tocharian languages in the middle of the first millennum AD certainly allows for a reversal of this general Indo-European drift similar to that which had taken hold in Indic by the Classical Sanskrit era. There was unquestionably, then, a similar process

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6 The derivations of two of these items appears straightforward: *tek* <Proto-Albanian **tai ku* < Proto-Indo-European **toi $k^{w}u$* 'to that which'; *ke* < **$k^{w}ōi$* or **$k^{w}ōd$* (Orel 1998:450, 174; see also Demiraj 1997:383). However, the source of *nga* is more controversial. Orel draws it from **en* 'in' + **kom* 'to, towards', reflected also in Slavic *ku* (292). Demiraj sees the second member of this compound as **kata*, with cognates in Greek and Hittite (1993:199). Neither of these analyses, however, accounts for the preposition's ability to govern the nominative case, as Mann's derivation from **en-qu̯o* (i.e., **$k^{w}o$*) is able to do (1977:204-5).

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at work in all these language groups after they had become independent of one another, and it must logically be supposed that it had begun while they were still in some contact as a single, if large and varied, group.

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Chapter VII:
Conclusions

1.0 Recapitulation

The present effort has been concerned with the development of adpositions in various Indo-European languages, from their origins through the changes in their syntactic alignment. The research undertaken for this investigation has turned up a remarkable diversity of courses of evolution for this part of speech. In the earliest-attested languages, Old Hittite and, to a lesser extent, Luwian, one finds, by all appearances, nouns in the process of taking on the functions of adpositions as well as preverbs. They are inflected for case; they govern genitive complements, while in most languages the fundamental members of the adpositional inventory have objects in the more oblique cases; and they may be modified by enclitic possessive pronouns. On the other hand, in languages more distant from their Indo-European roots, such as Umbrian and Old Lithuanian, adpositions carry none of these nominal characteristics and are instead derived from local adverbs.

Of especially great interest to this investigation is the change in word order in the adpositional phrase, and of the languages or language groups examined most closely, three, Luwian-Lycian, Umbrian, and Lithuanian, show great evidence of a shift from a postpositional to a prepositional syntax. These too were spurred by a variety of causes. G. Holland, in his dissertation on word order change in selected Indo-European languages, concludes that "The changes described came about through the exploitation of marked word order patterns"

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(1980:185). These shifts in adpositional alignment, however, evidently were motivated not by any gravitation towards a marked word order; marked word order of any sort is extremely rare in adpositional phrases. Rather, these changes were set in motion by various reanalyses involving existing word-order constraints and local adverbs, as will be indicated in the following summaries.

1.1. Hittite

The second chapter examined the adpositional system in Hittite. The earliest attested Indo-European language, Hittite shows most clearly the nominal origins of many of that language's postpositions. Some common Indo-European adpositions have cognates occurring in Hittite as nouns; a prominent example is Latin *ante*, Greek *ἀντί*, and Sanskrit *ánti*, cognate with Hittite *ḫanza* 'forehead, front.' Old Hittite postpositions also frequently take possessive pronouns, as in:

appam-mit	'behind me'
appan-tit	'behind you'
appan-šit	'behind him/her/it', etc.

Moreover, doublets such as *píran* 'in front, before'/*parā* 'forward'; *kattan* 'at the side'/*katta* 'downward'; and *appan* 'in back, behind'/*appā* 'away' certainly reflect the patterning of a declensional paradigm. The former item in each pair appears with an accusative case desinence, and the latter with that of the directive case in Old Hittite, as befits a sense of "moving in various directions

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from one place to another" (Goetze 1963:98). The other item in each set is somewhat more problematic in that its function tends to be indicative of location, which does not fit the accusative case well; the semantic change is accounted for in Chapter 2, Section 4.2). The noteworthy feature, in any case, is the difference in sense corresponding to different case forms. Further evidence of the recently nominal status of these items is their tendency to take genitive objects (W. Lehmann 1974:212).

While the seeming overlap in behavior of the lexemes under discussion, especially in the Older stage of the language, between nominal and postpositional characteristics is of great interest to this investigation, so too is the similar vagueness in demarcating between adpositional, preverbal, and adverbial functions. The overtly nominal qualities of the postpositions tend to fade between Old and Middle Hittite, but the somewhat blurry boundaries among the latter set remain. The application of Fuzzy Set Theory to these locative auxiliaries has proven useful in assessing the full range of their functions and in demonstrating that, especially in the Older stage of the language, these items often do not fit neatly under one label such as 'adposition', 'preverb', 'adverb', or indeed 'noun'. On the other hand, by the time of Middle Hittite, these items had lost most of their nominal qualities, and resembled their counterparts in the other Indo-European languages to a much greater extent, including governing of complements in the dative-locative case, and the collapse of the directional-locational distinction according to *-a* and *-an* endings.

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1.2. Luwian and Lycian

It is not at all clear that Hittite and Luwian are especially close in their genetic relationship beyond their common geography and antiquity. Nevertheless, they are typologically quite similar and nearly contemporary with one another. However, in the relatively few attested Luwian texts, some remarkable features new to the Anatolian languages appear. By far the most striking is the occurrence of a few prepositions, which are, as established, unknown in Hittite. Some of these prepositions are, however, apparently cognate with Hittite (*anta*, *šarra*, and *šarri* in Luwian, as compared to Hittite *anda*, *šarā*, and *šēr*). Some postpositions also are found in Luwian evidently cognate with Hittite counterparts (*appan* and *parran*, alongside Hittite *appan* and *píran*). The prepositions also correlate with an increasing flexibility of word order in Luwian in contrast with Hittite, whereby the sentence-final position of the verb, while still common, is no longer obligatory as in the latter language.

These developments represent a precursor to the syntactic characteristics of a later Anatolian tongue, Lycian. In Lycian, the successor to Luwian across southern Anatolia, sentential word order is by no means rigid, but the verb almost never comes at the end of the sentence. More directly relevant to the present study, postpositions are evidently unknown; the syntax has become entirely prepositional. And the Lycian prepositions etymologically relatable to Luwian forms are derived from prepositions in the older language, not postpositions.

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In short, the change in word order underway in Luwian is complete in Lycian. Unfortunately, within the Luwian texts available to us, there is no evidence to suggest how this shift may have started, or which aspect of it, the spread of prepositions or the retraction of the verb, began earlier; the end of the process is visible, but not the beginning. While there is nothing to contradict the hypothesis supported by the Italic and Baltic languages that the establishment of prepositions is an especially early step in the change from an OV to a VO syntax, there is in the slender data known to us nothing to corroborate it, either.

1.3. Umbrian and other Italic Languages

The state of affairs found among the Italic languages is especially striking. Latin in itself of course seems at first glance unremarkable, relying almost entirely on prepositions. An extremely interesting exception, however, is the seemingly obligatory postposing of *cum* with personal pronouns, as in *mēcum*, *nōbīscum*, and so forth, although this may at least in part stem from the unusual history of **kom*, as discussed in Chapter IV, Section 4.1.1. In Latin's sister language Umbrian, on the other hand, many of the same particles which in Latin are preposed appear as postpositions. Corresponding to the Latin preposition *in, ad, prō,* and *cum* are the postposed *-en, -aŕ, -per,* and *-ko*. Even where, as between *cum* and *-ko,* the semantic correspondences are not exact, the phonological matches are unmistakable. Complicating matters further is the fact that many other Umbrian adpositions, including *da-, e(he), pusti,* and *anter-,*

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corresponding to Latin *de, ex, post,* and *inter* respectively, are preposed. Moreover, lest this opposing placement of adpositions be taken as evidence that Latin and Umbrian are not after all very closely related, it must be pointed out that in Umbrian's indisputably close relative Oscan **az** and **pru**, corresponding to the Umbrian postpositions ***-aŕ*** and *-per,* are preposed. In Oscan, however, *en* 'in' may be either preposed or postposed. Indeed, in at least one instance the postposition follows both noun and adjective, thus appearing to function as a case suffix: **húrtín Kerríiín** 'in luco Cereali', a construction found also in Umbrian.

In contrast with the evidence offered by the Southern Anatolian languages, Umbrian contains a very manifest clue as to the causes of its change in adpositional word order. The aforementioned postpositions, which, given their centrality to the Umbrian adpositional system and slender phonological shapes, may safely be regarded as having filled this role longer than their preposed counterparts, have an additional constraint in their placement: they must occupy the second position in any adpositional phrase, regardless of its length. Furthermore, the Umbrian preposition *posti(n)* 'each, at each, per' clearly shows itself to be a compound of **posti* 'behind', cognate with Latin *post,* and **en.* It may be presumed that, before such a phrase as **posti-en* was ossified as an adposition, with the former member still functioning as a quasi-nominal adverb and the latter firmly entrenched in its second position, the resulting placement of this form at the beginning of the phrase it governs sparked a

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syntactic reanalysis in Umbrian. As a result, subsequent newly-derived adpositions will have been preposed by analogy with such compounds.

We are fortunate to have in Umbrian the record of a language whose syntactic shift took place relatively soon before attestation. The cause of such a change is not evident in Oscan or Latin. However, given the prepositional status in Oscan of items cognate with Umbrian postpositions, the reanalysis leading to this change took place earlier in Oscan. Furthermore, the prepositional ordering of Latin *in* suggests a still earlier change in that language. Umbrian, however, presents us at least with one possible mechanism for adpositional word order change.

1.4. Old Lithuanian and the Baltic Languages

Among the Baltic languages, and in Lithuanian in particular, one finds a predilection for prepositional constructions, but by no means exclusively so. Lithuanian certainly follows this pattern; modern Lithuanian relies virtually entirely on prepositions. Old Lithuanian, however, also shows a significant degree of postposing. Indeed, the most prominent of these postpositions, *-n(a)* and *-p(i)*, become so tightly bound to their preceding noun phrases that they come to form what many scholars consider case suffixes. On the other hand, *-n(a)* is cognate with the modern Lithuanian preposition *į*. As such, it is also cognate with Greek ἐν, Latin *in*, and English *in*. One thus observes within the history of the Lithuanian language the same striking twofold distribution of this

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adposition found in the Italic languages. Moreover, this same root **en* is also the source of the modern Lithuanian locative case suffix; this segment over the course of the history of the language has clearly been analyzed and reanalyzed as an adposition at many different times.

While the traces of the change in adpositional syntax are not as clear in Lithuanian as they are in Umbrian, V. Ambrazas and W. Schmalstieg in their respective works nevertheless offer a reasonable conjecture for its cause. Ambrazas offers clear evidence that Lithuanian throughout its attested history has been in the process of shifting from an OV to a VO syntax. In an OV syntactic system, modifiers, including adverbs and, thus, the local adverbs that tend to give rise to adpositions, normally precede their modified object. Consequently, the prepositional position would come to be regarded as a natural place for newly-reanalyzed adpositions. Indeed, Ambrazas argues that this change from postpositions to prepositions was one of the earliest steps in the overall switch in Lithuanian syntactic orientation and doubtless helped to spark subsequent reanalyses leading to new word orders. Certainly in both Lithuanian and Umbrian, prepositions are more established in the language than non-final verb placement, and as such the two languages suggest that the change in adpositional alignment is one of the earliest steps in a given language's wholesale syntactic remodeling.

Also of great significance to the present investigation from the data presented by Lithuanian are the multiple layers of **en* from various stages of the language's development. This same root has manifestly been analyzed as an

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adposition no less than three times, the first such event finally yielding the locative case suffix; the second well on its way also to becoming another inflectional affix; and the third resulting in the preposition in use today. This repeated layering of the same segment in the same role at different times offers a valuable insight into the development of such parts of speech. Adpositions are, it must be concluded, not necessarily coined once in the course of a language's history to remain in that same function over great periods of time. It is possible for this to happen, of course, given that most of the core prepositions of the modern Romance languages are descended from the same ones found in Old Latin. But it cannot be assumed that such a state of affairs holds in any given language. In Lithuanian, however, the same local adverb has been reanalyzed as a preposition repeatedly, presumably to fill a void left by the continuing evolution of its predecessor. This fact serves to remind us that the development of grammatical markers, be they affixes or independent words, is a continuing, cyclical process.

One implication of this state of affairs for Indo-European studies, incidentally, is that we cannot assume the adpositions we see in the various daughter languages were also adpositions in the proto-language. A root with adposed reflexes in many languages may have been a local adverb, a noun, or even some other part of speech in Proto-Indo-European and, by the same processes of grammaticalization that operate everywhere and at all times, yielded adpositions in many different branches as a result of parallel development rather than direct cognacy. These will be either postposed or preposed according to each language's syntax at the time of reanalysis.

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1.5. Other Indo-European Branches

The other branches of the Indo-European family offer varying degrees of support for the general direction of drift observed in the languages discussed in greater depth. In some cases, especially in Celtic and Germanic, there is evidence, not great in quantity but nevertheless clear, that the prepositional word order found in the attested languages was preceded by a postpositional alignment at some time in their respective prehistories. The appearance of a postposed *eni* 'in(to)' in Celtiberian in particular represents an exciting parallel to the structures found in Umbrian, Old Lithuanian, and, in the more distant past, Slavic. In Germanic, on the other hand, the Runic texts show prepositions but also a continuing syntactic upheaval ending in a verb-medial pattern, while Old English retains some occasional postpositions.

The Vedic and Avestan texts show a great deal of flexibility of word order, including in the placement of the locative auxiliaries. These items are more likely to be postposed, but exceptions are permissible, and lexemes with locatival connotations, such as *ā* and *púraḥ* in Sanskrit, display the opposite tendency. Indeed, J.-M. Verpoorten demonstrates that prepositional items are in the majority in the Vedic prose work *Aitareya-Brāhmaṇa*, but most of these are clearly of recent derivation, and are derived from inflected substantives, usually via an adverbial state. This is, of course, the normal path of adpositional derivation proposed for most Indo-European languages apart from Old Hittite

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and Proto-Luwian, but these newer forms lack the spatial or directional sense that characterizes the core set of adpositions in all of these tongues. In any event, the beginnings of a drift towards prepositions is seen in Indo-Iranian, but the influence of neighboring language communities had the effect of arresting this tendency before it could become as pervasive preposed forms are in Germanic or Celtic.

The consequences of outside influences are even more pervasive in Tocharian, where one finds a postposing syntax as well as a suffixing, rather agglutinative morphology. In Tocharian, the residue of earlier formations indicates a prepositional period, in stark contrast with the evidence from other branches of Indo-European. Tocharian, like Lithuanian, presents multiple derivations of **en*, for example, but here, postposed *-na* is of more recent origin, while the prefixes *e(n)-* and *y(n)-* reflect earlier prepositions. One would hope for the sake of reconciling this testimony with that of the other branches to detect remnants of a still-earlier postpositional syntax, but this is not discernible at present. Nevertheless, the most logical supposition in light of the facts from these other subgroups is that Tocharian switched to prepositions, or at least was beginning to, probably to a greater extent though than Indo-Iranian, and then changed again as a result of contact with Turkic or other agglutinating languages.

Classical Armenian and Albanian do not hold out significant hints of postpositions in the time before the prepositions that characterize their adpositional formations took hold. Nevertheless, some of the Armenian

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prepositions appear to have become grammaticalized to an exceptional degree, marking such concepts as definite direct object and occasionally taking on characteristics of case prefixes. Further study of these items may well offer an exciting opportunity to consider whether, or perhaps to what extent, prepositions and prefixes develop similarly to or differently from postpositions and suffixes, which are far more likely to develop into declensional morphology.

2.0. Summary and Final Thoughts

The driving force behind all the steps observed in the origin and development of adpositions and their attendant phrases among the Indo-European languages is reanalysis, although this reanalysis may take place in various fashions. In Old Hittite and, most likely, in the pre-history of Luwian, substantives with a directional or spatial sense ('the back', the top', the side', and so on) began to serve a more grammatical function. But along the way to this state, they continued to govern complements in the genitive case or, when this complement was a pronoun, to take possessive enclitic pronouns. Only after subsequent reanalyses left them devoid of nominal characteristics did they shed these enclitics and take objects in the more locationally-oriented dative-locative or ablative cases. Such reanalyses also rendered them more fit to modify verbs as preverbs.

Indeed, the Anatolian languages offer especially valuable evidence for the nominal origins of most Indo-European adpositions. A. Meillet had already proposed that the attested Indo-European adpositions share with all nouns in the proto-language the reflections of a declensional paradigm, as outlined in Chapter

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I, Section 2.3. The Anatolian items further substantiate Meillet's schema. From Meillet's root **perə* 'devant' (1937:350) one finds the Old Hittite directive form *parā* and the inanimate nominative/accusative *píran*, and in Luwian, another nominative/accusative form, *parran*, stands alongside a locative *parī*. All the more striking is that, in Old Hittite, forms like *píran* are, in formal terms, still nouns, as demonstrated above all by their ability to take enclitic possessive pronouns like any other noun in the language. While formally nouns, however, they have taken on the function of supplementing the Old Hittite declensional system in expressing case relations as postulated by C. Fillmore, by binding a substantive, in this instance the noun or pronoun governed by an item such as *píran, appan* or *šēr*, in a syntactic-semantic relationship to the sentence as a whole (1968:21). This represents the transition, which involves a certain degree of overlap, between relational noun, such as 'back, 'front', or 'head', and adposition.

In the other Indo-European languages, too, adpositions are ultimately derived from substantives, but they first pass through an adverbial stage before becoming analyzed as adpositions. Here, too, however, there are further stages of reanalysis. Morphological suffixes often disappear as the memory of the item's substantival function recedes and it comes to be regarded more purely as an adposition, as in Lithuanian *ta tarp* 'between', formerly a noun *tárpu* in the instrumental case. The difference, therefore, between the Anatolian group and the other Indo-European languages in this regard is that, in the larger set,

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evolution followed a course of *noun/adjective* > *adverb* > *adposition* or *preverb*, in that both adpositions and preverbs derive from adverbs rather than one from the other. In Anatolian, on the other hand, the course or cline is generally *noun* > *postposition* > *adverb* > *preverb*, where an inflected nominal lexeme takes on adpositional functions and subsequently, with its nominal qualities fading away, those of an adverb. Why there are evidently two different, language-specific courses of development for these locative auxiliaries is unquestionably well deserving of further investigation.

Word order change in adpositional phrases is also the result of more than one type of reanalysis. One type, exemplified by Umbrian, is the product of compounding of local adverbs, most likely at different stages of development. In Umbrian, this change grew out of compounds such as *postin*, comprised of **en*, a long-standing postposition, and **posti*, at this point still an adverb retaining a locative case suffix. Since postpositions such as **en* were required to take the second position in adpositional phrases, a construction **posti en [NOUN]* would eventually result in a reanalysis of the new compound as a preposition. Elsewhere, as in Lithuanian, a tendency for modifiers such as adverbs, including locational or directional adverbs, to be preposed to their modified complements would, after their reanalysis as prepositions, result in a shift in orientation for the adpositional phrases.

In these or other possible reanalytic processes not yet observed, there is no one-time or occasional minting of new adpositions. Rather, it takes places

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continuously, as some adpositions undergo still further reanalyses and become case affixes or possibly derivational morphemes, or simply fall out of use. At times, there may be a redundancy of expressions for the same concept, possibly even, as in Lithuanian or Tocharian, derived from the same source, and at other times a void arises from such events, impelling the reanalysis of a replacement. But it is worth stating once again that the adpositions we see in the known Indo-European languages are unlikely to be the same adpositions that undoubtedly existed in the proto-language, and as the attestations grow later, the odds of such a survival grow vanishingly small. Cognate local particles in the various daughter languages are instead the products of commonly-held local adverbs that later became adpositions, as different constructions involving **en* and other segments testify. Different placements in Umbrian and Latin, or in Celtiberian and Old Irish, demonstrate that these reanalyses took place at different points in each language's syntactic development.

Certainly further research on this subject will be most welcome. Fruitful areas for deeper investigation include language groups whose adpositional grammar has been widely thought to be essentially settled, but where textual evidence indicates that as late as the relatively recent prehistory, this was not the case. Included especially in this set are Indo-Iranian, Tocharian, and also Greek; while these languages lack the syntactic ambiguity in their adpositional phrases found in Luwian, Umbrian, or Old Lithuanian, further investigation of their prehistories should reveal additional insights into the evolution of such formations. In addition, complexities arising from the myriad adjoining dialects

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in the Baltic languages remain to be explored, and newly-published texts in the Anatolian languages may continue to be expected. But the basic course of development for Indo-European adpositions and their phrases is clear.

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